



Quantum Universality at an Absorbing-State Phase Transition

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Abstract

The study investigates quantum universality at an absorbing-state phase transition using the Truncated Wigner Approximation (TWA) for spins method [1, 2]. To achieve this goal, the study focuses on a system of Rydberg atoms trapped in a lattice, which exhibits a nonequilibrium phase transition [3, 4]. The Non-Equilibrium Phase Transition (NEPT) occurs through a contact process that models the stochastic spreading and decay of activity, where active sites can activate neighboring inactive ones and spontaneously deactivate. Rydberg atoms are particularly suitable for this purpose due to a physical property called *Rydberg facilitation*, a phenomenon in which an excited Rydberg atom assists nearby atoms to become excited to the Rydberg state at a specific distance through an interaction-induced energy level shift [5].

A theoretical foundations for nonequilibrium critical phenomena and Absorbing-State Phase Transitions (ASPT) [1] is established, together with a review of previous attempts to extract critical exponents, which are fundamental quantities characterizing groups of physical systems that exhibit the same critical behavior near phase transitions, known as universality classes. One of the most interesting universality classes, Directed Percolation (DP), appears in the dynamics of dissipative atomic Rydberg lattices [6].

This research advances previous work by addressing some limitations faced by numerical methods such as Monte Carlo simulations, which require the Rydberg system to be in a regime dominated by dephasing [7, 8]. This restriction suppresses meaningful quantum mechanical effects such as quantum fluctuations, correlations, and entanglement between Rydberg atoms. To overcome these limitations, a more advanced method, TWA, is employed [2, 9]. This method performs a mapping of the Lindblad master equation from Hilbert space to a set of Wigner functions in phase space, allowing the dynamics of the Rydberg system to evolve semiclassically.

Within this framework, TWA enables the extraction of critical exponents in the low-dephasing regime by studying the scaling laws near system criticality, and results from both regimes are discussed. Moreover, the study also investigates an equilibrium phenomenon — the Transverse Field Ising Model (TFIM), which is well known — to address certain concerns about the method when applied to equilibrium phase transitions [10]. Lastly, the study outlines a plan to further benchmark the new method using different models.

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Abbreviations

2LS	2-Level-System
ASPT	Absorbing-State Phase Transition
DCTWA	Discrete Continuous Truncated Wigner Approximation
DP	Directed Percolation
DTWA	Discrete Truncated Wigner Approximation
EM	Euler Maruyama
EPT	Equilibrium Phase Transition
FPE	Fokker-Planck Equation
FWF	Flatten Wigner Function
NEPT	Non-Equilibrium Phase Transition
OBE	Optical Bloch Equations
ODE	Ordinary Differential Equation
PPO	Phase-Point Operator
SDE	Stochastics Differential Equation
TFIM	Transverse Field Ising Model
TWA	Truncated Wigner Approximation

Introduction

In both Equilibrium Phase Transition (EPT) and Non-Equilibrium Phase Transition (NEPT), the behavior of systems near a phase transition is characterized by *critical exponents*, which govern how physical quantities change as the system approaches the critical point [1, 11]. In EPT, these exponents and their associated scaling laws are well understood for various models, such as the transverse-field Ising model (TFIM), where analytical and numerical methods have been successfully applied [11]. In contrast, the determination of critical exponents in NEPT remains an active area of research due to the inherent nonequilibrium nature of these systems [12].

There are three main critical exponents. The exponent β describes how the stationary density of active sites vanishes near the critical point

$$\rho \sim (p - p_c)^\beta,$$

where p is the control parameter of the system, for example, the probability that activity spreads from an active site to a neighboring site, and p_c is the critical value of p at which the system undergoes a phase transition from the absorbing state (no activity) to the active state (sustained activity).

Two other important correlation exponents are the spatial correlation exponent $\nu_\perp$, which governs the divergence of the spatial correlation length

$$\xi_\perp \sim |p - p_c|^{-\nu_\perp},$$

and the temporal correlation exponent $\nu_\parallel$, which governs the divergence of the temporal correlation length

$$\xi_\parallel \sim |p - p_c|^{-\nu_\parallel}.$$

These three exponents are crucial in identifying the universality class of different physical models.

In the past, Monte Carlo simulations have been shown to extract these critical exponents with a good degree of accuracy for dissipative Rydberg cold gases [6, 13]. However, this approach is limited to the classical regime, where the dephasing satisfies $\gamma \gg \Gamma, \Omega$, with Γ being the decay rate and Ω the driving strength of the light field. In this regime, the system can be simulated using classical rate equations [14]. With TWA for spins, this limitation is addressed. This work presents a systematic approach for the semiclassical treatment of many-body dynamics of interacting, open spin systems. The method works by embedding the Discrete Truncated Wigner Approximation (DTWA) in a continuous phase space, thereby forming a hybrid discrete-continuous variant [2, 9].

This continuous embedding allows the extension of the method to open spin systems with dissipative properties such as dephasing, losses, and incoherent driving, while preserving the positive-definite Wigner distribution of state initialization from its predecessor, DTWA. This is particularly useful for Rydberg lattices that are coherently driven in a Markovian environment containing the aforementioned dissipative properties [6]. Ultimately, this method allows the critical exponents of the Rydberg system in the classical regime to be extracted with high accuracy, enabling simulations at low dephasing to extract the bulk exponents and gain insight into the quantum absorbing-state phase transition.

This work is structured as follows. First, it discusses the theoretical foundations of nonequilibrium critical phenomena, specifically the mechanism of phase transitions between the absorbing state and the active state, and the behavior of collective systems near criticality. Then, it investigates the properties of critical exponents and the methods used to extract them through the mean-field approximations. The next section discusses the physics of Rydberg atoms, in particular the mechanism of Rydberg facilitation and the different types of interactions that Rydberg atoms exhibit, i.e., dipole-dipole and van der Waals interactions. This section then reviews past results related to the extraction of critical exponents in similar Rydberg lattices.

The following section discusses the TWA method in more detail, first addressing the operator-differential mappings that yield the exact equations of motion for the continuous $SU(2)$ Wigner function of spins, and then explaining how the continuous embedding method allows the truncation of higher-order derivative terms present in many-spin interactions.

Finally, the results of critical exponents obtained in the low-dephasing regime through finite-size extrapolation with increasing system sizes are discussed. The work also addresses some limitations of the method in a system without coupling to external noise, such as the TFIM in a quantum phase transition. Overall, the results show that the critical exponents in the low-dephasing regime behave similarly to those in the classical case. The work then attempts to justify these results from a numerical perspective.

Part I

Foundations

1 Non-equilibrium Phase Transition and Universality Class

1.1 Universality Class and Directed Percolation

This section follows the review of Hye Hinrichsen, *Nonequilibrium Critical Phenomena and Phase Transitions into Absorbing States*, due to its comprehensive overview of the theoretical foundations of Non-Equilibrium Phase Transition (NEPT), Directed Percolation (DP), and the determination of critical exponents [1]. Furthermore, all results presented in this thesis, including the extraction of β , δ , and $z = \nu_{\parallel}/\nu_{\perp}$ through power-law scaling, the analysis of active-state density decay and finite-size expansion, are obtained following Hinrichsen's paper.

In many-body physics, a phase transition refers to a sudden qualitative change in the macroscopic behavior of a system as some control parameter, such as temperature or external field, is varied. For instance, heating up water can turn it into steam, or a magnet losing its magnetization when applying an opposing external magnetic field. These phenomena are *equilibrium* phase transitions; they happen in systems that are in thermal equilibrium, where energy exchange has settled and microscopic configurations follow Boltzmann statistics.

However, in nature, most systems are not in thermal equilibrium, constantly changing instead of remaining in a steady, balanced state. Usually, a system starts from some initial state far from equilibrium and then gradually relaxes (or evolves) toward a stationary state, where its properties stop changing in time. Some examples include traffic flow models [15], where cars interact but there's continuous motion, or biological systems like bacterial colonies [16], and absorbing state models, where activity, such as chemical reactions or infection spreading, can die out permanently [17].

Since there's no thermal equilibrium, the steady state cannot be described by free energy or the Boltzmann distribution. Instead, stochastic dynamics, random processes that evolve in time, are governed by microscopic rules that specify probabilistic time evolution. These rules describe processes such as particle diffusion, annihilation, creation, or interaction, and are often represented by the master equation or stochastic field theories.

The dynamics are time-dependent and irreversible, often involving both deterministic drift terms and stochastic noise components that account for microscopic fluctuations, which describe how probability changes over time based on assumed transition probabilities between states. Sometimes, systems are kept far away from equilibrium by external drive, such as an electric current flowing through a resistor. Even though such

a system may reach a stationary state, its probability distribution is not the same as the Gibbs ensemble (the standard equilibrium distribution). For example, the thermal noise in a resistor carrying current is not Gaussian, unlike equilibrium.

A nonequilibrium phase transition occurs when a control parameter is tuned through a critical threshold, leading to a qualitative change in the stationary state. In other words, theoretical interest in models for spreading stems from the emerging transition between survival and extinction. In spreading models, sites can either be active or inactive. Depending on the parameters controlling the spreading processes, observable activities can be such that the system either remains dormant or propagates, eventually reaching a steady state. For the case that is entirely stuck in the inactive state, this phenomenon is coined the absorbing state of the model. The term “absorbing” reflects the fact that once the system reaches that state, it can’t leave it. The other phase, “active”, means that the system maintains a finite level of dynamic activity. In short, the above absorbing state represents a nonequilibrium process and exhibits a continuous phase transition.

This phase transition happens in many spreading processes. They are encountered in many natural situations, such as epidemics [18], forest fires [19], and others. These processes are governed by two competing mechanisms. For example, in a forest fire, the spreading agent, the fire, may spread and burn neighboring trees or plants, while at the same time burning might stop in some regions, decreasing the total number of spreading agents. The outcome of the process depends on the relative rates of spreading and recovery. If the spreading rate is higher, this could lead to the steady state of the forest where the fire consumes the forest. Conversely, if the recovery dominates, the fire gradually diminishes and eventually dies out.

Similarly, continuous phase transitions occur in equilibrium statistical mechanics. One of the best-known examples is the paramagnetic-ferromagnetic phase transition in the Ising model, with criticality occurring at some temperature T_c [20]. In most cases, symmetry properties of the model under study specify the emergence of long-range correlations, which are independent of the microscopic details. This leads to the classification of models into *universality classes*, which group different systems together under the same critical behavior near the phase transition [21]. This makes universality a central theme in the theory of equilibrium phase transitions. Since nonequilibrium phenomena also exhibit phase transitions, the concept of *universality* can also be applied here with much more diversity since the dynamics are governed by various symmetry properties of the evolution [12].

Although the critical behavior for nonequilibrium phase transitions shares some similarities with that of the equilibrium model, for example, it’s possible to use scale invariance and determine various critical exponents, the critical behavior is highly nontrivial to derive analytically. Similarly to equilibrium statistical mechanics, where identifying critical exponents helps categorize models into universality classes, this approach can also be applied in nonequilibrium systems. An important universality class very common to absorbing-state phase transitions is Directed Percolation (DP).

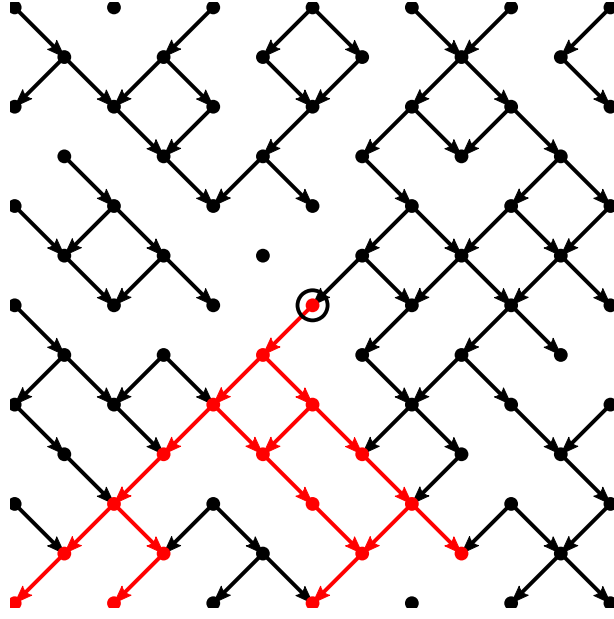


Figure 1.1: Schematic illustration of directed bond percolation, showing the directionality of connectivity in space-time.

Directed Percolation is the most robust universality class of nonequilibrium phase transitions. It describes systems where there is a single absorbing state, meaning when the system's inactivity stay permanent once reached. The dynamics are local and Markovian (no memory effects). The control parameter tunes the system from an active to an absorbing phase. This universality class is illustrated in Fig. 1.1. Each site can be active (red) or inactive (black). Activity can spread only in a preferred direction, forward in time or along a spatial axis, here the direction is indicated by the downward arrows. If activity stops, the system becomes frozen (absorbing). Examples of DP-type behavior include epidemic models without immunity [22], catalytic reaction models where reaction front can die out, and flow through a random medium with a directional bias [23]. Because of this directional bias, the critical properties and exponents, are nontrivial. More specifically, the critical exponents for DP cannot be found exactly using analytical methods. Instead, their values appear to be irrational numbers are determined mainly numerically or through approximations.

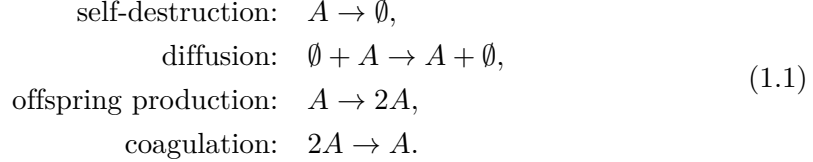
1.2 Directed Percolation and its Properties

First, it is useful to state the conditions that a model must satisfy in order to belong to the DP universality class. These conditions were listed by Janssen and Grassberger as follows [24, 25]:

1. The model displays a continuous phase transition from a fluctuating active phase into a unique absorbing state.
2. The transition is characterized by a positive one-component order parameter.

3. The dynamic rules involve only short-range processes.
4. The system has no special attributes such as additional symmetries or quenched randomness.

DP can be considered as a reaction–diffusion process of interacting particles. Referring to Fig. 1.1, denoting active sites by A (dots with color red) and empty sites by $\emptyset$ (dots with color black), the DP process corresponds to the reaction–diffusion scheme



For example, in the case of directed bond percolation, self-destruction occurs when both bonds are closed, indicated by black lines (valves). If only one bond is open, indicated by red line, the particle diffuses either to the left or to the right with equal probability. Offspring production occurs when both bonds are open. Finally, when two particles reach the same target site, they merge into a single particle in the case of coagulation.

The dynamical rule for DP is straightforward. Fig. 1.1 shows a lattice with a $(1+1)$ -dimensional directed bond percolation process, where the spreading flows from top to bottom with the direction of time pointing downward. At each site (either a red or black dot), a local variable $s_i(t)$ is assigned, where $s_i = 1$ means the site is active (occupied in red), whereas $s_i = 0$ means the site is inactive (empty in black). The set $s = \{s_i\}$ at a given time t specifies the configuration of the system at that moment.

The next configuration at time $t+1$ can be determined from the configuration at time t as follows. For each bond between the sites $(i \pm 1, t)$ and $(i, t+1)$, two random numbers $z_i^\pm \in (0, 1)$ are generated. A bond is considered open if $z_i^\pm < p$, with p being the bond-opening probability. The spreading process is then governed by the following update rule

$$s_i(t+1) = \begin{cases} 1 & \text{if } s_{i-1}(t) = 1 \text{ and } z_i^- < p, \\ 1 & \text{if } s_{i+1}(t) = 1 \text{ and } z_i^+ < p, \\ 0 & \text{otherwise.} \end{cases} \tag{1.2}$$

Common choices for the initial state include a fully occupied lattice, random initial conditions, or a single active seed (a single particle at the origin) as in Fig. 1.1.

A few lattice models for DP are the Domany–Kinzel cellular automaton [26], the Ziff, Gulari, Barshad model for heterogeneous catalysis [27], and the contact process [28]. For the first, the Domany–Kinzel model is a probabilistic cellular automaton defined on a one-dimensional lattice with discrete time, where each site can be either active ($s_i = 1$) or inactive ($s_i = 0$). The state of a site at the next time step depends only on the states of its two nearest neighbors at the previous time step, according to the transition probabilities p_1 (activation from one active neighbor) and p_2 (activation from two active neighbors).

The second model, the Ziff, Gulari, Barshad model, describes heterogeneous catalysis, e.g., the oxidation of CO on a platinum surface. The lattice sites represent catalytic surface sites, which can be occupied by a CO molecule, an O atom, or remain empty. The reaction rules are stochastic adsorption, reaction, and desorption events. Depending on the adsorption probabilities, the system exhibits absorbing states or an active catalytic phase. This phase transition also belongs to the DP universality class.

Finally, the contact process is a continuous-time stochastic process defined on a lattice, where each site is either occupied or empty. An occupied site becomes empty at rate 1, while an empty site becomes occupied at a rate proportional to the number of occupied neighbors, with the control parameter λ determining whether the system evolves into its absorbing state or remains active. It is also a classic example of a DP process.

Now that the fundamental properties of universality class of DP such as its anisotropic behavior and spreading process are discussed. The next step is to understand the way in which these critical exponents are extracted.

1.3 Critical Exponents for Directed Percolation Models

In equilibrium statistical physics, models such as the TFIM are characterized by universal scaling laws. For instance, the magnetization order parameter in the ordered phase of the one-dimensional Ising chain vanishes close to the critical point as

$$m \sim |T - T_c|^\beta,$$

where β is a universal exponent. Similarly, the correlation length ξ , which represents the characteristic length scale over which spins are correlated, diverges as

$$\xi \sim |T - T_c|^{-\nu}.$$

These examples illustrate how physical quantities exhibit power-law behavior near criticality, governed by universal exponents. Interestingly, many of the concepts of scaling laws can also be applied to nonequilibrium phenomena.

The three notable scaling exponents for DP are β , $\nu_{\parallel}$, and $\nu_{\perp}$. Here, β is the critical exponent associated with the stationary particle density. Since spreading processes are described by correlation lengths in both time and space, unlike equilibrium models without dynamical aspects, it is necessary to distinguish between temporal and spatial properties. These are denoted by $\parallel$ and $\perp$, respectively. Ultimately, nonequilibrium phase transitions are characterized by two correlation lengths: $\xi_{\perp}$ (spatial length scale) and $\xi_{\parallel}$ (temporal length scale). The scaling behavior of these three exponents can be expressed as

$$\rho^{\text{stat}} \sim (p - p_c)^\beta, \quad \xi_{\perp} \sim |p - p_c|^{-\nu_{\perp}}, \quad \xi_{\parallel} \sim |p - p_c|^{-\nu_{\parallel}}. \quad (1.3)$$

Keep in mind that

$$\rho(t) = \left\langle \frac{1}{N} \sum_i s_i(t) \right\rangle, \quad (1.4)$$

where $\langle \cdot \rangle$ denotes an ensemble average. In the active phase, $\rho(t)$ decays and eventually saturates to ρ^{stat} , with $(p - p_c)$ varying. Here, p is the control parameter (spreading probability) that represents the probability that activity spreads from the active site to its neighbor in the next time step. p_c is the critical threshold (critical probability) that is a specific value of p at which the system undergoes a phase transition between two quantitatively different steady states. For $p < p_c$, the system is in the absorbing phase, whereas $p > p_c$, the system is in the active phase. Thus $(p - p_c)$ measures how far the system is from the critical point. Near p_c , various quantities such as steady-state density $\rho(t)$, and the spatial and temporal correlation lengths $\xi_{\perp}$ and $\xi_{\parallel}$ follow power-law scaling, which characterizes the critical behavior of the DP universality class.

In the scaling regime, the two correlation lengths are related by

$$\xi_{\perp} \sim \xi_{\parallel}^{1/z},$$

where $z = \nu_{\parallel}/\nu_{\perp}$ is the dynamic exponent. In many cases, the three exponents β , $\nu_{\parallel}$, and $\nu_{\perp}$ form the fundamental set of critical exponents that define a universality class.

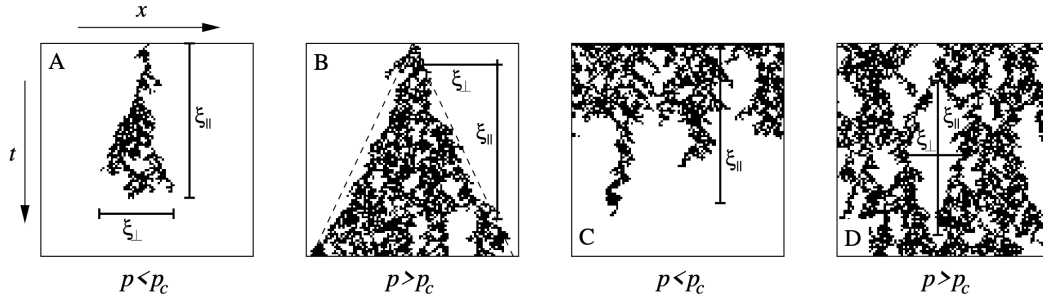


Figure 1.2: Interpretation of correlation lengths $\xi_{\parallel}$ and $\xi_{\perp}$ in the $(1 + 1)$ -dimensional case of below and above criticality. The two leftmost figures grow from a single active site, whereas the two on the right start from fully active states. Illustration taken from Hinrichsen [1].

The properties of the correlation lengths are illustrated in Fig. 1.2, the inactive phase clusters starting from a single active seed resemble droplets. Averaging over many simulations, the lateral size and temporal extent of such droplets are proportional to $\xi_{\perp}$ and $\xi_{\parallel}$, respectively, as shown in Fig. A. In the spreading cone of clusters (Fig. B), the opening angle is determined by the ratio $\xi_{\perp}/\xi_{\parallel}$. In the case of homogeneous decay, the temporal correlation length $\xi_{\parallel}$ governs the system's relaxation. For the stationary state in the active phase (Fig. D), the correlation lengths are determined by the average size of inactive regions.

Near the critical point, these correlation lengths obey scale invariance, which characterizes the universal behavior of spreading processes. Specifically, a spreading process should be invariant under dilatations $x \rightarrow \Lambda x$, accompanied by a rescaling of time and of the deviation from criticality $\Delta = p - p_c$ [29, 30]:

$$x \rightarrow \Lambda x, \quad t \rightarrow \Lambda^z t, \quad \Delta \rightarrow \Lambda^{-1/\nu_{\perp}} \Delta, \quad \rho \rightarrow \Lambda^{\beta/\nu_{\perp}} \rho. \quad (1.5)$$

This scaling invariance leads to the construction of dimensionless combinations such as t/x^z , $\Delta t^{1/\nu_{\parallel}}$, and $\Delta x^{1/\nu_{\perp}}$.

In the case of an infinitely large system, an important quantity emerge: $\rho(t)$, the particle density (starting from a fully occupied lattice).

$\rho(t)$ scale as expected, following the standard scaling concepts known from equilibrium statistical mechanics

$$\rho(t) \simeq t^{-\delta} f(\Delta t^{1/\nu_{\parallel}}) \quad (1.6)$$

where δ is certain critical exponents for decay [31]. f is the *universal scaling function*, i.e., it has the same functional form in all DP models.

The scaling theory outlined above assumes the system size to be infinite. For finite systems sizes, the scaling functions also depend on the invariant ratio $\xi_{\perp}^d/N = t^{d/z}/N$, where $N = L^d$ is the total number of sites. The generalized scaling forms read

$$\rho(t) \sim t^{-\beta/\nu_{\parallel}} f(\Delta t^{1/\nu_{\perp}}, t^{d/z}/N), \quad (1.7)$$

1.4 Mean-field Theory for Non-Equilibrium Absorbing-State Phase Transition

1.4.1 Mean-Field Approximation

To understand how the critical exponents behave, first, one starts with the mean-field analysis. In the mean-field approach, spatial correlations are neglected and the system is described solely by the average density of active sites $\rho(t)$. The essential features of Absorbing-State Phase Transition (ASPT) in DP can be captured through a simple deterministic equation. At this level of approximation, fluctuations are ignored and each site is assumed to experience the average influence of its neighbors.

Mean-Field Equation

The evolution of the order parameter $\rho(t)$ can be expressed phenomenologically as

$$\frac{d\rho}{dt} = \lambda\rho - (\lambda - 1)\rho^2, \quad (1.8)$$

where λ is the control parameter associated with the spreading rate. The first term, $\lambda\rho$, represents the creation of activity due to the spreading process, while the nonlinear term $-(\lambda - 1)\rho^2$ accounts for the saturation of activity, ensuring that $\rho(t)$ remains bounded. Eq. 1.8 is analogous to the logistic growth equation and simplistic enough to describe an absorbing-state phase transition.

Stationary States and Critical Point

The stationary states are obtained from the condition $d\rho/dt = 0$, leading to

$$\lambda\rho - (\lambda - 1)\rho^2 = 0. \quad (1.9)$$

Factoring out ρ gives two steady-state solutions,

$$\rho_{ss} = 0, \quad \rho_{ss} = \frac{\lambda}{\lambda - 1}. \quad (1.10)$$

The first solution, $\rho_{ss} = 0$, corresponds to the *absorbing state*, where all activity has ceased. The second, nonzero solution represents the *active phase*, which exists only for $\lambda > 1$. The critical point separating these two phases is therefore identified as $\lambda_c = 1$.

Order Parameter and Critical Behavior

To characterize the continuous transition near the critical point, the distance from criticality is written as $\Delta = \lambda - \lambda_c$. Close to the threshold, the stationary density behaves as a power law,

$$\rho_{ss} \sim \Delta^\beta, \quad (1.11)$$

where β is the order-parameter critical exponent. From the mean-field solution, the steady-state density is $\rho_{ss} \propto \Delta$, hence $\beta_{MF} = 1$.

Time-Dependent Behavior at Criticality

At the critical point $\lambda = \lambda_c$, Eq. 1.8 simplifies to

$$\frac{d\rho}{dt} = -\rho^2. \quad (1.12)$$

Integration yields an algebraic decay of the activity density, $\rho(t) \sim t^{-\delta_{MF}}$, with the mean-field decay exponent $\delta_{MF} = 1$.

As for the temporal scaling and correlation time, Eq. 1.8 implies that in the inactive phase, the density decays exponentially as

$$\rho(t) \sim e^{-|\lambda - \lambda_c|t}. \quad (1.13)$$

This equation describes correlation time $\xi_{||}$, which scales as $\xi_{||} \sim |\lambda - \lambda_c|^{-1}$, leading to the mean-field exponent $\nu_{||} = 1$.

These exponents are exact for dimensions $d \geq d_c = 4$, where fluctuations are irrelevant and the mean-field description becomes quantitatively accurate. Although the mean-field approximation captures the qualitative features of the active-to-absorbing transition, it fails in low dimensions where fluctuations dominate. In particular, correlations between neighboring sites and the formation of spatial clusters strongly modify the critical exponents in one, two, and three dimensions.

1.4.2 Field-Theoretic Formulation

The mean-field approximation neglects spatial correlations and fluctuations, which play a decisive role near criticality, especially in low-dimensional systems. To capture these effects, one introduces a coarse-grained description of the DP process in terms of a continuous density field $\rho(\mathbf{x}, t)$, representing the local activity at position $\mathbf{x}$ and time t .

Coarse-Grained Langevin Equation

At the mesoscopic level, the dynamics of $\rho(\mathbf{x}, t)$ can be described by a stochastic partial differential equation of the Langevin type,

$$\partial_t \rho(\mathbf{x}, t) = D \nabla^2 \rho(\mathbf{x}, t) + (\lambda - 1) \rho(\mathbf{x}, t) - \lambda \rho^2(\mathbf{x}, t) + \eta(\mathbf{x}, t), \quad (1.14)$$

where D is the diffusion constant, $(\lambda - 1)$ controls the proximity to the critical point, and the nonlinear term $-\lambda \rho^2$ limits the unbounded growth of activity. The stochastic noise $\eta(\mathbf{x}, t)$ accounts for the inherent randomness of the local creation and annihilation processes. The absorbing state corresponds to the configuration $\rho(\mathbf{x}, t) = 0$ for all $\mathbf{x}$. In this state, both stochastic and deterministic terms vanish, preventing any spontaneous reactivation. The first term on the right-hand side describes diffusive spreading of activity, the second term represent local growth, and the nonlinear term $-\lambda \rho^2$ accounts for local saturation due to competition or annihilation processes.

The noise is assumed to be Gaussian with zero mean,

$$\langle \eta(\mathbf{x}, t) \rangle = 0, \quad (1.15)$$

and correlations that depend on the local activity,

$$\langle \eta(\mathbf{x}, t) \eta(\mathbf{x}', t') \rangle = \Gamma \rho(\mathbf{x}, t) \delta(\mathbf{x} - \mathbf{x}') \delta(t - t'), \quad (1.16)$$

where Γ is a noise amplitude. The multiplicative form of Eq. 1.16 ensures that fluctuations vanish when $\rho = 0$, thereby preserving the absorbing-state condition: once the system reaches $\rho = 0$, it remains inactive indefinitely.

Neglecting the nonlinear and noise terms near criticality, the Langevin equation reduces to

$$\partial_t \rho(\mathbf{x}, t) = D \nabla^2 \rho(\mathbf{x}, t) + \tau \rho(\mathbf{x}, t), \quad (1.17)$$

where $\tau = \lambda - 1$ measures the distance from the critical point. Performing a Fourier transform,

$$\partial_t \tilde{\rho}(\mathbf{k}, t) = (-Dk^2 + \tau) \tilde{\rho}(\mathbf{k}, t), \quad (1.18)$$

the stationary condition $\partial_t \tilde{\rho} = 0$ yields the dispersion relation $Dk^2 \sim \tau$. Identifying the correlation length $\xi_\perp$ via $k \sim \xi_\perp^{-1}$ gives $\xi_\perp \sim \tau^{-1/2}$, which corresponds to the mean-field exponent $\nu_\perp^{\text{MF}} = \frac{1}{2}$.

Response-Field Functional Formalism

Using the Martin–Siggia–Rose–Janssen–De Dominicis (MSRJD) formalism [32], one introduces a *response field* $\tilde{\rho}(\mathbf{x}, t)$ and expresses the generating functional as a path integral [33],

$$Z = \int \mathcal{D}[\rho, \tilde{\rho}] e^{-S[\tilde{\rho}, \rho]}, \quad (1.19)$$

with the effective action

$$S[\tilde{\rho}, \rho] = \int d^d x dt \left\{ \tilde{\rho} \left[\partial_t \rho - D \nabla^2 \rho - (\lambda - 1) \rho + \lambda \rho^2 \right] - g \tilde{\rho}^2 \rho \right\}. \quad (1.20)$$

Here, the coupling constant g is the strength of the fluctuations derived from the noise correlations in Eq. 1.16. The first term in Eq. 1.20 corresponds to the deterministic dynamics, while the second term $-g\tilde{\rho}^2\rho$ arises from the stochastic noise and represents the nonlinear fluctuation corrections. Equation 1.20 is formally equivalent to the *Reggeon field theory* (RFT) [34], which serves as the field-theoretic representation of the directed percolation universality class.

Renormalization Group Analysis

The field theory defined by Eq. 1.20 can be analyzed using renormalization group (RG) methods. By integrating out short-wavelength fluctuations and rescaling space and time, one identifies fixed points of the RG flow that determine the asymptotic scaling behavior near the critical point. To study critical behavior, a scale transformation is represented as

$$\mathbf{x} \rightarrow b\mathbf{x}, \quad t \rightarrow b^z t, \quad \rho \rightarrow b^\chi \rho, \quad \tilde{\rho} \rightarrow b^{\tilde{\chi}} \tilde{\rho}. \quad (1.21)$$

where χ and $\tilde{\chi}$ are the scaling dimensions of ρ and $\tilde{\rho}$. By assigning canonical dimensions to the terms in the action, the upper critical dimension is found to be $d_c = 4$. Above the upper critical dimension $d_c = 4$, fluctuation corrections are irrelevant, and the mean-field critical exponents remain valid. Below d_c , fluctuations become relevant, and the RG analysis predicts nontrivial power-law exponents for the order parameter and correlation functions. These renormalized exponents have been confirmed by extensive numerical simulations.

For the DP action, the 1-loop beta functions (schematic) are:

$$\frac{dr}{dl} = 2r - A\lambda g, \quad (1.22)$$

$$\frac{d\lambda}{dl} = \epsilon\lambda - B\lambda^2, \quad (1.23)$$

$$\frac{dg}{dl} = \epsilon g - C\lambda g, \quad (1.24)$$

where r is the “mass” term controlling the distance from criticality, $d = 4 - \epsilon$ is the deviation from the upper critical dimension, and A, B, C are numerical constants arising from loop integrals. The RG fixed point determines the scaling behavior near phase transition. The critical exponents near $d_c = 4$ are:

$$\nu_\perp = \frac{1}{2} + \mathcal{O}(\epsilon), \quad \beta = 1 - \frac{\epsilon}{6}, \quad z = 2 - \frac{\epsilon}{12}. \quad (1.25)$$

These exponents correspond to the DP universality class near the upper critical dimension.

Mean-Field Critical Exponents

Within the mean-field approximation, the critical exponents take the following values:

Order-parameter exponent	β^{MF}	$= 1$
Temporal correlation exponent	$\nu_\parallel^{\text{MF}}$	$= 1$
Spatial correlation exponent	$\nu_\perp^{\text{MF}}$	$= 1/2$
Decay exponent	δ^{MF}	$= 1$

1.5 Monte Carlo Simulations with Fully Occupied States

Because of the difficulty in extracting critical exponents from the directed percolation universality class, numerous numerical and analytical techniques have been developed to address this problem. Examples include transfer matrix methods, where the system is mapped onto a matrix whose eigenvalues provide access to correlation lengths and scaling properties; series expansions, where physical quantities are expanded in powers of $(p - p_c)$ and the convergence is analyzed to estimate critical exponents; and field-theoretical approaches, where DP is mapped to a stochastic field theory (Reggeon field theory) and renormalization group techniques are applied to obtain scaling relations as seen previously. These methods, together with improved simulations, continue to yield increasingly accurate estimates of critical exponents.

In the past, the study of nonequilibrium phase transitions has become far more productive through the use of powerful numerical methods such as Monte Carlo simulations, which typically yield results with only a few percent error. Moreover, the rapid growth of computational power, both in hardware and software tools, has made such simulations increasingly convenient and accessible. Before introducing the main method of this research, TWA for spin-systems, it is useful to first discuss Monte Carlo simulations, their basic principles, and what can be expected from them.

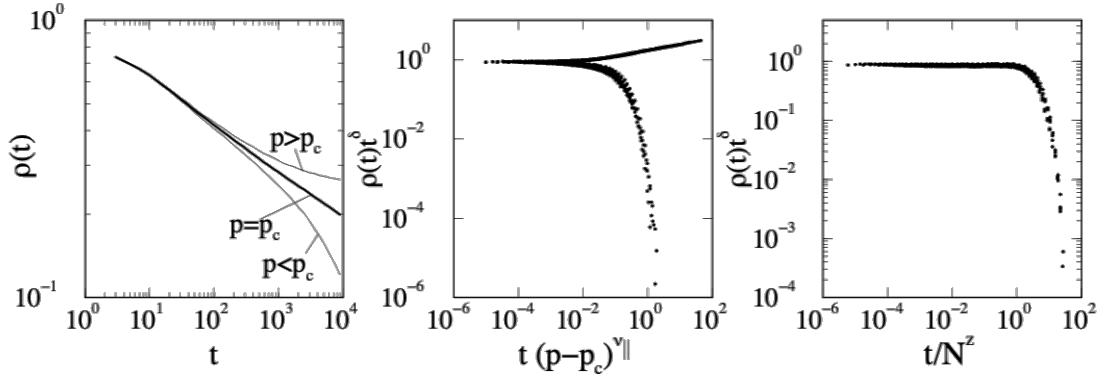


Figure 1.3: Left figure shows the particle density as a function of time, where the critical point can be determined by looking at the divergence of the curves above and below the critical line. Middle figures shows data collapses for off-critical simulation by varying β and δ . Right figure shows data collapses for finite-size simulation by varying z . Illustration taken from Hinrichsen [1].

For the case of a homogeneous initial state, where all sites are occupied, one starts in the fully occupied state (all active sites, $s_i(t = 0) = 1$), and then evolves the system using the microscopic update rules of the model (e.g., directed bond percolation from Eq. 1.2). At each time step, random numbers are generated to decide bond opening/closing or site activation. The dynamics progress until the system reaches a steady state, Eq. 1.4, or exhibits power-law decay, Eq. 1.3. Critical exponents can be extracted via finite-size scaling by repeating the simulation with different lattice sizes.

The critical point can be estimated by varying the control parameter p around the critical point p_c . As mentioned, if $p < p_c$, the system is in the absorbing state, whereas if $p > p_c$, the system saturates at some finite value $\rho(t = t_{final}) > 0$. Fig. 1.3 shows the results obtained from following the described procedure.

Considering the leftmost figure, the critical percolation threshold is determined by measuring deviations from the asymptotic power-law decay $\rho(t) \sim t^{-\delta}$ for a sufficiently large system. Here, $\rho(t)$ is plotted against the simulation time t on a logarithmic scale for both axes. Curves lying above the critical line indicate that the system is in the active phase, while curves bending downward correspond to the inactive phase. One can use finite-size scaling with different lattice sizes to estimate the correct critical line.

After determining the critical point p_c and exponent δ , the next step is to find β . This is done by measuring the stationary density of active sites $\rho^{stat} \sim \Delta^\beta$ in the active phase. From the second figure, which plots $\rho(t)t^\delta$ versus $t\Delta^{\nu_\parallel}$, the collapse of the curves can be observed by adjusting $\nu_\parallel$. The exponent β is then given by $\beta = \delta\nu_\parallel$.

Finally, the third figure shows the plot of $\rho(t)t^\delta$ versus t/N^z to extract the exponent $\nu_\perp$. By tuning z , the curves from different lattice sizes collapse onto each other, giving $\nu_\perp = \nu_\parallel/z$. The table below summarizes the critical exponents that have been previously obtained from various DP models. In this research, three critical exponents that are extracted are δ , β and z .

critical exponent	MF	IMF	$d = 1$	$d = 2$	$d = 3$
β	1	1/2	0.276850(8)	0.584(4)	0.81(1)
$\nu_\perp$	1/2	1	1.09685(4)	0.734(4)	0.581(5)
$\nu_\parallel$	1	3/2	1.733847(6)	1.295(6)	1.105(5)
δ	1	1/2	0.159464(6)	0.451	0.73

Table 1.1: Critical exponents for various spatial dimensions d and theoretical approaches. Mean-field (MF), improved mean-field (IMF), and numerical or perturbative results for $d = 1, 2$ and 3 , are shown. The listed exponents (β , $\nu_\perp$, $\nu_\parallel$, and δ) characterize the scaling behavior near the critical point [35–39].

Though Monte Carlo is an extremely powerful tool for numerical simulations [40], especially in the study for extracting critical exponents of Rydberg system in dissipative cold gases in the classical regime [6], several aspects of the method itself come under scrutiny. Since this research focuses on interacting Rydberg spin-system when quantum mechanical fluctuations become relevant, first, simulating the density matrix for a large quantum system can be extremely cumbersome using this method. For a quantum system of size N , the number of state vector components scales as 2^N , and thus the density matrix has $2^N \times 2^N = 4^N$ elements [41]. For large system sizes, roughly 100 sites, this exhausts most of the computational resources currently available, let alone performing such calculations on a commercial computer. For example, with $N = 20$,

the density matrix has $4^{20} \approx 1.1 \times 10^{12}$ elements, making storage of these values extremely challenging.

Furthermore, the time evolution of the density matrix is governed by the master equation, or the Liouville–von Neumann equation. Evaluating the commutator and any dissipative terms scales as $O(4^N)$, which again grows exponentially with system size [42]. These issues alone create tremendous challenges for Monte Carlo simulations.

One promising candidate to address these issues is the TWA, a semi-classical method that uses truncation of higher-order derivative terms from the spin interactions. The mapping of operators from Hilbert space to the Wigner phase space proves particularly useful in this approach.

Now that the directed percolation universality class from non-equilibrium phenomena has been addressed in detail, the next step is to discuss Rydberg physics, specifically Rydberg facilitation, and to show that this particular system exhibits DP behavior.

2 Rydberg atoms and the Underlying Physics

Rydberg physics is one of the most interesting topics that has been heavily researched recently due to its potential applications in quantum computing and many-body physics [3, 43]. By taking advantage of the Rydberg blockade and its long lifetime, the realization of quantum gates has demonstrated that Rydberg atoms are very promising candidates for constructing a neutral atom quantum computer [3, 44, 45]. Building on these advances, recent experiments have achieved precise control over large arrays of neutral atoms, successfully trapping thousands of Rydberg atoms on optical lattices using tweezer arrays [45, 46]. Their range of interactions with neighboring atoms—from dipole-dipole to Van der Waals type—also highlights their versatility [3]. In this research, the focus is not on Rydberg blockade, but rather on another intriguing property: Rydberg facilitation, where neighboring atoms assist each other in being excited to the Rydberg state by shifting nearby energy levels into resonance with the driving laser, thereby facilitating excitation [14].

In this section, the topics of discussion include the fundamental physics of Rydberg atoms, from Rydberg blockade to Rydberg facilitation. Above all, the section discusses the usefulness of Rydberg facilitation in understanding NEPT and its relation to the DP universality class. Finally, the section concludes with previous work that used Rydberg properties to extract critical exponents using Monte Carlo as a numerical method [6]. Ultimately, it highlights the limitations of these results and outlines a pathway to address these issues through a new numerical method, TWA [9].

2.1 Rydberg Physics

The fundamental principles of Rydberg physics date back as far as 1862 with the identification of spectral lines of hydrogen [47]. In 1890, the Swedish physicist Johannes Rydberg penned a more general formula for the spectral lines emitted by atoms that have a single outer-shell electron. The Rydberg formula describes the energy of the n th energy level of hydrogen as follows [48]:

$$E_n = -\frac{R_H}{n^2}, \quad (2.1)$$

where R_H is the Rydberg constant and n is the principal quantum number. In hydrogen, the Rydberg formula arises because there is only one electron interacting with the nucleus through the Coulomb potential. More interestingly, this concept extends beyond the hydrogen atom. Any atomic or ionic system with a single weakly bound valence electron outside a more tightly bound, positively charged core can exhibit hydrogen-like

properties [49]. In such systems, the positively charged ionic core (localized) provides an effective Coulomb potential for the outermost electron. The outer electron (delocalized) is spatially extended for high quantum number n , and experiences the core almost as if it were a point charge. This means that the energy spectrum resembles that of hydrogen, forming Rydberg series of states converging toward an ionization threshold. The main difference between Rydberg atoms and hydrogen is that, while the long-range part of the electron–core interaction is Coulombic and supports a hydrogen-like Rydberg series, deviations from hydrogen arise due to short-range interactions (core penetration, exchange, and polarization effects), which are encoded in the quantum defect [50]. Over the years, Rydberg atoms have been shown to have large sizes, with electron radii large enough to enable micrometer-scale wavefunctions overlapping in traps, long lifetimes, and high polarizability, making them versatile candidates for many applications [3].

2.1.1 Background

Rydberg states with large principal quantum numbers n can be found in extreme environments, including interstellar space and terrestrial plasmas [51]. Over the years, research on Rydberg atoms has led to numerous breakthroughs in areas such as quantum optics, quantum simulation, and precision sensing, opening up promising avenues for neutral-atom quantum computing [3, 52]. Consequently, it is crucial to grasp the fundamental concepts underlying Rydberg physics.

Rydberg lifetimes

When a Rydberg atom is excited to a very high principal quantum number n , the radius or size of the electron’s orbit, or more precisely, the extent of its wavefunction in a hydrogen-like atom, scales as $r \propto n^2$ [52]. This means that as n increases, the electron is much further from the nucleus, and the electron cloud becomes extremely large, sometimes micrometers across, which is huge on the atomic scale. Since many physical quantities depend on how extended the electron cloud is, many properties of Rydberg atoms scale with powers of n , such as electric dipole moments, transition rates, interaction strengths with fields, and lifetimes, each with its own dependence on n .

For example, the lifetime of a Rydberg state scales as n^3 to n^5 , depending on the angular quantum number l . The ground state wavefunction is compact, meaning that the electron is close to the nucleus, whereas the Rydberg state wavefunction for very large n is spread out, meaning that the electron is far from the nucleus. Because these two wavefunctions occupy very different regions in space, their spatial overlap is very small. The integral of the transition dipole moment between the two states, $\langle \psi_{\text{ground}} | \hat{d} | \psi_{\text{Rydberg}} \rangle$, is small if the wavefunctions hardly overlap, implying weak coupling.

This has two consequences. First, laser excitation from the ground state to a high- n Rydberg state is difficult, since the atom does not interact strongly with the light field. Second, spontaneous decay is also slow, because the same weak coupling reduces the spontaneous emission rate. These consequences give Rydberg states very long life-

times. Because Rydberg atoms are long-lived and highly sensitive to electric fields and other Rydberg atoms, they are valuable in quantum technologies, such as quantum information processing (e.g., Rydberg quantum gates) [3], quantum simulation of strongly interacting systems as in this research, precision measurement [53], quantum sensing [54], and many more.

2.2 Interactions between Rydberg Atoms

This introduction focuses mainly on the van der Waals and dipole-dipole interactions. For dipole-dipole interaction, which originates from strong interactions between two atoms excited into Rydberg states via (virtual) photon exchange between them [55]. Fig. 2.1 (left) shows a two-level system with states $|(n-1)P\rangle$ and $|nS\rangle$, involving two Rydberg atoms separated by spacing R . Initially, if one atom is in the state $|nS\rangle$, where S is the label used for a zero angular momentum state, and the other is in the state $|(n-1)P\rangle$, the first atom (in the up state) can emit a photon, decaying to $|(n-1)P\rangle$, while the second absorbs that photon and is excited to $|nS\rangle$ in the process. This effect is a transfer of state and energy between the atoms: $|nS, (n-1)P\rangle \rightarrow |(n-1)P, nS\rangle$. For the first decay process (spontaneous emission), the rate is proportional to the optical dipole operator that couples these two states via decay to the vacuum mode, $\langle nS | \mathbf{e} \cdot \mathbf{r} | (n-1)P \rangle \propto n^2$. Similarly, the inverse process (absorption of a single photon from a given mode) is determined by the same operator. Thus, the interaction scales with the principal quantum number as n^4 . The overall strength of these dipole-dipole interactions is $\propto n^4/R^3$, since the dipole-dipole interaction scales as R^{-3} .

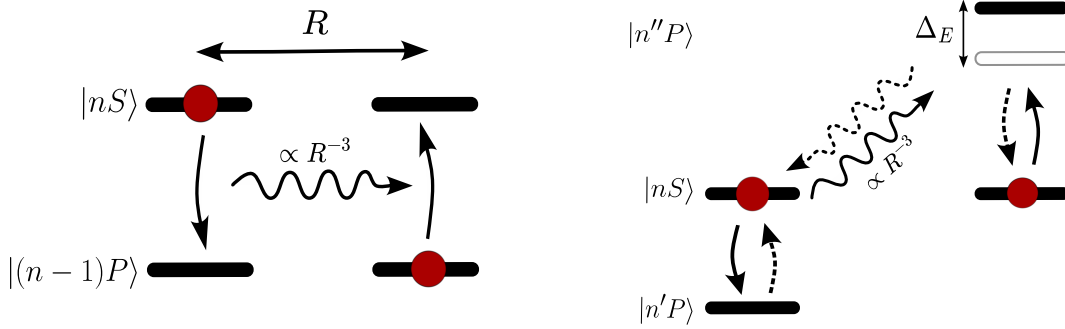


Figure 2.1: Illustrations of dipole-dipole interaction (left) and van der Waals interaction (right).

The second type of interaction is the van der Waals interaction. In this case, consider two atoms in the same Rydberg state $|nS\rangle$, as illustrated in Fig. 2.1 (right). Suppose the leftmost atom decays into the state $|n'P\rangle$, effectively emitting a photon, which is then absorbed by the rightmost atom, exciting it to the state $|n''P\rangle$. However, there exists an energy defect $\Delta_E = E(n'P, n''P) - E(nS, nS)$ which prevents exact energy matching between the pair of states $|n'P, n''P\rangle$ and the original pair state $|nS, nS\rangle$.

If the coupling is sufficiently strong, the process can still occur, since the energy is

only imprecisely defined over a short time. In this case, the system can temporarily access the virtual state $|n'P, n''P\rangle$, after which the atoms can return to the original pair state $|nS, nS\rangle$. This corresponds to a second-order process involving two photon emission–absorption events, each with a rate scaling as $\propto n^2$. Using second-order perturbation theory, the strength of this process is found to scale as $n^8/\Delta_E R^6$ where the R^6 dependence arises from the interatomic distance, with each virtual photon exchange (depicted as solid and dashed curved lines) contributing a factor of R^{-3} [56]. This effective second-order interaction is known as the van der Waals interaction: even when two atoms occupy the same Rydberg state, they can still interact through photon exchange via virtual intermediate states. Since the neighboring energy defect scales as $\Delta_E \propto n^{-3}$, the overall interaction strength scales as $\propto n^{11}/R^6$.

For resonant dipole–dipole interactions, two atoms can exchange energy via real photon exchange, that is, from $|nS, nP\rangle \rightarrow |nP, nS\rangle$ and back. Here, the state of the atoms changes, conserving total energy, with one atom going down while the other goes up. On the other hand, for off-resonant van der Waals interactions, due to the energy defect Δ_E , the atoms interact only virtually through second-order perturbation processes, while remaining in their original states. Since the van der Waals potential scales as $V_{\text{vdW}}(R) \propto C_6/R^6$, where C_6 depends strongly on the principal quantum number n , two atoms close together experience a strong Rydberg level shift, whereas the shift becomes negligible at large distances.

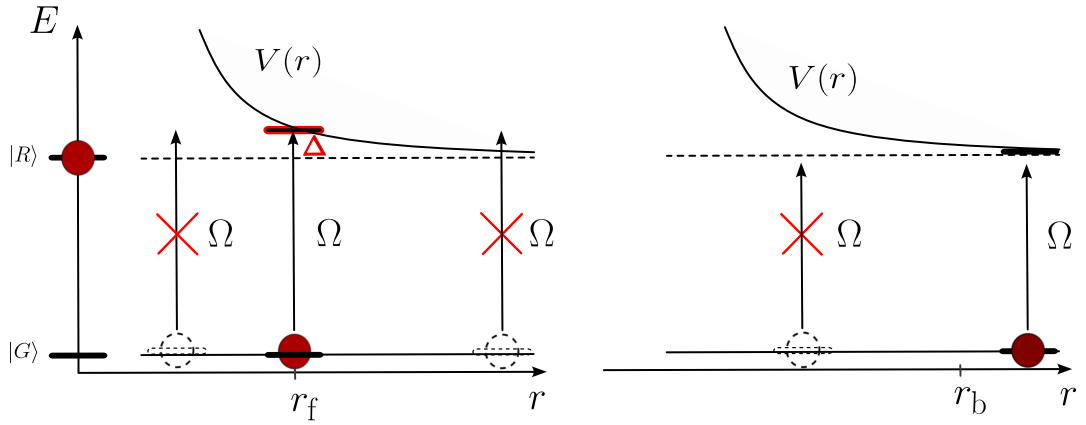


Figure 2.2: Illustration shows Rydberg facilitation (left) and Rydberg blockade (right).

Because of this intricacy, a pair of Rydberg atoms can exhibit two phenomena: *blockade* and *facilitation* [3, 57]. In the case of Rydberg *blockade*, illustrated in Fig. 2.2 (right), strong interactions between atoms in highly excited Rydberg states prevent more than one atom within a certain radius (the blockade radius) from being excited simultaneously. The working mechanism is straightforward: consider a laser tuned to the Rydberg resonance for a single atom with detuning $\Delta = 0$. For a pair of atoms, if one is already in the Rydberg state and the other remains in the ground state, the ground-state atom experiences an energy shift V due to the interaction. This interaction may arise from dipole–dipole coupling ($V \propto n^4/R^3$) or van der Waals coupling

($V \propto n^{11}/R^6$). At large distance R , the shift in energy is small compared to the transition linewidth Γ , so the second atom can still be excited to the Rydberg state. However, when the two atoms are closer than the blockade radius R_B , the interaction strength V shifts the energy level of the second atom by more than the transition linewidth Γ . Below this blockade radius, the second atom is effectively decoupled from the light field and cannot be excited to the Rydberg state.

As for the second case, *facilitation*, illustrated in Fig. 2.2 (left), the presence of a Rydberg excitation makes it resonant to excite nearby atoms at specific distances. In this regime, the driving laser is detuned in the direction of the interaction-induced level shift, which defines the facilitation condition. Here, the second atom is not coupled to the driving laser except at a specific distance R_F , where the interaction-induced level shift brings it into resonance with the light field, i.e., $V(R_F) = \Delta$. This mechanism is the key ingredient in studies of Rydberg lattice collective dynamics: excitations spread in a correlated, distance-dependent way, forming *facilitation shells* around existing excitations. This process leads to avalanche-like excitation growth, self-organized structures, and is directly linked to NEPT phenomena such as DP.

2.3 Universality Class for the Absorbing-State Phase Transition in Rydberg Facilitation

Critical exponents extraction of directed percolation

Previous work by Marcuzzi, *et al.* has utilized the physics of Rydberg facilitation to study interacting ensembles of highly excited Rydberg atoms, where the dynamics are governed by the rules of a DP process [6, 8]. In this study, the setup consists of an ensemble of N atoms trapped in a lattice with spacing a in one- and two-dimensional cases. The atoms are modeled as two-level systems, with a ground state $|g\rangle$ and a Rydberg state $|r\rangle$, coupled by a laser with Rabi frequency Ω and detuning Δ .

As mentioned, when two atoms are in the Rydberg state at positions $\mathbf{r}_i$ and $\mathbf{r}_j$, they experience a long-range interaction with strength V_{ij} given by

$$V_{ij} = \frac{C_\alpha}{|\mathbf{r}_i - \mathbf{r}_j|^\alpha}, \quad (2.2)$$

with $\alpha = 6$ for the van der Waals interaction.

The Hamiltonian of this system is expressed as

$$H = \frac{\Omega}{2} \sum_i \sigma_i^x - \Delta \sum_i \sigma_i^r + \frac{1}{2} \sum_{i \neq j} V_{ij} \sigma_i^{rr} \sigma_j^{rr}, \quad (2.3)$$

where

- $\sigma_i^x = |r\rangle_i \langle g| + |g\rangle_i \langle r|$ is the Pauli- x operator acting on atom i ,
- $\sigma_i^{rr} = \frac{\mathbb{I} + \sigma_i^z}{2}$ is the projector onto the Rydberg state of atom i ,

- and V_{ij} encodes the pairwise interaction between atoms i and j .

The system is coupled to an external Markovian environment with spontaneous radiative decay rate Γ and dephasing of atomic superposition states at rate γ . The Lindblad terms in the master equation then read

$$\mathcal{L}_\Gamma \rho = \sum_i \frac{\Gamma}{2} \left(2\sigma_i^- \rho \sigma_i^+ - \{\sigma_i^+ \sigma_i^-, \rho\} \right), \quad (2.4)$$

and

$$\mathcal{L}_\gamma \rho = \sum_i \frac{\gamma}{2} (2\sigma_i^{rr} \rho \sigma_i^{rr} - \{\sigma_i^{rr}, \rho\}), \quad (2.5)$$

respectively.

To make it possible to simulate the system using Monte Carlo, some constraints have to be imposed. In the limit of strong dephasing, i.e. $\gamma \gg \Gamma, \Omega$, a perturbative treatment of the full dissipative quantum dynamics allows for a description of the system via a classical rate equation, whose components are the statistical weights of classical spin configurations [6]. This reduces quantum effects between the spins, such as correlations and long-range correlations, thereby allowing the system to be treated classically.

To create a system that follows the DP conjecture, four dynamical processes must be defined. Specifically, the system exhibits self-destruction, in which a spin transitions as $|\downarrow\downarrow\uparrow\downarrow\downarrow\rangle \rightarrow |\downarrow\downarrow\downarrow\downarrow\downarrow\rangle$ with rate $1 + \lambda/\nu^2$. Here, $\lambda = 4\Omega^2/(\gamma\Gamma)$ and $\nu = 2|\Delta|/\gamma$ are two dimensionless parameters measured in units of the decay rate Γ . The second process is offspring production, where a spin becomes excited due to a neighboring excitation, $|\downarrow\uparrow\downarrow\downarrow\downarrow\rangle \rightarrow |\downarrow\uparrow\uparrow\downarrow\downarrow\rangle$, with rate λ . The third process is coagulation, where an excited spin de-excites itself, $|\downarrow\uparrow\uparrow\downarrow\downarrow\rangle \rightarrow |\downarrow\uparrow\downarrow\downarrow\downarrow\rangle$, with rate $1 + \lambda$. Finally, the system includes self-activation, in which a spin excites spontaneously, $|\downarrow\downarrow\downarrow\downarrow\downarrow\rangle \rightarrow |\downarrow\downarrow\uparrow\downarrow\downarrow\rangle$, with rate λ/ν^2 .

Results

Monte Carlo dynamical simulations were performed for small one-dimensional (1D) and two-dimensional (2D) systems with open boundary conditions. While the model can be simulated for large lattices, the focus here is on system sizes and boundary conditions relevant to current Rydberg experiments.

The dynamics in 1D were analyzed for systems nearest-neighbor (NN) interactions in a chain of $N = 50$ atoms. By changing the rate λ (changing Ω), one observe a qualitative change in the behavior of the system. At small λ , the average excitation density decays exponentially toward zero, whereas at large λ , it relaxes exponentially to a finite stationary value. At the critical point $\lambda = \lambda_c$, the dynamics exhibit an algebraic decay, $\langle n(t) \rangle \sim t^{-\delta}$, consistent with DP critical behavior. These data show that the critical exponents

$$\delta^{(1D)} = 0.159, \quad \beta^{(1D)} = 0.276$$

fit the numerical simulations well.

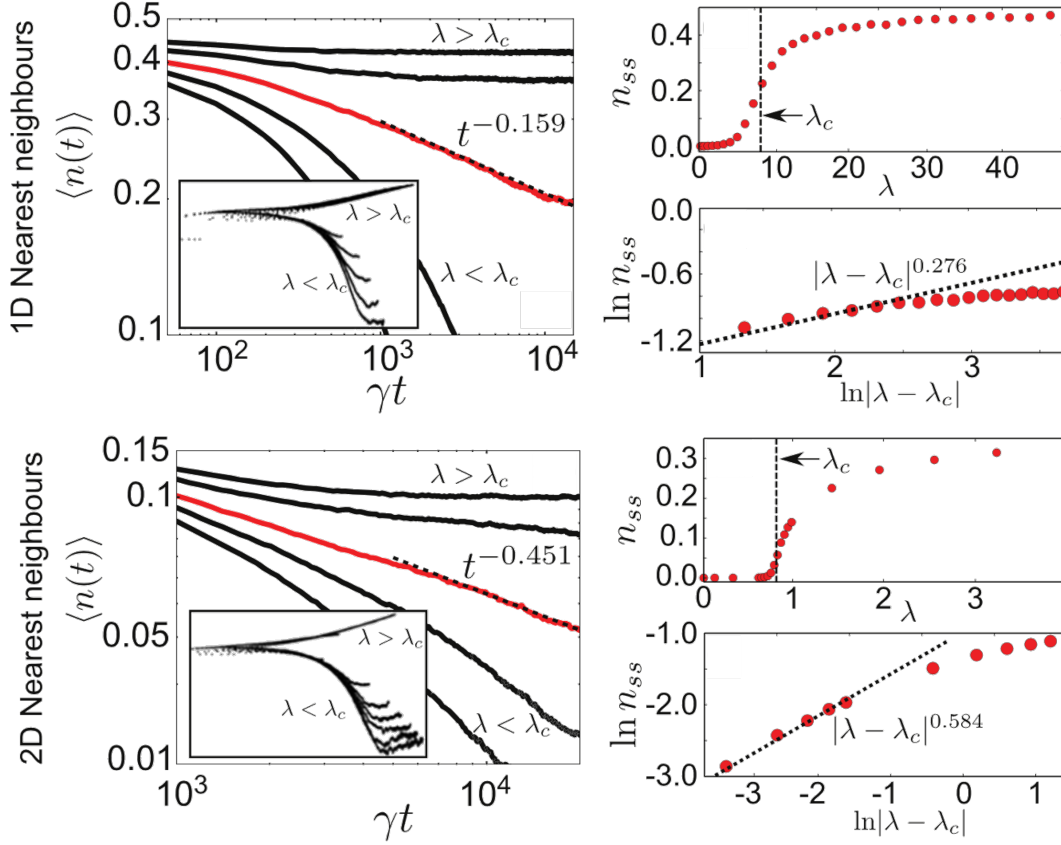


Figure 2.3: The critical exponents β, δ are fitted for the 1D and 2D case with nearest neighbors interaction to find the critical point λ_c . Figures from Marcuzzi, et al. [6].

Assuming DP scaling, the data collapse can be obtained by plotting $t^{\delta^{(1D)}} \langle n(t) \rangle$ vs. $t |\lambda - \lambda_c|^{\beta^{(1D)}/\delta^{(1D)}}$. The resulting curves display a tendency to collapse in the intermediate-time regime, confirming scaling behavior characteristic of a continuous phase transition. Deviations at short times arise from neglected initial-slip effects, while those at long times $t \gg (\gamma/\Omega^2)$ result from suppressed activation processes.

For 2D system, the NN case provides a faithful approximation even in higher dimensions. The 2D data presented here correspond to a system of $N = 20 \times 20$ atoms, considering only NN interactions.

These data show that the critical exponents

$$\delta^{(2D)} = 0.451, \quad \beta^{(2D)} = 0.584$$

fit the numerical simulations well. The corresponding scaling collapse again reveals distinct master curves, consistent with continuous phase transition behavior within the DP universality class.

One of the main issues in this study arises from the requirement that the dephasing rate γ must be significantly larger than both Γ and Ω in order to obtain a perturbative

treatment of the system dynamics and describe the system in terms of classical rate equations. This approach, however, faces limitations because the behavior of dissipative quantum systems with low dephasing remains largely unexplored. As a result, quantum effects such as long-range correlations and spin entanglement are neglected. A complete understanding of such systems, which is fundamentally motivated by quantum behavior, cannot be achieved using the Monte Carlo random sampling approach.

In the next chapter, a new method, the TWA for spin-systems, is introduced. This approach enables a semi-classical treatment of fully dissipative quantum systems, including regimes with low γ , with $\gamma \ll \Gamma, \Omega$. Ultimately, this allows the system dynamics to be studied in far greater detail than is possible with the Monte Carlo approach.

3 Truncated Wigner Approximation for Spins

The many-body dynamics of dissipative spin systems present significant challenges in numerical simulations. One can either obtain exact solutions for small systems or use approximate methods such as Monte Carlo simulations, which require sacrificing certain quantum effects by taking the system to the classical limit through strong dephasing.

The Wigner representation is a general phase-space representation, and the TWA was originally developed for bosonic systems [17]. A quantum mechanical state of a spin-1/2 cannot be represented by a single point in phase space. A first approach to extend the TWA to spin systems was introduced by Schachenmayer *et al.* [42], who used a discrete basis of four points, hence the name DTWA. More recently, Mink *et al.* further developed this method to construct a continuous extension, which they first called Discrete Continuous Truncated Wigner Approximation (DCTWA) [9]. The TWA for spin systems has gained attention for describing effects such as superradiance and correlations in the emitted light, and it continues to be developed further [58].

The construction of the TWA method [9], which is based on embedding the discrete Wigner representation into the continuous SU(2) phase space. This method improves upon its predecessor, the DTWA, by allowing the treatment of both dephasing and decay, and by providing a more accurate truncation approximation. From this motivation, a set of operator-differential mappings is introduced to derive the equation of motion for the continuous SU(2) Wigner function. The TWA thus captures quantum fluctuations both through the stochastic sampling of initial conditions and through the noise terms arising during the time propagation.

The chapter is structured as follows. First, an introduction to the Wigner representation for spins and its formulation for two-level systems is presented, starting from establishing the connection between the Hilbert space and the Wigner phase space, to choosing the appropriate Phase-Point Operator (PPO). Next, the method for constructing correspondence rules to translate the master equation of the density operator $\hat{\rho}$ into an equation of motion for the Wigner function $W_{\hat{\rho}}(\mathbf{\Omega})$ is outlined. The chapter concludes with the derivation of the mapping from the Lindblad master equation of the Rydberg collective spin system to a set of Stochastic Differential Equation (SDE)s suitable for numerical simulation.

3.1 Wigner Representation for Spins

The connection between the Hilbert space and the Wigner phase space is established through the representation of an operator $\hat{O}$ as [2]

$$\hat{O} = \int d\mathbf{\Omega} W_{\hat{O}}(\mathbf{\Omega}) \hat{\Delta}(\mathbf{\Omega}), \quad (3.1)$$

where $\mathbf{\Omega}$ denotes the set of c -number variables that define the mapping between the two spaces, and $\hat{\Delta}(\mathbf{\Omega})$ is called the PPO, or Wigner kernel, with $\int d\mathbf{\Omega} = \int_0^\pi d\theta \sin \theta \int_0^{2\pi} \frac{d\phi}{2\pi}$ and $\hat{\Delta} = \hat{\Delta}^\dagger$.

The Weyl symbol $W_{\hat{O}}(\mathbf{\Omega})$ of the operator $\hat{O}$ can then be expressed as

$$W_{\hat{O}}(\mathbf{\Omega}) = \text{Tr}[\hat{\Delta}(\mathbf{\Omega}) \hat{O}]. \quad (3.2)$$

Any observable $\hat{O}(\hat{\sigma})$ in Hilbert space for a spin- $\frac{1}{2}$ system can be expressed as a linear combination of the identity and Pauli matrices:

$$\hat{O}(\hat{\sigma}) = a_0 \hat{1} + a_x \hat{\sigma}_x + a_y \hat{\sigma}_y + a_z \hat{\sigma}_z \quad (3.3)$$

and its Weyl symbol is obtained by replacing the spin operators with the corresponding components of the classical spin vector:

$$\mathcal{O}(\mathbf{s}(\theta, \phi)) = \text{Tr}[\hat{\Delta}(\theta, \phi) \hat{O}(\hat{\sigma})] = O(s). \quad (3.4)$$

where the c -number vector

$$\mathbf{s}(\theta, \phi) = \sqrt{3} (\sin \theta \cos \phi, \sin \theta \sin \phi, \cos \theta)^T, \quad (3.5)$$

represents the Weyl symbol of the Pauli spin matrices and corresponds to points on the surface of a sphere with radius $\sqrt{3}$.

For this study, the Weyl symbol of the density operator $\hat{\rho}$ defines the Wigner function as

$$\int d\mathbf{\Omega} W_{\hat{\rho}}(\mathbf{\Omega}) = \text{Tr}(\hat{\rho}) = 1, \quad (3.6)$$

which is also referred to as the Wigner quasiprobability distribution, since $W_{\hat{\rho}}(\mathbf{\Omega}) \in \mathbb{R}$ and the trace of the density operator equals one, although the function itself can take negative values.

To choose the PPO, one can adopt the continuous representation of a spin-1/2 state $\hat{\rho}$ through rotations of the discrete phase-point operator $\hat{\Delta}_0 = \frac{1}{2} (\hat{1}_2 + \sqrt{3} \hat{\sigma}^z)$ such that

$$\hat{\Delta}(\theta, \phi) = U(\theta, \phi, \psi) \hat{\Delta}_0 U^\dagger(\theta, \phi, \psi) \quad (3.7)$$

where $U(\theta, \phi, \psi)$ is a unitary rotation operator acting on the reference kernel $\hat{\Delta}_0$, with the rotation operator are defined as

$$U(\theta, \phi, \psi) = e^{i\hat{\sigma}_z\phi/2} e^{i\hat{\sigma}_y\theta/2} e^{i\hat{\sigma}_z\psi/2} \quad (3.8)$$

where $U(\theta, \phi, \psi)$ is an $SU(2)$ rotation operator parameterized by the Euler angles (θ, ϕ, ψ) , which span the continuous phase space on the surface of a sphere, with $\theta \in [0, \pi]$ and $\phi \in [0, 2\pi)$.

In Eq. 3.7, the dependence on the angle ψ drops out, yielding

$$\begin{aligned} \hat{\Delta}(\theta, \phi) &= \frac{1}{2} [\hat{1}_2 + \mathbf{s}(\theta, \phi) \cdot \hat{\boldsymbol{\sigma}}] \\ &= \frac{1}{2} \begin{pmatrix} 1 + \sqrt{3} \cos \theta & \sqrt{3} e^{-i\phi} \sin \theta \\ \sqrt{3} e^{i\phi} \sin \theta & 1 - \sqrt{3} \cos \theta \end{pmatrix}, \end{aligned} \quad (3.9)$$

Consequently, the continuous Wigner representations for fully polarized spin states can be obtained by taking the trace of the phase-point operator $\hat{\Delta}(\theta, \phi)$ with the density matrix of the quantum state of interest. This yields

$$W_{\uparrow}(\theta, \phi) = \text{Tr} [\hat{\Delta}(\theta, \phi) |\uparrow\rangle\langle\uparrow|] = \frac{1}{2} (1 + \sqrt{3} \cos \theta), \quad (3.10a)$$

$$W_{\downarrow}(\theta, \phi) = \text{Tr} [\hat{\Delta}(\theta, \phi) |\downarrow\rangle\langle\downarrow|] = \frac{1}{2} (1 - \sqrt{3} \cos \theta). \quad (3.10b)$$

It turns out that these Wigner functions are not positive definite and, therefore, cannot be directly approximated by Monte Carlo sampling. However, by relating the continuous and discrete Wigner representations of spins using a gauge freedom [9], the relevant states $|\downarrow\rangle$ and $|\uparrow\rangle$ can be represented by a positive Wigner function.

$$W_{|\psi\rangle\langle\psi|}(\theta, \phi) = \delta(\theta_\psi - \theta), \quad (3.11a)$$

$$\theta_\psi = \begin{cases} \arccos\left(\frac{1}{\sqrt{3}}\right), & \psi = \uparrow, \\ \pi - \arccos\left(\frac{1}{\sqrt{3}}\right), & \psi = \downarrow, \end{cases} \quad (3.11b)$$

3.2 Correspondence Rules and Operators Mapping

3.2.1 Time evolution of continuous Wigner function

To perform the mapping of the Lindblad master equation for a single-spin density operator $\hat{\rho}$ in a phase-space representation, the first step is to specify it as [59]

$$\frac{d\hat{\rho}}{dt} = -i[\hat{H}, \hat{\rho}] + \frac{1}{2} \sum_{\mu} \left(2\hat{L}_{\mu} \hat{\rho} \hat{L}_{\mu}^{\dagger} - \{\hat{L}_{\mu}^{\dagger} \hat{L}_{\mu}, \hat{\rho}\} \right), \quad (3.12)$$

where $\hat{H}$ is the system Hamiltonian and $\hat{L}_{\mu}$ are Lindblad operators describing Markovian dissipation. Both $\hat{H}$ and $\hat{L}_{\mu}$ may involve interactions between spins that cannot, in general, be reduced to one-dimensional topologies, which complicates exact numerical treatments.

To translate the master equation from Hilbert space into a differential equation for the Wigner function $W_{\hat{\rho}}(\Omega)$ in phase space, we first express the Weyl symbol of a composition of operators, $\hat{H}\hat{\rho} \rightarrow W_{\hat{H}\hat{\rho}}$, in terms of the individual symbols $W_{\hat{H}}$ and $W_{\hat{\rho}}$ to decompose the density operator and the rotation spin operators. However, the Weyl symbol of operator products is generally not equivalent to the product of individual Weyl symbols. Instead, the decomposition of operators can be captured by the star (Moyal) product, which can be represented in differential form via correspondence rules. These rules allow operator products involving spin operators to be systematically converted into differential operators acting on the Wigner function. The star product of two operators $\hat{A}$ and $\hat{B}$ is

$$W_{\hat{A}\hat{B}}(\Omega) = W_{\hat{A}} \star W_{\hat{B}} = \iint d\Omega' d\Omega'' W_{\hat{A}}(\Omega') W_{\hat{B}}(\Omega'') \text{Tr}[\hat{\Delta}(\Omega)\hat{\Delta}(\Omega')\hat{\Delta}(\Omega'')]. \quad (3.13)$$

To obtain the phase-space equation, one need to construct a basis in the Hilbert space. For a single-spin continuous phase-space representation, the matrices

$$\hat{\Delta}(\theta, \phi), \quad \partial_{\theta}\hat{\Delta}(\theta, \phi), \quad \partial_{\phi}\hat{\Delta}(\theta, \phi), \quad \partial_{\phi}^2\hat{\Delta}(\theta, \phi) \quad (3.14)$$

are linearly independent and thus form a complete basis for 2×2 matrices in $\mathbf{C}^2$. Consequently, any operator acting on $\hat{\rho}$ can be expressed in terms of its action on a phase-point operator via an operator-differential identity. Note that this basis is not unique can other basis can be chosen to suit different problems. Every spin operator can be decomposed into the set of Pauli matrices in the Hilbert space

$$\begin{aligned} \mathbb{1} &= 2\hat{\Delta} + 2\cot\theta\partial_{\theta}\hat{\Delta} + 2\csc^2\theta\partial_{\phi}^2\hat{\Delta}, \\ \sigma_x &= -\frac{2\csc\theta}{\sqrt{3}}\left(\sin\phi\partial_{\phi}\hat{\Delta} + \cos\phi\partial_{\phi}^2\hat{\Delta}\right), \\ \sigma_y &= \frac{2\csc\theta}{\sqrt{3}}\left(\cos\phi\partial_{\phi}\hat{\Delta} - \sin\phi\partial_{\phi}^2\hat{\Delta}\right), \\ \sigma_z &= \frac{2}{\sqrt{3}}\left(-\csc\theta\partial_{\theta}\hat{\Delta} - \csc\theta\cot\theta\partial_{\phi}^2\hat{\Delta}\right) \end{aligned} \quad (3.15)$$

or in more compact form

$$\begin{aligned} \sigma_{\pm} &= \frac{1}{2}(\sigma_x \pm i\sigma_y) = \frac{1}{\sqrt{3}}\csc\theta e^{\pm i\phi}\left(\pm i\partial_{\phi}\hat{\Delta} - \partial_{\phi}^2\hat{\Delta}\right), \\ \sigma_z &= -\frac{2}{\sqrt{3}}\csc\theta\left(\partial_{\theta}\hat{\Delta} + \cot\theta\partial_{\phi}^2\hat{\Delta}\right) \end{aligned} \quad (3.16)$$

Thus, the action of $\hat{\sigma}_z$ on the kernel with the Pauli matrices and use Eq. 3.15

$$\hat{\sigma}^z\hat{\Delta}(\theta, \phi) = \left[\sqrt{3}\cos\theta + \frac{2\csc\theta - 3\sin\theta}{\sqrt{3}}\frac{\partial}{\partial\theta} + i\frac{\partial}{\partial\phi} + \frac{2\cot\theta\csc\theta}{\sqrt{3}}\frac{\partial^2}{\partial\theta^2}\right]\hat{\Delta}(\theta, \phi), \quad (3.17)$$

$$(3.18)$$

which can be verified by direct evaluation of both sides.

Using these mappings, the master equation is transformed into a partial differential equation for $W(\theta, \phi)$. After performing partial integration, one obtains a dynamical equation describing the evolution of the Wigner function. Unlike typical quantum-optical Fokker-Planck equations for bosonic systems, the resulting equation is not of standard Fokker-Planck type due to the nonlinear integration measure on the sphere, $d\Omega = \sin \theta d\theta d\phi/2\pi$. To address this, the Flatten Wigner Function (FWF) is introduced:

$$\chi(\theta, \phi) = \frac{\sin \theta}{2\pi} W(\theta, \phi), \quad (3.19)$$

defined on $\theta \in [0, \pi]$ and $\phi \in [0, 2\pi)$, with periodicity in ϕ of 2π . The distinction between $W(\theta, \phi)$ and $\chi(\theta, \phi)$ is important: while the evolution of $W(\theta, \phi)$ generally contains terms that cannot be expressed solely as derivatives, the evolution of the FWF can be written in a Fokker-Planck form, provided terms of order higher than second are negligible.

More formally, one can introduce the contravariant coordinate vector $(x_1, x_2) = (\theta, \phi)$ and the metric tensor $g_{\mu\nu}$ of the spherical phase space:

$$g = \frac{1}{2\pi} \begin{pmatrix} 1 & 0 \\ 0 & \sin^2 \theta \end{pmatrix}. \quad (3.20)$$

The infinitesimal volume element is $\sqrt{\det(g)} = \sin \theta/2\pi$. Using covariant derivatives

$$\nabla_\mu = \frac{1}{\sqrt{\det(g)}} \frac{\partial}{\partial x_\mu} \sqrt{\det(g)}, \quad (3.21)$$

would yield diffusion processes on the spherical surface without introducing the FWF. Since both approaches lead to identical results, we choose the FWF method.

Including the time dependence in the Wigner function while keeping the phase-point operators constant:

$$\hat{\rho}(t) = \iint d\theta d\phi \chi(\theta, \phi; t) \hat{\Delta}(\theta, \phi), \quad (3.22)$$

and substituting into the Lindblad equation, one has

$$\hat{\sigma}_z \hat{\rho} = \int d\theta d\phi \hat{\Delta}(\theta, \phi) \left[\sqrt{3} \cos \theta + \frac{\partial}{\partial \theta} \frac{3 \sin \theta - 2 \csc \theta}{\sqrt{3}} + \frac{\partial}{\partial \phi} i - \frac{\partial^2}{\partial \phi^2} \frac{2 \cot \theta \csc \theta}{\sqrt{3}} \right] \chi(\theta, \phi). \quad (3.23)$$

Abbreviating, this maps Hilbert space to phase space as

$$\hat{\sigma}_z \hat{\rho} \longleftrightarrow \left[\sqrt{3} \cos \theta + \frac{\partial}{\partial \theta} \frac{3 \sin \theta - 2 \csc \theta}{\sqrt{3}} + \frac{\partial}{\partial \phi} i - \frac{\partial^2}{\partial \phi^2} \frac{2 \cot \theta \csc \theta}{\sqrt{3}} \right] \chi(\theta, \phi). \quad (3.24)$$

Now that the mapping of the operator $\hat{\sigma}^z$ on the kernel is shown, one can perform a translation of the 2-Level-System (2LS) of single driven atom.

3.2.2 Many spins and truncated Wigner approximation

Once the mapping for a single spin is formalized, the next step is to consider the interaction between the spins. The formalism for $N > 1$ spins can be constructed via the phase-space representation of the many-body density operator:

$$\hat{\rho}(t) = \left[\prod_{n=1}^N \int d\Omega_n \hat{\Delta}(\theta_n, \phi_n) \right] W(t, \boldsymbol{\theta}, \boldsymbol{\phi}), \quad (3.25)$$

where $d\Omega_n = d\theta_n \sin \theta_n d\phi_n / 2\pi$ corresponds to the n th spin, $\boldsymbol{\theta} = (\theta_1, \dots, \theta_N)^T$, and $\boldsymbol{\phi} = (\phi_1, \dots, \phi_N)^T$.

Using the FWF, we define

$$\chi(\boldsymbol{\theta}, \boldsymbol{\phi}) = \sum_{n=1}^N \frac{\sin \theta_n}{2\pi} W(\boldsymbol{\theta}, \boldsymbol{\phi}). \quad (3.26)$$

The spin interactions, which is what the Rydberg system needed, can be represented in the form of an Ising Hamiltonian:

$$\hat{H} = -\frac{1}{2} \sum_{m < n} J_{mn} \hat{\sigma}_m^z \hat{\sigma}_n^z, \quad (3.27)$$

From Eq. (3.27), mapping the spin interactions to phase space gives the rate equation for the PPO as

$$\begin{aligned} \frac{d}{dt} \hat{\Delta}(\boldsymbol{\theta}, \boldsymbol{\phi}) &= -\frac{i}{2} \sum_{m < n} J_{mn} [\hat{\sigma}_m^z \hat{\sigma}_n^z, \hat{\Delta}(\boldsymbol{\theta}, \boldsymbol{\phi})] \\ &= \sum_n \sum_{m \neq n} J_{mn} \left[-\sqrt{3} \cos \theta_m \frac{\partial}{\partial \phi_n} + \frac{2 \csc \theta_m - 3 \sin \theta_m}{\sqrt{3}} \frac{\partial^2}{\partial \theta_m \partial \phi_n} \right. \\ &\quad \left. - \frac{2 \cot \theta_m \csc \theta_m}{\sqrt{3}} \frac{\partial^3}{\partial \phi_m^2 \partial \phi_n} \right] \hat{\Delta}(\boldsymbol{\theta}, \boldsymbol{\phi}). \end{aligned} \quad (3.28)$$

One can observe that higher-order derivatives appear in this expression. However, to obtain a Fokker-Planck Equation (FPE), the terms with mixed derivatives—such as the second and third terms, which entangle the spins—must be neglected. Thus, to achieve a valid TWA, all mixed-derivative terms must be truncated.

Next, one obtains the FPE from the FWF $\chi(\boldsymbol{\theta}, \boldsymbol{\phi}, t)$. For the case of an Ising-type Hamiltonian, this reduces to a set of Ordinary Differential Equation (ODE)s for the spin angles:

$$\frac{d}{dt} \theta_n(t) = 0, \quad (3.29a)$$

$$\frac{d}{dt} \phi_n(t) = -\sqrt{3} \sum_{m \neq n} J_{mn} \cos \theta_m(t). \quad (3.29b)$$

To check that this spin angles representation is true, transforming to Cartesian coordinates yields

$$\frac{d}{dt} \begin{pmatrix} s_n^x \\ s_n^y \\ s_n^z \end{pmatrix} = \sum_{m \neq n} J_{mn} s_m^z \begin{pmatrix} +s_n^y \\ -s_n^x \\ 0 \end{pmatrix}, \quad (3.30)$$

which is the mean-field equation of motions for the Ising Hamiltonian in DTWA.

3.2.3 Dephasing, decay, and incoherent pump

Now that the spin operator and the Ising-type spin interaction mapping have been presented, the next step is to couple the system to an external Markovian reservoir in which dissipation is present. The TWA formalism for spins allows the mapping of Eq. 3.24. Using this relation, for the dephasing Lindbladian we have

$$\hat{\sigma}_z \hat{\rho} \hat{\sigma}_z - \hat{\rho} \longleftrightarrow 2 \frac{\partial^2}{\partial \phi^2} \chi, \quad (3.31)$$

and from the corresponding FPE, the transformation to Cartesian coordinates give a set of SDEs

$$ds^x = -2s^x dt - 2s^y dW_\phi, \quad (3.32a)$$

$$ds^y = -2s^y dt + 2s^x dW_\phi, \quad (3.32b)$$

$$ds^z = 0. \quad (3.32c)$$

where s^x, s^y, s^z are the components of the spin on the Bloch sphere, and dW_ϕ is a Wiener increment (Gaussian white noise) representing random phase fluctuations. The deterministic terms $(-2s^x dt, -2s^y dt)$ cause an exponential decay of the transverse spin components, which is dephasing. The stochastic terms $\propto dW_\phi$ introduce random rotation in the $x - y$ plane due to phase noise, and s^z remains constant, which is consistent with pure dephasing, not energy relaxation.

Similarly, the mapping for incoherent gains and losses gives

$$\begin{aligned} \Gamma_0 \hat{\sigma}^\pm \hat{\rho} \hat{\sigma}^\mp - \frac{\Gamma_0}{2} \{ \hat{\sigma}^\mp \hat{\sigma}^\pm, \hat{\rho} \} &\longleftrightarrow -\Gamma_0 \frac{\partial}{\partial \theta} \left(\cot \theta \mp \frac{\csc \theta}{\sqrt{3}} \right) \chi \\ &+ \frac{\Gamma_0}{2} \frac{\partial^2}{\partial \phi^2} \left(1 + 2 \cot^2 \theta - \frac{2 \cot \theta \csc \theta}{\sqrt{3}} \right) \chi, \end{aligned} \quad (3.33)$$

where Γ_0 is the Einstein A coefficient. The corresponding SDEs in θ, ϕ parametrization become

$$d\theta = \Gamma_0 \left(-\frac{\cot \theta \mp \csc \theta}{\sqrt{3}} \right) dt, \quad (3.34a)$$

$$d\phi = \sqrt{\Gamma_0 \left(\frac{1 + 2 \cot^2 \theta - 2 \cot \theta \csc \theta}{\sqrt{3}} \right)} dW. \quad (3.34b)$$

3.2.4 Fockker-Planck Equations to Stochastic Differential Equations

Once this correspondence is established, one can perform efficient simulations using numerical tools such as `DifferentialEquations.jl` in `Julia` to obtain the full dynamics of dissipative spin systems. Numerical simulation using the relation between the FPE and the Itô SDEs, describing dynamics with deterministic drift and stochastic noise, is straightforward. Including the Lindblad operators $\hat{L}_\mu$, one finds that

$$\int d\theta d\phi \hat{\Delta}(\theta, \phi) \frac{\partial}{\partial t} \chi = \int d\theta d\phi \hat{\Delta}(\theta, \phi) \mathcal{L} \chi, \quad (3.35)$$

so that the corresponding FWF satisfies the partial differential equation

$$\frac{\partial}{\partial t} \chi(\theta, \phi, t) = \mathcal{L} \chi(\theta, \phi, t), \quad (3.36)$$

which is exact. When the operator $\mathcal{L}$ is of Fokker-Planck type, this equation can be written as

$$\frac{\partial}{\partial t} \chi(\mathbf{x}, t) = \left(- \sum_n \frac{\partial}{\partial x_n} A_n(\mathbf{x}, t) + \sum_{m,n} \frac{\partial^2}{\partial x_m \partial x_n} D_{mn}(\mathbf{x}, t) \right) \chi(\mathbf{x}, t), \quad (3.37)$$

where $\mathbf{x} = (\theta, \phi)^T$, $\mathbf{A}(\mathbf{x}, t)$ is the drift vector describing deterministic processes, and $\mathbf{D}(\mathbf{x}, t) = \frac{1}{2} \mathbf{B}(\mathbf{x}, t)^T \mathbf{B}(\mathbf{x}, t)$ is the diffusion matrix, which must be positive semidefinite. This equation corresponds to the multi-dimensional FPE

$$\frac{\partial}{\partial t} W(\boldsymbol{\Omega}, t) = \left(- \sum_{n=1}^N \frac{\partial}{\partial x_n} A_n(\boldsymbol{\Omega}, t) + \sum_{m,n=1}^N \frac{\partial^2}{\partial x_m \partial x_n} D_{mn}(\boldsymbol{\Omega}, t) \right) W(\boldsymbol{\Omega}, t), \quad (3.38)$$

where $\boldsymbol{\Omega}$ denotes all phase-space variables $\{\theta_n, \phi_n\}$ for N spins.

The corresponding Itô SDE that reproduces this FPE is given by

$$dx_n(t) = A_n(\boldsymbol{\Omega}, t) dt + \sum_m B_{mn}(\boldsymbol{\Omega}, t) dW_m, \quad (3.39)$$

where $d\mathbf{W} = (dW_1, \dots, dW_{2N})^T$ represents a multivariate differential Wiener process.

The procedure for performing numerical simulations of the time evolution via these SDEs is as follows. First, one must specify the number of trajectories n_T for the system. For each trajectory, the phase-space coordinates $\Omega^{(n)} = (\theta^{(n)}, \phi^{(n)})$ of all N spins are initialized according to the Wigner distribution $W_0(\Omega)$. For example, Eq. 3.11b shows the initialization of spins in the down state $|\downarrow\rangle$ and the up state $|\uparrow\rangle$. Since a quantum state is not represented by a single point in phase space, each spin and each trajectory begin with different initial conditions.

Next, the time evolution of each trajectory proceeds independently according to the stochastic equations (3.39). The stochastic terms introduce independent noise realizations from the Wiener process, ensuring that the trajectories are statistically independent of one another. During the time evolution, the Weyl symbol $\mathcal{A}(\Omega(t))$ of the desired observable is computed for each trajectory at every time step.

Finally, the expectation value of the observable is obtained by averaging over all trajectories,

$$\langle \hat{A}(t) \rangle \approx \frac{1}{n_T} \sum_{n=1}^{n_T} \mathcal{A}(\Omega^{(n)}(t)), \quad (3.40)$$

where $\Omega^{(n)}(t)$ denotes the time-evolved phase-space variables in the n th trajectory.

In the limit of an infinite number of trajectories ($n_T \rightarrow \infty$), this stochastic approach reproduces the exact time evolution described by the Fokker–Planck equation (3.38). The number of trajectories required for a good approximation depends on the strength of the interactions and the relative magnitude of the stochastic noise compared to the deterministic terms.

3.3 Mapping of the Lindblad Equation to the Phase Space for the Rydberg System

For this work, the master Lindblad equation needed is

$$\frac{d}{dt} \hat{\rho} = i[\hat{\rho}, \hat{H}] + \sum_l \mathcal{L}_l \hat{\rho} \mathcal{L}_l^\dagger - \frac{1}{2} \{ \mathcal{L}_l^\dagger \mathcal{L}_l, \hat{\rho} \}, \quad (3.41)$$

with the Hamiltonian

$$\hat{H} = \sum_i \left[\Omega (\hat{\sigma}_i^{gr} + \hat{\sigma}_i^{rg}) + \left(\sum_{j \neq i} \frac{c_6}{r_{ij}^6} \hat{\sigma}_j^{rr} - \Delta \right) \hat{\sigma}_i^{rr} \right], \quad (3.42)$$

where Ω is the laser driving strength, Δ is the detuning from the ground-Rydberg resonance frequency, and $\sum_{j \neq i} \frac{c_6}{r_{ij}^6}$ represents the van der Waals interaction, with $r_{ij} = |r_j - r_i|$ being the distance between atom i and j . The dissipative processes are governed by Lindblad terms, with $\mathcal{L}_i = \sqrt{\Gamma} \hat{\sigma}_i^{gr}$ for spontaneous decay with $\sigma_i^{\mu\nu} = |\mu\rangle_{ii}\langle\nu|$ and $\mathcal{L}_\perp^{(i)} = \sqrt{\gamma} \hat{\sigma}_i^{rr}$ for dephasing and Doppler broadening with $\hat{\sigma}_i^{rr} = (\hat{\mathbf{1}} + \hat{\sigma}_i^z)$.

For the first part of the Hamiltonian involving driving the atom, the FPE can be expressed for the unitary dynamics of a single spin is of the form

$$-i[\hat{\sigma}^x, \hat{\rho}] \leftrightarrow -2 \left(\frac{\partial}{\partial \theta} \sin \phi + \frac{\partial}{\partial \phi} \cot \theta \cos \phi \right) \chi \quad (3.43)$$

with the corresponding SDEs to be

$$\begin{aligned} d\theta_i &= [2 \Omega \sin \phi_i] dt, \\ d\phi_i &= [2 \Omega \cot \theta_i \cos \phi_i] dt. \end{aligned} \quad (3.44)$$

For the Lindblad dissipative terms, the mapping as already showed in Eq. 3.34, here the corresponding SDEs are

$$\begin{aligned} d\theta_i &= \left[\Gamma \left(\cot \theta_i + \frac{\csc \theta_i}{\sqrt{3}} \right) \right] dt, \\ d\phi_i &= \sqrt{\Gamma \left(1 + 2 \cot^2 \theta_i + \frac{2 \cot \theta_i \csc \theta_i}{\sqrt{3}} \right) + 4\gamma} dW_i(t), \end{aligned} \quad (3.45)$$

with the dephasing γ involving the $\hat{\sigma}^z$ term, the mapping of that is simply $-i[\hat{\sigma}^z, \hat{\rho}] \leftrightarrow -2\frac{\partial}{\partial \phi}\chi$. The detuning Δ , is coupled to a single $\hat{\sigma}^{rr}$ term, thus expressing it in terms of $\hat{\sigma}^z$ give the mapping to FPE is $\Delta\frac{\partial}{\partial \phi}\chi$.

So far, the set of SDEs are

$$\begin{aligned} d\theta_i &= \left[2\Omega \sin \phi_i + \Gamma \left(\cot \theta_i + \frac{\csc \theta_i}{\sqrt{3}} \right) \right] dt, \\ d\phi_i &= [2\Omega \cot \theta_i \cos \phi_i + \Delta] dt \\ &\quad + \sqrt{\Gamma \left(1 + 2 \cot^2 \theta_i + \frac{2 \cot \theta_i \csc \theta_i}{\sqrt{3}} \right) + 4\gamma} dW_i(t), \end{aligned} \quad (3.46)$$

In regard to the Rydberg-Rydberg interaction, similar approach can be taken from the case of the Ising interaction Hamiltonian, Eq. 3.28, and consider the case of nearest neighbor interaction, the scaling of the interaction is simply reduced from the C_6 interaction to the nearest neighbor interaction strength as V ,

$$\frac{d}{dt}\hat{\Delta}(\boldsymbol{\theta}, \boldsymbol{\phi}) = - \sum_{j \neq i} V_{ij} [\hat{\sigma}_i^{rr} \hat{\sigma}_j^{rr}, \hat{\Delta}(\boldsymbol{\theta}, \boldsymbol{\phi})] \quad (3.47)$$

after truncation of higher-order derivative terms for the FPE gives us the ODE for Rydberg-Rydberg interaction

$$\frac{d}{dt}\theta_n(t) = 0, \quad (3.48a)$$

$$\frac{d}{dt}\phi_n(t) = - \sum_{j \neq i} V_{ij} (1 + \sqrt{3}) \cos \theta_j(t). \quad (3.48b)$$

which gives

$$d\phi_i = \left[-V \sum_{j \neq i} (1 + \sqrt{3} \cos \theta_j) \right] dt \quad (3.49)$$

Finally, the complete set of Itô SDEs for our Rydberg system in the phase space is

$$\begin{aligned} d\theta_i &= \left[2\Omega \sin \phi_i + \Gamma \left(\cot \theta_i + \frac{\csc \theta_i}{\sqrt{3}} \right) \right] dt, \\ d\phi_i &= \left[2\Omega \cot \theta_i \cos \phi_i - V \sum_{j \neq i} (1 + \sqrt{3} \cos \theta_j) + \Delta \right] dt \\ &\quad + \sqrt{\Gamma \left(1 + 2 \cot^2 \theta_i + \frac{2 \cot \theta_i \csc \theta_i}{\sqrt{3}} \right) + 4\gamma} dW_i(t), \end{aligned} \quad (3.50)$$

this set of stochastic differential equations is ready for simulations.

Part II

Results

4 Evaluating TWA for spins: Limits and Reliability

4.1 Case Studies for Assessment

The benchmark results of the TWA methods are presented for various models and cases, showing agreement with exact theoretical results and direct computation from the Hamiltonian. The models presented here start with the simplest case, a 2LS for a single spin. The second case is the Dicke model for superradiance of interacting spins. Both cases mentioned above show high overlap between the TWA and the exact data, showcasing the strength of the method.

The last case, however, highlights some weaknesses of the method. The case study is the TFIM, where the order parameter m_z (magnetization) is determined from the long-range spin correlations. From this, the critical exponent β of the phase transition is extracted. The results reveal a disagreement between the predictions of the TWA and the thermodynamic limit of the model.

The section concludes by presenting an argument on why this method fails in some aspects while remaining valid in others. Furthermore, the differences between EPT and NEPT have distinct starting conditions, thus still making NEPT valid for using TWA as the primary method.

4.1.1 Two-Level System

In a 2LS, atoms have two states: the ground state $|g\rangle$ and the excited state $|e\rangle$, equivalent to a spin- $\frac{1}{2}$ system with states $|+\frac{1}{2}\rangle$ and $|-\frac{1}{2}\rangle$, belonging to the SU(2) group. The dynamics are described using Pauli matrices:

$$\hat{\sigma}^z = |e\rangle\langle e| - |g\rangle\langle g|, \quad (4.1)$$

$$\hat{\sigma}^+ = |e\rangle\langle g|, \quad (4.2)$$

$$\hat{\sigma}^- = |g\rangle\langle e|. \quad (4.3)$$

The spin population is governed by $\hat{\sigma}^z$, while $\hat{\sigma}^+$ and $\hat{\sigma}^-$ drive transitions between the up and down states. The operators $\hat{\sigma}^x$ and $\hat{\sigma}^y$ are expressed as:

$$\hat{\sigma}^x = \hat{\sigma}^- + \hat{\sigma}^+, \quad (4.4)$$

$$\hat{\sigma}^y = i(\hat{\sigma}^- - \hat{\sigma}^+), \quad (4.5)$$

$$\hat{\sigma}^l \hat{\sigma}^m = \delta_{lm} \hat{1} + i \sum_{k=1}^3 \epsilon_{lmk} \hat{\sigma}^k, \quad (4.6)$$

where $\hat{1}$ is the identity matrix.

The collective spin operators for an ensemble are denoted by $\hat{S} = \hat{S}_N$, with N omitted, and calculated as:

$$\hat{S}^i = \sum_n^N \hat{\sigma}_n^i, \quad (4.7)$$

The collective states are characterized by the spin quantum number $0 \leq j \leq \frac{N}{2}$ and the projection on the z -axis $|m| \leq j$:

$$\hat{S}^2 |j, m\rangle = j(j+1) |j, m\rangle, \quad (4.8)$$

$$\hat{S}^z |j, m\rangle = m |j, m\rangle, \quad (4.9)$$

$$\hat{S}^\pm |j, m\rangle = \sqrt{(j \mp m)(j \pm m + 1)} |j, m \pm 1\rangle. \quad (4.10)$$

4.1.2 Master Equations

The dynamics of a quantum system coupled to a reservoir are described by the Hamiltonian [60]:

$$\hat{H}' = \sum_n^N \frac{\omega_a}{2} \hat{\sigma}_n^z + \sum_m^N \omega_m \hat{a}_m^\dagger \hat{a}_m + \sum_n^N \sum_m^N g_m \left(\hat{a}_m \hat{\sigma}_n^+ + \hat{a}_m^\dagger \hat{\sigma}_n^- \right), \quad (4.11)$$

where ω_a is the atomic transition frequency, ω_m is the m -th reservoir mode frequency, $\hat{a}_m^\dagger$ and $\hat{a}_m$ are creation and annihilation operators, and g_m is the spin-reservoir coupling strength.

Since a full description of the bosonic degrees of freedom is complex, an approximation is made to eliminate the light field. A few assumptions are introduced: first, the system and the reservoir are independent at the initial time and are coupled only through the interaction term in the Hamiltonian. The Born approximation can be applied due to the weak coupling $g_m \ll \omega_a, \omega_m$ between the electric field and the system. Under this assumption, the total density operator factorizes as

$$\hat{\rho}_{\text{tot}}(t) = \hat{\rho}_{\text{sys}}(t) \otimes \hat{\rho}_{\text{res}}(t). \quad (4.12)$$

The bosonic light field is now regarded as a reservoir. The reduced density operator is obtained by tracing out the reservoir degrees of freedom from the total density operator:

$$\hat{\rho}(t) = \text{Tr}_{\text{res}} \{ \hat{\rho}_{\text{tot}}(t) \}. \quad (4.13)$$

The Markov approximation assumes an instantaneous decay of system-reservoir correlations, such that the reservoir state does not change on the timescale of the system [61]:

$$\hat{\rho}_{\text{tot}}(t) = \hat{\rho}(t) \otimes \hat{\rho}_{\text{res}}(0). \quad (4.14)$$

Under these assumptions, the time evolution of the system can be expressed by the Lindblad master equation [62]:

$$\frac{d}{dt} \hat{\rho}(t) = -i [\hat{H}, \hat{\rho}] + \frac{1}{2} \sum_i \left(2\hat{L}_i \hat{\rho} \hat{L}_i^\dagger - \hat{L}_i^\dagger \hat{L}_i \hat{\rho} - \hat{\rho} \hat{L}_i^\dagger \hat{L}_i \right), \quad (4.15)$$

where the operators $\hat{L}_i$ are called *jump operators* and describe the non-unitary effects in the time evolution of the open system, such as dissipation and dephasing.

For the specific system considered here, the master equation takes the form

$$\frac{d}{dt}\hat{\rho} = -i[\hat{H}, \hat{\rho}] + \sum_{m,n}^N \frac{\Gamma_{mn}}{2} \left(2\hat{\sigma}_m^- \hat{\rho} \hat{\sigma}_n^+ - \{ \hat{\sigma}_m^+ \hat{\sigma}_n^-, \hat{\rho} \} \right), \quad (4.16)$$

where Γ_{mn} represents the decoherence rate between spins m and n .

4.2 Driven Single Atom

A single two-level atom interacting with a driving field represents the simplest open quantum system that can be analyzed using the Lindblad master equation formalism. As the interaction between the spins can be neglected, what remains are the influences from the Rabi oscillations Ω , and the coupling of the qubit to the external environment, including noise terms. For this system, the exact solution of the Optical Bloch Equations (OBE) is compared with $\hat{S}_z$, the population term, computed from the TWA. For this system, the atom has a ground state $|g\rangle$, an excited state $|e\rangle$, and is driven by a near-resonant laser. Dissipation enters the model through spontaneous emission into the surrounding electromagnetic vacuum at a rate Γ_0 .

Master Equation

The system dynamics are governed by the Lindblad master equation

$$\frac{d}{dt}\hat{\rho} = -i[\hat{H}_0, \hat{\rho}] + \frac{\Gamma_0}{2} \left(2\hat{\sigma}^- \hat{\rho} \hat{\sigma}^+ - \{ \hat{\sigma}^+ \hat{\sigma}^-, \hat{\rho} \} \right), \quad (4.17)$$

where $\hat{\rho}$ is the atomic density operator and $\hat{\sigma}^\pm$ are the raising and lowering operators of the 2LS. The Hamiltonian in the rotating frame reads

$$\hat{H}_0 = -\Omega_0 (\hat{\sigma}^+ + \hat{\sigma}^-) - \frac{\Delta}{2} \hat{\sigma}^z, \quad (4.18)$$

where Ω_0 is the Rabi frequency and $\Delta = \omega_L - \omega_0$ is the detuning between the driving laser frequency ω_L and the atomic transition frequency ω_0 .

Optical Bloch Equations

Writing the density matrix in the atomic basis $\{|e\rangle, |g\rangle\}$,

$$\hat{\rho} = \begin{pmatrix} \rho_{ee} & \rho_{eg} \\ \rho_{ge} & \rho_{gg} \end{pmatrix}, \quad (4.19)$$

the master equation yields the coupled differential equations known as the OBE:

$$\frac{d}{dt}\rho_{ee} = -\Gamma_0\rho_{ee} + i\Omega_0(\rho_{ge} - \rho_{eg}), \quad (4.20)$$

$$\frac{d}{dt}\rho_{gg} = \Gamma_0\rho_{ee} + i\Omega_0(\rho_{eg} - \rho_{ge}), \quad (4.21)$$

$$\frac{d}{dt}\rho_{ge} = \left(-i\Delta - \frac{\Gamma_0}{2}\right)\rho_{ge} + i\Omega_0(\rho_{ee} - \rho_{gg}), \quad (4.22)$$

$$\frac{d}{dt}\rho_{eg} = \left(i\Delta - \frac{\Gamma_0}{2}\right)\rho_{eg} + i\Omega_0(\rho_{gg} - \rho_{ee}). \quad (4.23)$$

These equations describe the coupled dynamics of the populations (ρ_{ee}, ρ_{gg}) and the coherences (ρ_{ge}, ρ_{eg}) . Because of the constraints $\rho_{ee} + \rho_{gg} = 1$ and $\rho_{eg} = \rho_{ge}^*$, only two equations are independent. The expression for the Rabi frequency is

$$\Omega = \frac{\vec{d}_{g,e} \cdot \vec{E}_0}{\hbar}, \quad (4.24)$$

where $\vec{d}_{g,e}$ is the transition dipole moment of the $g \rightarrow e$ transition and $\vec{E}_0 = \hat{e}E_0$ is the vector electric field amplitude, including the polarization vector $\hat{e}$. The detuning

$$\delta = \omega - \omega_0 \quad (4.25)$$

measures the difference between the light frequency ω and the atomic transition frequency ω_0 .

From the OBE, one can compute observable quantities such as the atomic inversion:

$$\langle \hat{\sigma}^z \rangle(t) = \text{Tr}\{\hat{\sigma}^z \hat{\rho}(t)\} = \rho_{ee}(t) - \rho_{gg}(t). \quad (4.26)$$

This quantity measures whether the atom is predominantly in the excited state ($\langle \hat{\sigma}^z \rangle > 0$) or in the ground state ($\langle \hat{\sigma}^z \rangle < 0$). At steady state, the inversion reflects the balance between coherent laser driving and spontaneous emission.

Stochastic Formulation

An alternative approach to solving the population of the single atom is provided by stochastic methods. The mapping from the Hilbert space of the Hamiltonian Eq. 4.18 and the master equation Eq. 4.17 to the phase space using the kernel $\hat{\Delta}(\theta, \phi)$ yields a set of FPE. From here, the conversion to a set of Itô SDEs is trivial. To reiterate, the system comprises a single spin coupled to an external environment with decoherence and has two distinct equations, $d\theta$ and $d\phi$, that are derived from the mappings.

$$d\theta = \left[2\Omega_0 \sin \phi + \Gamma \left(\cot \theta + \frac{\csc \theta}{\sqrt{3}} \right) \right] dt, \quad (4.27)$$

$$d\phi = 2\Omega_0 \cot \theta \sin \phi dt + \sqrt{\Gamma \left(1 + 2 \cot^2 \theta + \frac{2 \cot \theta \csc \theta}{\sqrt{3}} \right)} dW_\phi, \quad (4.28)$$

where dW_ϕ is a Wiener increment representing quantum noise and is only present in $d\phi$, dt is the evolution of the Bloch vector inside the sphere. The driving Ω and decoherence Γ in this Itô SDEs are present in both equations, with influence on both θ and ϕ . Additionally, no higher-order derivative terms are neglected as there is no spin interaction, so the FPE is a direct mapping from the Hilbert space without any loss of information.

Numerical Illustration

The simulations are fairly straightforward. The spin is initialized in the down state $|g\rangle$, with the ODE solver calculating the time evolution of the OBE. For the case of the TWA, the spin is sampled near the region of the complete down state, that is, $\theta = \arccos(1/\sqrt{3})$, with 2×10^4 trajectories. Setting $\Delta = 0$ with $\Omega, \Gamma = 1$, the simulation is run until the spin has reached its steady state. Lastly, the result collected from the TWA the population term, θ , is compared to the expectation value of $\langle \hat{S}_z \rangle$ from the OBE.

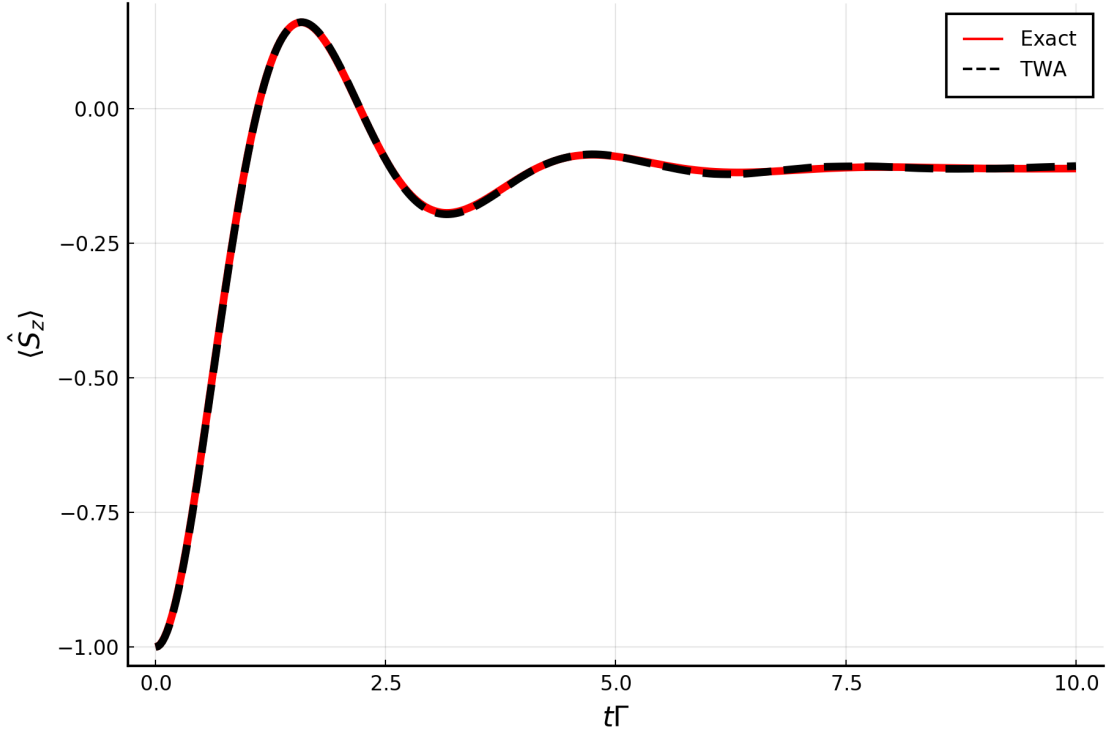


Figure 4.1: The dynamics of a single spin system from the TWA compared to the exact solution from the OBE. The TWA method for single spin shows complete agreement with the OBE. Numerical simulations are carried out independently and compared with Jens Hartmann’s diploma thesis *Truncated Wigner Simulations of Collective Radiation Phenomena in Spin Systems*.

4.3 Dicke Superradiance

Independent spontaneous decay from the case of a driven single atom is largely different from the collective light emission from an ensemble of N two-level atoms. When atoms couple to a common radiation mode, they emit photons collectively and cooperatively. The result is a burst of radiation; Dicke coined this phenomenon *superradiance* in 1954 [63]. The cause of the collective burst effect is that the quantum correlations between atoms build up over time to modify the coherent dynamics and the dissipation. The burst occurs if the two-level systems are initialized in the excited state and placed at a distance much smaller than the photon’s wavelength. The pulse duration

is inversely proportional to the number of spins, N , and the maximal intensity of the emitted light scales as N^2 .

Master Equation for Collective Light Emission

The many-body system is described by the master equation

$$\frac{d}{dt}\hat{\rho} = -i[\hat{H}, \hat{\rho}] + \frac{1}{2} \sum_{mn} \Gamma_{mn} \left(2\hat{\sigma}_m^+ \hat{\rho} \hat{\sigma}_n^- - \hat{\sigma}_m^+ \hat{\sigma}_n^- \hat{\rho} - \hat{\rho} \hat{\sigma}_m^+ \hat{\sigma}_n^- \right), \quad (4.29)$$

where Γ_{mn} denotes the collective decay rate coupling atoms m and n . For $m = n$, this reduces to ordinary spontaneous emission, while for $m \neq n$, it describes correlated photon emission processes. For the ideal Dicke model, identical decay rates $\Gamma_{mn} = \Gamma_0$ and no coherent exchange interactions ($J_{mn} = 0$) are assumed, as proposed by Dicke for a strongly confined free gas or in cavity/waveguide QED.

The effective Hamiltonian governing the internal dynamics is

$$\hat{H} = -\frac{\Delta}{2} \sum_n \hat{\sigma}_n^z - \sum_n \left(\Omega_n \hat{\sigma}_n^+ + \text{h.a.} \right) + \sum_{m \neq n} J_{mn} \hat{\sigma}_m^+ \hat{\sigma}_n^-, \quad (4.30)$$

with detuning Δ , coupling to the external coherent drive Ω_n , and radiative coupling between the 2-level spins with rate J_{mn} .

Dicke States and Population Dynamics

To capture the essence of superradiance, assume that the atomic ensemble is initially in the fully inverted state $|e_1 e_2 e_3 \dots\rangle = |j = N/2, m = N/2\rangle$. Using the master equation

$$\frac{d}{dt}\hat{\rho} = \frac{1}{2} \sum_{mn} \Gamma_{mn} \left(2\hat{\sigma}_m^+ \hat{\rho} \hat{\sigma}_n^- - \hat{\sigma}_m^+ \hat{\sigma}_n^- \hat{\rho} - \hat{\rho} \hat{\sigma}_m^+ \hat{\sigma}_n^- \right), \quad (4.31)$$

with the ansatz

$$\hat{\rho} = \sum_{m=-N/2}^{N/2} \rho_m |j, m\rangle \langle j, m|, \quad (4.32)$$

yields the coupled rate equations

$$\begin{aligned} \frac{d}{dt}\rho_m &= -\Gamma_0(j+m)(j-m+1)\rho_m \\ &\quad + \Gamma_0(j+m+1)(j-m)\rho_{m+1} \\ &= \Gamma_0[g(j, m+1)\rho_{m+1} - g(j, m)\rho_m], \end{aligned} \quad (4.33)$$

where $-\Gamma_0(j+m)(j-m+1)$ is the transition rate from $|j, m\rangle$ to $|j, m-1\rangle$. When $m = 0$, the transition rate peaks, and it is at its minimum when $m = -j, +j$.

Stochastic Formulation

Because the exact mapping of the master equation is beyond a Fokker–Planck description, the asymptotic correspondence rules

$$W_{\mathcal{S}(\{c_n\})\hat{\rho}}(\Omega) \approx \mathcal{S}(\{c_n\})W_{\hat{\rho}}(\Omega) = \sum_{n=1}^N c_n (s_n + L_n) W_{\hat{\rho}}(\Omega), \quad (4.34)$$

with

$$L_n = i\nabla_{\theta_n} \begin{pmatrix} +\sin \phi_n \\ -\cos \phi_n \\ 0 \end{pmatrix} + i\nabla_{\phi_n} \begin{pmatrix} \cot \theta_n \cos \phi_n \\ \cot \theta_n \sin \phi_n \\ -1 \end{pmatrix}. \quad (4.35)$$

are applied to reduce it and obtain the expression

$$\frac{\partial}{\partial t} W_{\hat{\rho}}(\Omega, t) = \left(-\mathcal{L}_1 + \frac{1}{2} \mathcal{L}_2 \right) W_{\hat{\rho}}(\Omega, t), \quad (4.36)$$

$$\mathcal{L}_1 = \sum_{n=1}^N \left\{ \nabla_{\theta_n} \left[\frac{\Gamma_{nn}}{2} \cot \theta_n + \sqrt{3} \sum_{m=1}^N \sin \theta_m \left(J_{mn} \sin \phi_{mn} + \frac{\Gamma_{mn}}{2} \cos \phi_{mn} \right) \right] \right. \quad (4.37)$$

$$\left. + \nabla_{\phi_n} \sqrt{3} \cot \theta_n \sum_{m=1}^N \sin \theta_m \left(-J_{mn} \cos \phi_{mn} + \frac{\Gamma_{mn}}{2} \sin \phi_{mn} \right) \right\},$$

$$\mathcal{L}_2 = \sum_{m,n=1}^N \Gamma_{mn} \left(\nabla_{\theta_m} \nabla_{\theta_n} \cos \phi_{mn} + \nabla_{\phi_m} \nabla_{\phi_n} \cot \theta_m \cot \theta_n \cos \phi_{mn} \right. \quad (4.38)$$

$$\left. - \nabla_{\theta_m} \nabla_{\phi_n} \cot \theta_n \sin \phi_{mn} + \nabla_{\phi_m} \nabla_{\theta_n} \cot \theta_m \sin \phi_{mn} \right).$$

where $\phi_{mn} = \phi_m - \phi_n$, the equivalent set of Itô SDEs is expressed as

$$d\theta_n = \left[\frac{\Gamma_{nn}}{2} \cot \theta_n + \sqrt{3} \sum_{m=1}^N \sin \theta_m \left(J_{mn} \sin \phi_{mn} + \frac{\Gamma_{mn}}{2} \cos \phi_{mn} \right) \right] dt \\ + \sum_{m=1}^N G_{nm} (-\cos \phi_n dW_{\theta m} + \sin \phi_n dW_{\phi m}), \quad (4.39)$$

$$d\phi_n = \sqrt{3} \cot \theta_n \sum_{m=1}^N \sin \theta_m \left(-J_{mn} \cos \phi_{mn} + \frac{\Gamma_{mn}}{2} \sin \phi_{mn} \right) dt \\ + \sum_{m=1}^N G_{nm} \cot \theta_n (\sin \phi_n dW_{\theta m} + \cos \phi_n dW_{\phi m}), \quad (4.40)$$

where $dW_{\theta m}$ and $dW_{\phi m}$ denote independent Wiener processes, and $G_{mn} = \sqrt{\Gamma_0} \delta_{n,1}$ couples the system to two noise processes.

Numerical Illustration

For a completely mixed initial state, the steady-state population is

$$\langle \hat{S}_z(t \rightarrow \infty) \rangle = \sum_{j=0}^{N/2} \frac{(2j+1)d_j}{2^N} (-j), \quad d_j = \frac{(2j+1)N!}{(N/2+j+1)!(N/2-j)!}. \quad (4.41)$$

where the sum computes the weighted contribution of each spin j state, where j ranges from 0 to $N/2$. The d_j term is the degeneracy factor for spin j , accounting for the number of ways to distribute N spins into states with total $S_z = -j$, along with the normalization factor. The factor $2N$ scales the sum to match the system size. From this, the exact solution of $\langle S_z \rangle / N$ over time is computed using a trivial ODE solver. Using the relation $\gamma(t) = \frac{d}{dt} \langle \hat{S}^z \rangle$, one obtains the total radiated intensity

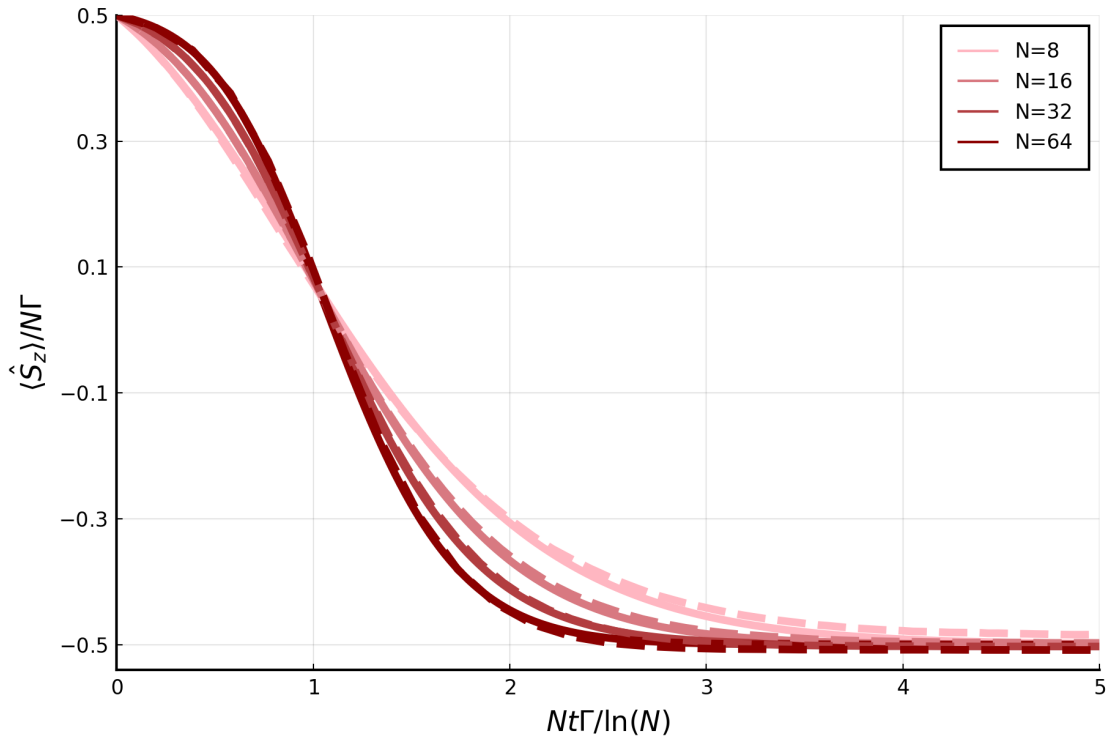


Figure 4.2: Collective decay of the Dicke model, with the solid lines being exact calculations, and dashed lines as the TWA. Numerical simulations are carried out independently and compared with the already available results from Mink, *et al.* [58].

$$I(t) = -\frac{\sqrt{3}}{2} \Gamma_0 \sum_n \overline{\cos \theta_n} - \frac{3}{4} \sum_{mn} \Gamma_{mn} \overline{\sin \theta_m \sin \theta_n \cos(\phi_m - \phi_n)} \quad (4.42)$$

Numerical simulations of Dicke superradiance reveal the characteristic radiation burst followed by rapid decay, as shown in Fig. 4.3.

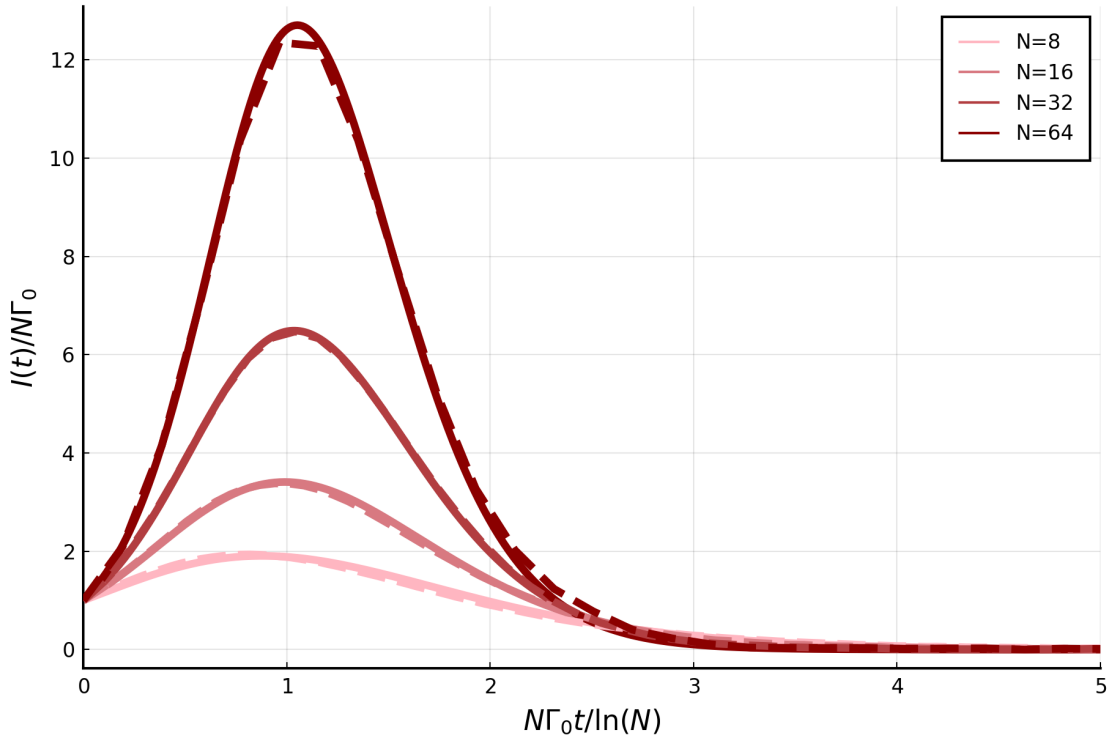


Figure 4.3: Radiated intensity of the superradiance burst, with the solid lines being exact calculations, and dashed lines as the TWA. Numerical simulations are carried out independently and compared with the already available results from Mink, *et al.* [58].

The high degree of accuracy of the TWA method with a small error margin showcases the strength of the method for such a complex system as the Dicke model, even at the truncated approach from the correspondence rules to obtain the Itô SDEs.

4.4 Equilibrium Quantum Phase Transition

Benchmark cases of the single driven atom and Dicke model highlight the strengths of the TWA. However, weaknesses arise in the TFIM case, particularly with the EPT, where calculations from the truncation method misalign with exact diagonalization of the Hamiltonian. The following section briefly outlines the theoretical foundation of the model [64], followed by the solution for phase transition derived from exact simulation and TWA. Results from both approaches are compared directly, with further discussion on the weaknesses and reasons for the failure to capture the true physics.

To illustrate quantum phase transitions, the transverse-field Ising model TFIM, known for its paramagnetic properties, serves as a good example. A well-known experimental realization is the compound LiHoF_4 , an ionic crystal where magnetic degrees of freedom originate from the spins of Holmium atoms [65]. These spins preferentially align along a specific crystal axis, known as the “easy axis,” and can be modeled using Ising spin variables. In the absence of an external field, dipolar interactions between spins result in a ferromagnetically ordered ground state.

In 1996, Bitko, Rosenbaum, and Aeppli demonstrated that applying a magnetic field perpendicular to the easy axis drives the system through a phase transition [66]. At finite temperatures, thermal fluctuations disrupt ferromagnetic order at a critical temperature of approximately $T_c \approx 1.5$ K, marking a classical continuous phase transition. At zero temperature, a sufficiently strong transverse magnetic field induces spin flips, leading to quantum fluctuations. When the field strength exceeds a critical value of $h_c \approx 50$ kOe, the long-range ferromagnetic order vanishes. This behavior, known as a *quantum phase transition*, is driven purely by quantum fluctuations, independent of thermal effects.

Critical Behavior and Quantum Phase Transition

Phase transitions are of two types: first-order, where two phases (e.g., water and ice) coexist, and continuous, where they do not (e.g., ferromagnetic phase transitions). The transition point of a continuous phase is called a critical point. Similar to the approach used with the NEPT, an order parameter emerges near this critical point, being zero in the disordered phase and non-zero in the ordered phase. In this study, the total magnetization of the ferromagnetic transition is the order parameter under investigation.

As the critical point is approached, two key correlation length scales—spatial and temporal—significantly influence the transition's behavior. Spatial correlations of order parameter fluctuations become long-ranged. Near the critical point, the correlation length ξ diverges as

$$\xi \propto |t|^{-\nu} \quad (4.43)$$

where ν is the correlation length critical exponent and t is a dimensionless measure of the distance from the critical point. Temporal correlations, expressed as

$$\tau_c \propto \xi^z \propto |t|^{-\nu z} \quad (4.44)$$

where z is the dynamic critical exponent, also become long-ranged. Near the critical point, no characteristic length scales exist other than ξ and τ_c .

In more detail, for the case of ferromagnet with order parameter being magnetization, there is a homogeneity relation for the singular part of the free energy density,

$$f(t, B) = b^{-d} f(tb^{1/\nu}, Bb^{y_B}) \quad (4.45)$$

with y_B being another critical point. b is an arbitrary positive number. The table below list out relevant critical exponents.

	exponent	definition	conditions
specific heat	α	$C \propto t ^{-\alpha}$	$t \rightarrow 0, B = 0$
order parameter	β	$m \propto (-t)^\beta$	$t \rightarrow 0$ from below, $B = 0$
susceptibility	γ	$\chi \propto t ^{-\gamma}$	$t \rightarrow 0, B = 0$
critical isotherm	δ	$B \propto m ^\delta \text{sign}(m)$	$B \rightarrow 0, t = 0$
correlation length	ν	$\xi \propto t ^{-\nu}$	$t \rightarrow 0, B = 0$
correlation function	η	$G(r) \propto r ^{-d+2-\eta}$	$t \rightarrow 0, B = 0$
dynamic	z	$\tau_c \propto \xi^z$	$t \rightarrow 0, B = 0$

for this study, the scaling relations β is of interest. Similar to NEPT, universality of one of the most relevant feature for recognizing classes for different physical systems. Hence, it's reasonable to imply, that every critical exponents of a phase transition in nature can be determined exactly by looking at its simpler models within the same universality class.

For the TFIM, several important critical exponents are

$$\alpha = 0, \beta = 1/8, \gamma = 7/4, \delta = 15, \eta = 1/4, \nu = 1 \quad (4.46)$$

Quantum–Classical Mapping

One of the most fascinating results arises from the differences between classical and quantum phase transitions. To establish the mapping and scaling, it is simplest to start from the partition function

$$Z = \text{Tr} e^{-H/k_B T} \quad (4.47)$$

where $H = H_{\text{kin}} + H_{\text{pot}}$ is the Hamiltonian of the system. In a classical system, the kinetic and potential terms commute, whereas in a quantum system they do not. This distinction leads to the fact that, in the classical case, the partition function can be factorized so that statics and dynamics decouple, while in the quantum case such factorization is not possible, implying that statics and dynamics are coupled.

Further analysis shows that the classical homogeneity law, Eq. (4.45), for the free energy density can be extended to the case of a quantum phase transition at zero temperature, giving

$$f(t, B) = b^{-d+z} f(tb^{1/\nu}, Bb^{y_B}) \quad (4.48)$$

where $t = |r - r_c|/r_c$. This demonstrates that a quantum phase transition in d spatial dimensions is related to a classical transition in $(d + z)$ dimensions.

Ising model

Choosing the z -axis as the spin easy axis, the Hamiltonian of the transverse-field Ising model reads:

$$H = -J \sum_{\langle i,j \rangle} \sigma_i^z \sigma_j^z - h \sum_i \sigma_i^x, \quad (4.49)$$

where J is the ferromagnetic coupling, h is the strength of the transverse magnetic field, and $\sigma_i^{x,z}$ are spin components at lattice site i . For $h = 0$, this reduces to the classical Ising model. Increasing h introduces quantum spin flips through the operator identity $h\sigma_i^x = \frac{h}{2}(\sigma_i^+ + \sigma_i^-)$, thereby generating quantum fluctuations.

A few characteristic traits of the model are worth mentioning. The model exhibits a Z_2 symmetry, $\sigma_i^z \mapsto -\sigma_i^z$. Furthermore, a few limiting cases can be identified:

- (a) For no external field ($h \rightarrow 0$):
 - The ground state $|0\rangle$ is ferromagnetic.

- The Z_2 symmetry is spontaneously broken.
- (b) For vanishing coupling ($h \rightarrow \infty$):
 - The ground state $|0\rangle$ is the polarized state $\prod_i |\rightarrow\rangle_i$.
 - There is no spontaneous symmetry breaking.

Consider a one-dimensional classical Ising chain with the Hamiltonian

$$H_{\text{cl}} = -J \sum_{i=1}^N \sigma_i^z \sigma_{i+1}^z. \quad (4.50)$$

Its partition function is given by

$$Z = \text{Tr} e^{-H_{\text{cl}}/T} = \text{Tr} \exp\left(\frac{J}{T} \sum_{i=1}^N \sigma_i^z \sigma_{i+1}^z\right) = \text{Tr} \prod_{i=1}^N M_{i,i+1}, \quad (4.51)$$

where $\mathbf{M}$ is the so-called transfer matrix. It can be represented as

$$\mathbf{M} = \begin{pmatrix} e^{J/T} & e^{-J/T} \\ e^{-J/T} & e^{J/T} \end{pmatrix} = e^{J/T} (1 + e^{-2J/T} \sigma^x) \approx \exp\left(\frac{J}{T} + e^{-2J/T} \sigma^x\right). \quad (4.52)$$

The partition function of the classical Ising chain can be written as

$$Z = \text{Tr} e^{-H_Q/T_Q} = \text{Tr} e^{h\sigma^x/T_Q} = \text{Tr} \prod_{i=1}^N e^{h\sigma^x/(T_Q N)}. \quad (4.53)$$

Thus, a single quantum spin can be mapped onto a classical Ising chain. Here, a d -dimensional transverse-field (quantum) Ising model which can be mapped onto a $(d+1)$ -dimensional classical Ising model. Consequently, the dynamic exponent is $z = 1$ here.

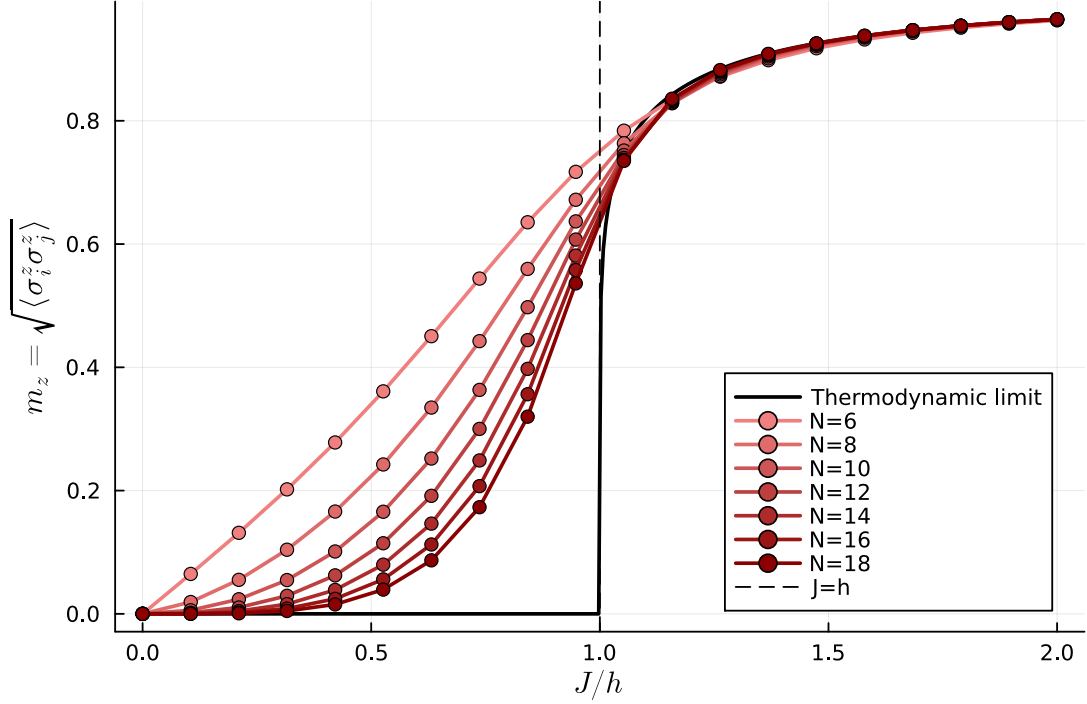


Figure 4.4: As the number of spins increases, the phase transition becomes sharper, approaching the thermodynamic limit.

From the Hamiltonian, one can calculate the order parameter magnetization m_z using the ground-state expectation value of the spin-spin correlation function along the z -direction

$$m_z = \sqrt{\langle \psi_0 | \sigma_i^z \sigma_j^z | \psi_0 \rangle} \quad (4.54)$$

where $|\psi_0\rangle$ is the ground state of the TFIM. This expression estimates the magnetization by measuring how strongly two distant spins are correlated. Figure 4.4 illustrates the competition between h and J . When the critical field $h_c \approx J$, a continuous phase transition occurs, driven by quantum fluctuations at zero temperature. The two-point correlation function $\langle \hat{\sigma}_i^z \hat{\sigma}_j^z \rangle$ measures the long-range order.

$m_z \approx 0$ for $J/h < 1$, the system is in the paramagnetic phase, whereas $m_z > 0$ for $J/h > 1$, the system is in the ferromagnetic phase. As the number of spins in the system increases, the curves converge to the thermodynamic limit, where mathematically, in the infinite-size limit ($N \rightarrow \infty$), the behavior of m_z is

$$m_z = \begin{cases} (1 - (h/J)^2)^{1/8}, & J > h, \\ 0, & J \leq h, \end{cases} \quad (4.55)$$

so the transition occurs at $J_c = h$, in which case $h = 1$. This model exhibits a quantum phase transition at zero temperature, a qualitative change in the ground state due to quantum fluctuations. At $J \gg h$, the ground state $\psi_0 \approx |\uparrow\uparrow \dots \uparrow\rangle$ or $|\downarrow\downarrow \dots \downarrow\rangle$, leads to symmetry breaking that makes $m_z > 0$. At $h \gg J$, $\psi_0 \approx |++ \dots +\rangle$, where

$|+\rangle$ a superposition of the up and down states. This transition belongs to the $(1+1)$ -dimensional Ising universality class, with critical exponent $\beta = 1/8$, which is where the power law (the first term of Eq. 4.55) comes from.

Phase Transition

One of the challenges when simulating the phase transition of the Ising model arises from its Z_2 symmetry. The degeneracy of the ground states is due to the fact that they transform into each other under the Z_2 symmetry. In the model, there exists a unitary operator U that performs the transformation

$$U\sigma_i^z U^\dagger = -\sigma_i^z. \quad (4.56)$$

Finding the true ground state of the model can be tricky when the spins are in the paramagnetic ground state. Suppose the system starts at $J = 0$, the Hamiltonian is simply

$$H = -h \sum_i \sigma_i^x, \quad (4.57)$$

where all spins are aligned along the $+x$ direction (paramagnetic). The state can be denoted as $|++++\rangle$ in the x -basis. Initially, the system has zero magnetization along the z direction.

When a sudden change occurs such that $J \gg h$, the Hamiltonian becomes dominated by

$$H \approx -J \sum_{\langle i,j \rangle} \sigma_i^z \sigma_j^z. \quad (4.58)$$

The new ground state is ferromagnetic, where all spins are either aligned along the $+z$ or $-z$ direction, with states denoted as $|\uparrow\uparrow \dots \uparrow\rangle$ and $|\downarrow\downarrow \dots \downarrow\rangle$. The initial state $|++++\rangle$ is no longer an eigenstate of the new Hamiltonian. Instead, it becomes a superposition of many eigenstates of the J -dominated Hamiltonian. Consequently, measuring σ^z immediately yields a mixture of outcomes corresponding to excited states rather than the true ground-state magnetization.

To understand this further, the Hamiltonian, similar in context to Eq. (4.49), can be expressed in terms of a parameter s that varies between 0 and 1.

$$H(s) = (1-s) \left(-\sum_i \sigma_i^x \right) + s \left(-\sum_i \sigma_i^z \sigma_{i+1}^z \right). \quad (4.59)$$

At $s = 0$, as mentioned, the Hamiltonian is dominated by the transverse field term $-\sum_i \sigma_i^x$, favoring a quantum paramagnetic phase. In this phase, the ground state is a product state such as $|++++\rangle$ (all spins aligned in the positive x -direction), with an energy $E_0 = -n = -5$ for $n = 5$ spins. Excited states, resulting from spin flips, are well-separated in energy due to the transverse field's strong alignment effect.

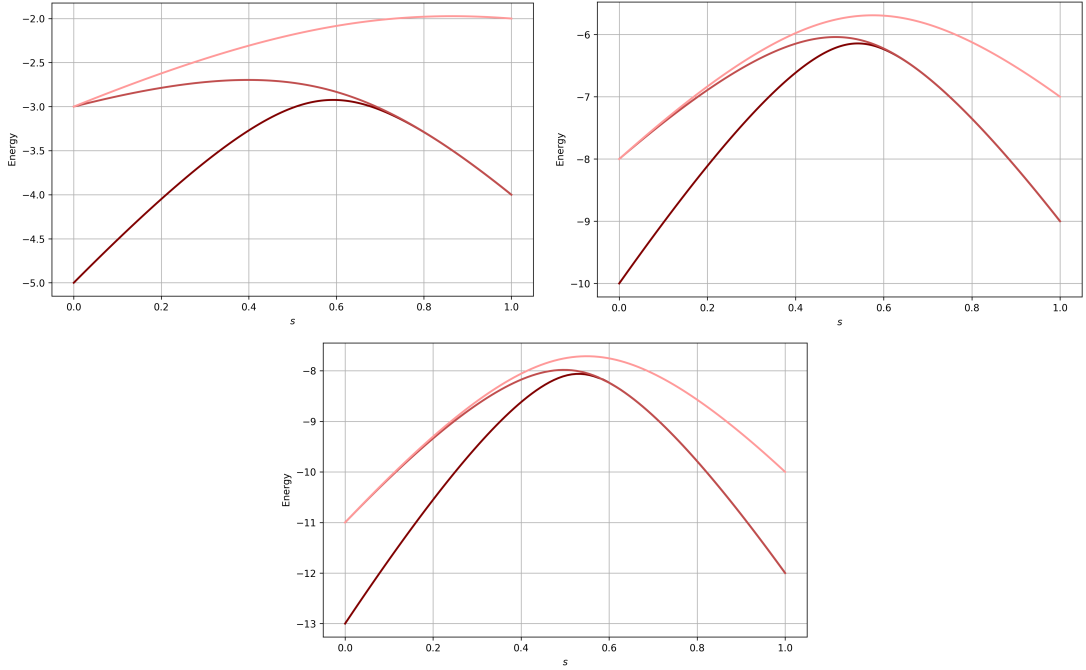


Figure 4.5: In the thermodynamic limit, as $N \rightarrow \infty$, the gap between the ground state and the first excited state reduces to 0 at $s = 0.5$. The figures from left to right have $N = 5, 10, 13$.

At $s = 1$, the Hamiltonian is dominated by the Ising interaction term $-\sum_i \sigma_i^z \sigma_{i+1}^z$, bringing the system into a ferromagnetic phase where spins align along the z -direction. The ground state exhibits two-fold degeneracy, with state like $|00000\rangle$ and $|11111\rangle$ both having the same energy $E_0 = -(n-1) = -4$ because of the Z_2 symmetry in the σ_z term.

Noticably, the figures show that when varying s from 0 to 1, the system undergoes a quantum phase transition at a critical point $s_c \approx 0.5$ and (exactly $s = 0.5$ in the infinite system, approaching the thermodynamic limit). As observed, near this critical point, the energy gap between the ground state and the first excited state begin to shrink significantly, and eventually share the same energy when s moves beyond 0.5.

More specifically, the alignment observe in the plot refers to the ground state energy (E_0) and the first excited state energy (E_1) first has distinct energy level as the spins are dominated by the transverse field term, as soon as s reaches its criticality, the gap between the energies shrink $\Delta E = E_1 - E_0$, because of the competition between the Ising interactions and the transverse field. For a small system size such as $n = 5, 10, 15$, the gap does not close completely, as it should in the thermodynamic limit, but minimized with increasing number of spins in the system. For finite $n = 5$, the gap remains small but non-zero near $s \approx 0.5$.

For the TWA, the issue with overlapping energy levels is addressed by performing an adiabatic sweep. Specifically, J is turned on slowly so that the system has time to adiabatically adjust, gradually rotating the spin directions and still remains in its

instantaneous ground state. So, slowly ramping J from 0 ensures the system evolve from the paramagnetic ground state to the ferromagnetic ground state without overlapping with any excitations. Ultimately, after the adiabatic sweep, order parameter m_z can now be obtained.

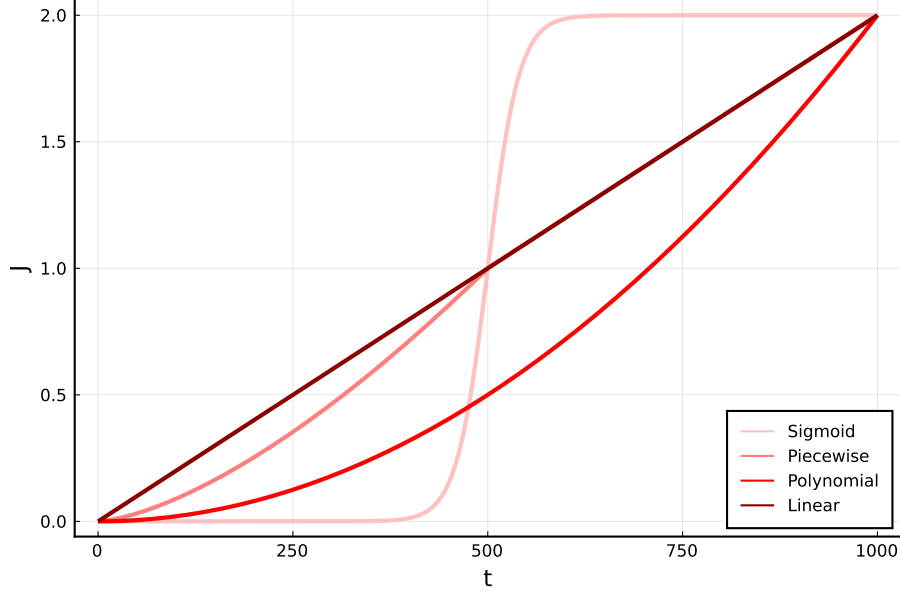


Figure 4.6: Multiple sweep rates are tested, however, linear and sigmoid are the most prominent candidate, with linear being $J_{max}t/t_{final}$ and sigmoid being $J_{max}/(1 + e^{-k(t-t_0)})$, where $k = 0.002$.

For the sweep rate options. A variety of methods are implemented, yet, the best seems to be sigmoid sweep rate from Fig. 4.6. Furthermore, because the results for the magnetization order parameter using TWA shows inaccuracies, it's necessary to outline a rough explanation for the numerical methods implemented.

The initial spin states uses the `sampleSpinXPlus` function, which turns the spins toward the $+x$ -direction. The function start with $\theta = \pi/2$, the Hadamard basis of the Bloch sphere, and introduces a small perturbation drawn from a normal distribution to help the ϕ adding stochasticity to the initial conditions, which is how the initial condition should be from random sampling. This approach creates a set of initial spin states that are slightly scattered around the $+x$ -axis, ensuring uncertainty in the initialization process.

Then, `prob_func` function initializes the state vector for each trajectory in the ensemble, allocating each spin a vector u_0 of length $2 \cdot nAtoms$ to store θ and ϕ for each of the spins. For every spin, θ_n and ϕ_n are sampled using `sampleSpinXPlus`, and these values are assigned to the elements of u_0 .

The adiabatic sweep simulates the time evolution of the spin system as the Hamiltonian transitions from a transverse-field-dominated regime to an Ising-interaction-dominated regime. This is managed by the `run_adiabatic_DTWA` function, which employs the

TWA method to model the dynamics.

The simulation parameters are, the number of spins, for example, ($nAtoms = 10$), the transverse field strength, setting to default and static ($h = 1.0$), the maximum Ising coupling ($J_{\max} = 10.0$), the total evolution time ($t_f = 1000$), the number of time steps ($nT = 400$), and the number of trajectories ($nTraj = 1000$). The coupling $J(t)$ is defined to increase using sigmoid sweep rate from 0 to $J_{\max}$, effectively driving the system slowly enough to achieve equilibrium phase transition between the two phases. As mentioned, this gradual sweep keeps the system in the instantaneous ground state, avoid excitation level hopping.

The TWA method evolves the spins classically, with the `drift!` function governing the deterministic changes in θ and ϕ based on the transverse field h and the time-dependent $J(t)$. Spins interact with their nearest neighbors (determined by `get_neighbors_1d` or could be higher dimensions, which uses periodic boundary conditions), and the evolution incorporates these interactions. The `diffusion!` function, set to zero, since the system isn't coupled to any open system that involve dissipation and no additional stochastic noise.

In summary, the simulation runs $nTraj$ independent trajectories, solved with the standard Euler Maruyama (EM) method. Each trajectory begins from a unique initial condition generated by `prob_func`, and the system evolves over the time span $(0, t_f)$, with solutions saved at specified time points. The adiabatic sweep increase J at a specific rate as t evolve.

Finally, the `compute_longrange_mz` function computes the long-range correlations between the spin at position i and $i + L/2$, by averaging the product of S_z values ($\cos(\theta)$) across all trajectories at each time step, ultimately, calculate the order parameter, magnetization m_z , for the TFIM.

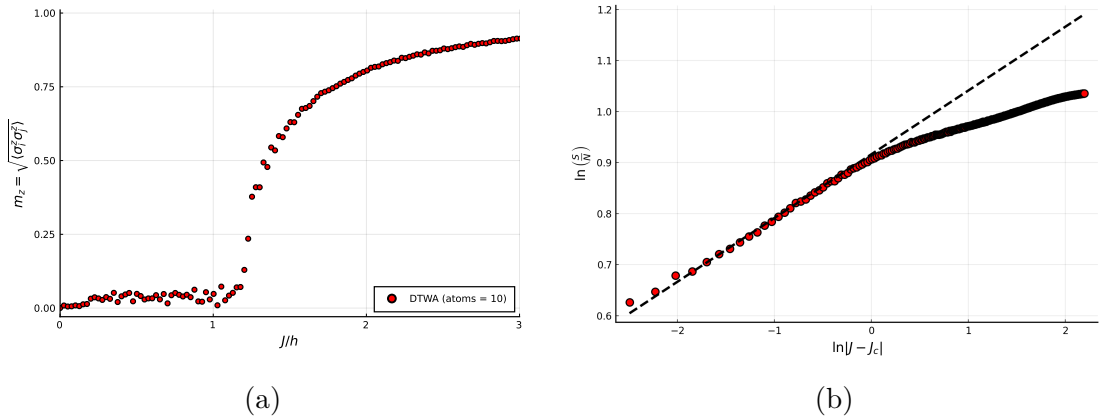


Figure 4.7: (a) Quantum phase transition of the transverse-field Ising model using TWA. (b) Logarithmic plot demonstrating scaling behavior near the critical point.

From calculating the long range correlation between the spins, the magnetization order parameter m_z shows a transition between the paramagnetic and ferromagnetic phase.

However, numerical error present in the critical point itself, at the thermodynamic limit, the magnetization strength and the coupling should be equal to each other, TWA fails to produce the correct critical point for the phase transition, in this case, the phase transition should happen at $h = J = 1$, in this semi-classical method, the critical point found is at $J \approx 1.3$. Though this critical point is not unreasonably far from the true critical point, some concern of why such behavior occur. Because of this inaccuracy, the critical exponent β is not correct. More specifically, if the critical point is places at 1.3, the scaling law actually show a $\beta \approx 1/8$. When the critical point is strictly placed $J = 1$, the critical point is off from that true value, in this case, the system is still in the paramagnetic phase and β shows no reliable result. Figure 4.7 illustrates the equilibrium phase diagram and the scaling behavior of observables near the quantum critical point.

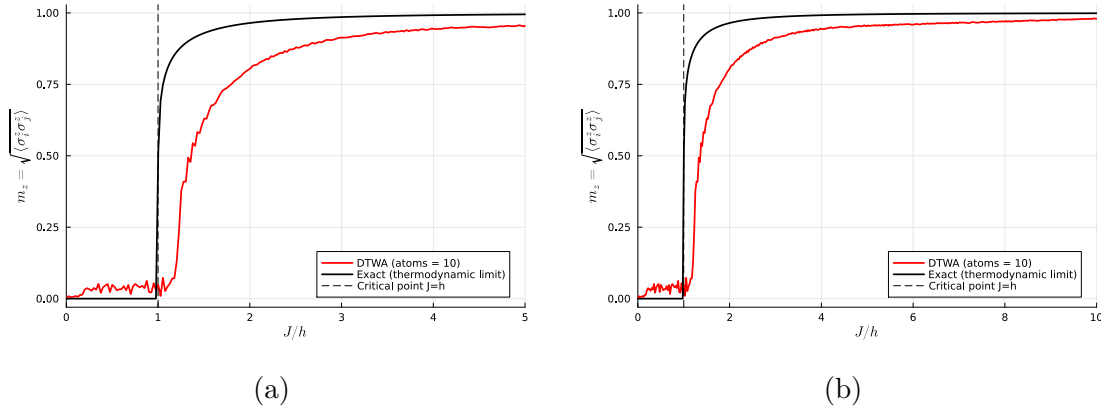


Figure 4.8: Magnetization order parameter approaching the upper limit of the thermodynamic limit (black line), showing that the equilibrium phase transition is captured, though not precisely at the correct critical point.

The simulation for the EPT yields several notable takeaways. Long-range correlations between spins at positions i and $i + N/2$ clearly indicate a phase transition between the paramagnetic and ferromagnetic phases. The shape of the transition closely resembles the true solution; as J increases, the order parameter m_z obtained from the TWA approaches the plateau expected in the thermodynamic limit. At the critical point of $J_c \approx 1.3$, the extracted critical exponent β attains a value that is reasonably close to the expected one with $\beta = 0.16$, even though the critical point itself is not located exactly where anticipated.

Despite these encouraging results, the method faces several challenges. One significant issue lies in the initialization of spins. The initial sampling is not ideal due to the non-positive-definite properties of the x -axis. To address this, the sampling for $\hat{\sigma}_i^x$ is clamped at the equator of the Bloch sphere, with added randomness to introduce stochasticity. In contrast, spin initialization along the z -axis is more straightforward, as the sampling revolves around the pole. Along the $+x$ -axis, however, initialization is restricted to bidirectional sampling within the xy -plane, which limits spin sampling for each trajectory.

Another issue arises from the truncation of the interaction term $\hat{\sigma}_i^z \hat{\sigma}_j^z$. From the mapping, only the first-order derivative term is retained. One could argue that with a larger number of spins in the system, this truncation is justified, since higher-order derivative terms scale as $1/N$ and vanish as $N \rightarrow \infty$. Yet, even in large collective spin ensembles, the critical point still deviates from the true value.

Additionally, this system is not coupled to an external environment, unlike in the cases of single driven atoms or Dicke superradiance, where randomness arises from dissipation and sampling. For future work, regarding spin initial conditions, one could start the simulation with all spins in the up or down state, where random sampling is straightforward, and then apply a $\pi/2$ pulse to rotate all spins to the $+x$ -direction before beginning the adiabatic sweep of J . Another approach is to initialize all spins in the up state and sweep J backward to zero, thereby preserving the positive-definite character of the z -direction.

Regarding the Hamiltonian, from the mapping one can see that $\hat{\sigma}_i^x$ does not require truncation of higher-order derivatives in phase space. For the interaction term, however, truncating at first order may cause the system to lose additional physical context. From the method itself, to translate the Lindblad equation From the method itself, to translate the Lindblad equation

$$\partial_t \hat{\rho}(t) = -i [H, \hat{\rho}(t)] + L_{\text{diss}} \hat{\rho}(t), \quad (4.60)$$

into an equation of motion for the Wigner function, one multiplies the kernel $\Delta(\Omega)$ with $\hat{\rho}(t)$ and takes the trace. However, this kernel is specifically suited for dissipative spin systems. Since the present system has no dissipation, one must question the validity of the constructed phase-space basis, which currently includes

$$\hat{\Delta}(\Omega), \quad \partial_\theta \hat{\Delta}(\Omega), \quad \partial_\phi \hat{\Delta}(\Omega), \quad \partial_\phi^2 \hat{\Delta}(\Omega). \quad (4.61)$$

This set of operators is not unique. While it is suitable for the two cases discussed above, another choice of operators might better capture the spin dynamics of the Ising chain with a higher degree of accuracy.

Now that some of the weaknesses of the TWA for spins have been addressed, it is worth considering whether these issues affect the NEPT case. For the non-equilibrium case, random sampling involves either all spins starting in the up state or, in the case of a seed particle, one spin starting in the up state while the others are initialized in the down state. Thus, spin initialization poses no particular challenges, and sampling is straightforward.

With regard to how the phase transition occurs, the EPT requires calculating the order parameter m_z , which introduces complications, as a suitable adiabatic sweep must be chosen to avoid level crossings between the ground state and the first excited state. In contrast, the phase transition in the NEPT is determined by taking the final value of the active-state density once the system reaches a steady state. Furthermore, as mentioned earlier, NEPT of DP is a dynamical process that can be described by the

stochastic differential equations with deterministic part for the system evolution and the noise part to mimic microscopic fluctuations. Thus NEPT fit with the descriptions of the TWA for spins that take advantage of the FPEs for describing absorbing-state phase transition.

Additionally, in the non-equilibrium case, the kernel works exactly as intended: the ensemble is coupled to an external Markovian reservoir, where dissipation is included. This allows the equation of motion to be formulated on the basis of the operators in Eq. 4.61. Previous work by Mink , *et al.* has shown that the TWA successfully describes the dynamics of laser-driven arrays of Rydberg atoms, giving validity to the method in this context [9].

5 Absorbing-State Phase Transition in the Low Dephasing Regime

In the following, the behavior of Rydberg facilitation of the absorbing state phase transition is studied when the system is in the coherent regime. In particular, the investigation focuses on whether the universal properties change compared to the purely classical regime of large dephasing.

5.1 Quantum Absorbing-state Phase Transitions for Rydberg Model

To determine the universality class in the quantum case, one needs to extract the critical exponents for three models with open boundary conditions: a 1D chain, a 2D lattice, and a 3D cube. For the 1D chain, the direction of percolation is bi-directional, meaning that each spin interacts with its nearest neighbors, except at the edges. In the 2D case, percolation occurs to the nearest neighbors to the right, with corner spins having two neighbors, edge spins having three, and all other spins having four. For the 3D cube with open boundary conditions, percolation occurs to the nearest neighbors, with corner spins having three neighbors, edge spins having four, face spins having five, and interior spins having six.

As the mapping is demonstrated in Sec. 3, for reiteration and convenience, the dynamics of the open quantum system are governed by the master equation:

$$\frac{d}{dt}\hat{\rho} = i[\hat{\rho}, \hat{H}] + \sum_l \mathcal{L}_l \hat{\rho} \mathcal{L}_l^\dagger - \frac{1}{2} \{ \mathcal{L}_l^\dagger \mathcal{L}_l, \hat{\rho} \}, \quad (5.1)$$

With the Hamiltonian,

$$\hat{H} = \sum_i \left[\Omega(\hat{\sigma}_i^{gr} + \hat{\sigma}_i^{rg}) + \left(\sum_{j \neq i} \frac{c_6}{r_{ij}^6} - \Delta \right) \hat{\sigma}_i^{rr} \right], \quad (5.2)$$

The corresponding SDEs in spherical coordinates (θ_i, ϕ_i) are:

$$\begin{aligned} d\theta_i &= \left[2\Omega \sin(\phi_i) + \Gamma \left(\cot \theta_i + \frac{\csc \theta_i}{\sqrt{3}} \right) \right] dt, \\ d\phi_i &= \left[2\Omega \cot \theta_i \cos(\phi_i) - V \sum_{j \in \text{neighbors}(i)} \left(1 + \sqrt{3} \cos \theta_j \right) + \Delta \right] dt \\ &\quad + \sqrt{\Gamma \left(1 + 2 \cot^2 \theta_i + \frac{2 \cot \theta_i \csc \theta_i}{\sqrt{3}} \right) + 4\gamma} dW_i(t), \end{aligned} \quad (5.3)$$

where $dW_i(t)$ is a Wiener increment representing stochastic noise, Γ and γ are dissipation rates related to dephasing and spontaneous decay, and V approximates the van der Waals interaction $\sum_{j \neq i} \frac{c_6}{r_{ij}^6}$.

The simulation begins by defining the input parameters. The system parameters include the number of atoms N (`nAtoms`), final time t_f (`tf`), number of time points N_t (`nT`), number of trajectories N_{traj} (`nTraj`), Rabi frequency Ω , detuning Δ , interaction strength V , coherent coupling Γ , and dephasing γ . Additionally, the lattice dimensionality is specified (1D, 2D, or 3D).

For each trajectory, the spin system is initialized with N spins. The initial spin orientations (θ, ϕ) for each atom corresponding to the initial state are sampled from a distribution biased toward the positive z -direction. Next, the neighbor configuration is established. Neighbor lists for each atom are computed or loaded based on the lattice dimensionality (1D chain, 2D square lattice, or 3D cubic lattice) with periodic boundary conditions.

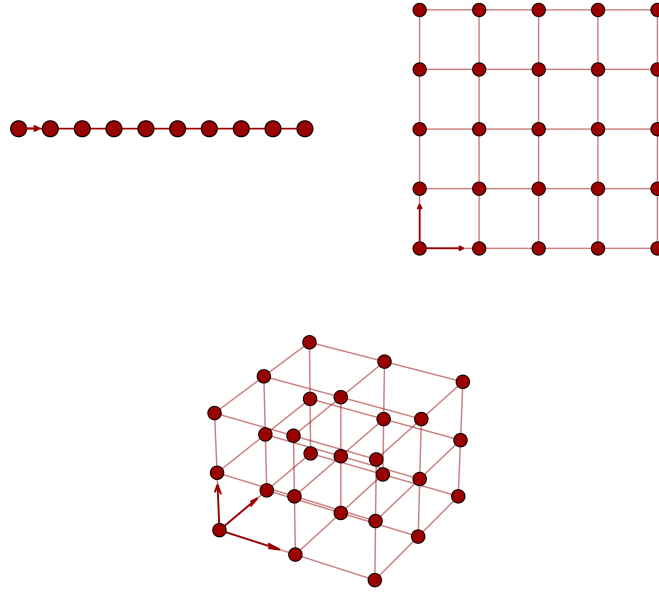


Figure 5.1: Visual representations of lattice structures with open boundary conditions: a 1D chain, a 2D square lattice, and a 3D cubic lattice. The shading highlights individual spins and their connectivity. In these examples, the 1D chain can extend to arbitrary length, while the 2D and 3D lattices are square and cubic, respectively. In all cases, large number of spins, $N \geq 3000$ is used to minimize finite-size effects, enhance connectivity and correlations, improve statistical sampling, and enhance the impact of quantum fluctuations.

The system dynamics are then defined. The SDEs are set up with drift and diffusion terms. The drift term incorporates Rabi oscillations (Ω), detuning (Δ), interactions

(V), and coherent coupling (Γ). The diffusion term accounts for dephasing (γ) and stochastic fluctuations. The SDEs are solved numerically (e.g., using the SRIW1 integrator) over the time span $[0, t_f]$ with N_t specified time points.

The ensemble simulation is performed by running N_{traj} independent trajectories in parallel, each with a unique initial spin configuration. For each trajectory, the time evolution of spin components is computed, and the z -component observable $\langle S_z \rangle$ is saved at the specified time points.

Data processing then follows. The ensemble average of $\langle S_z \rangle$ across all trajectories is computed at each time point. The mean $\langle S_z \rangle$ and the corresponding time points are saved to a file for further analysis.

For a single-spin excitation in a 2D lattice Fig. 5.2, an active seed is placed at the edge of the lattice. The system consists of 1089 Rydberg atoms arranged in a square lattice. The excitation spreads uniformly to nearest neighbors, in contrast to Monte Carlo simulations with random sampling, while the TWA model predicts consistent, uniform propagation. TWA approximates the dynamics of each spin semi-classically while preserving the average effects of interactions. Increasing the number of atoms improves the ensemble averaging, making the dynamics more accurate. In these simulations, strongly entangled states or long-range multi-spin correlations may not be fully captured. Additionally, as a semi-classical approximation, TWA treats quantum operators as classical variables with added quantum noise, which can underestimate quantum fluctuations.

The placement of the seed, whether at the edge or in the bulk, does affect the spreading. In the edge case, the seed initially affects three of its nearest neighbors, whereas when the active seed is placed at the center, the spreading is omnidirectional. However, the rate at which the spreading occurs in both cases appears to be comparable. The variable that largely influences the spreading in most cases is the driving strength of the external field. In this simulation, the driving is set within a reasonable range so that the spreading can still be observed. Lower driving suppresses the spreading more quickly; as mentioned, when the driving is below the critical point, the system cannot transition from the absorbing state to the active state. One also notices that the spreading process is not random, as the entire system is coherently driven by a single Ω , directed bond percolation has been defined as a Markov process, notably, in the spreading process, the parameter to determine the likelihood of spreading is the probability of opening and closing of valves. However, for this Rydberg lattice, no such probability is specified, as the system's dynamics is from physical properties and as observed, the spreading occurs in a uniform manner, in comparison to what has been theorized before.

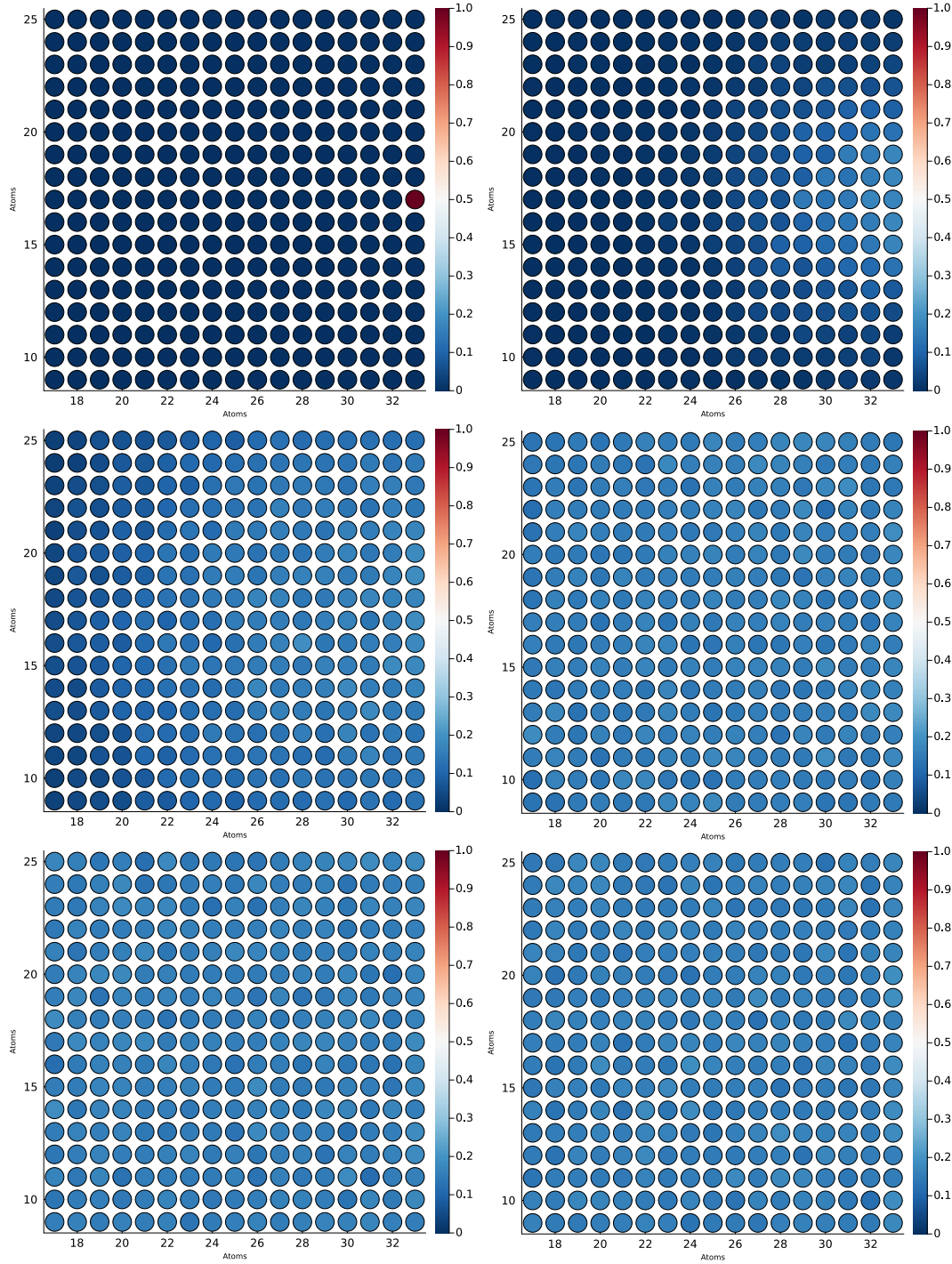


Figure 5.2: Selected images showing the spreading from an active seed in a square lattice of 1089 atoms, at times $t = 1, 4, 8, 12, 16, 20 \Gamma^{-1}$, with parameters $\Omega = 20\Gamma$, $\gamma = 0.1$, and $\Gamma = 1$. The spreading remains in the active state due to the driving field being much higher than the critical point.

5.2 2D Lattice

Multiple simulations with different system sizes show the critical point of the driving converging backward. With a larger number of spins present in the system, the critical point is found to be close to $\Omega_c \approx 11.2\Gamma$ within a 1% margin. Since the steady-state solution depends on the number of spins in the ensemble, larger systems require fewer trajectories to achieve low-error results. Therefore, it is reasonable to avoid using systems with fewer than 3×10^3 spins when determining the critical exponents. Smaller systems often require a high number of trajectories and, although this does not necessarily influence the outcomes, these systems tend to contain significant noise from the Wiener process, which is not well mitigated after averaging over the steady state of individual spins.

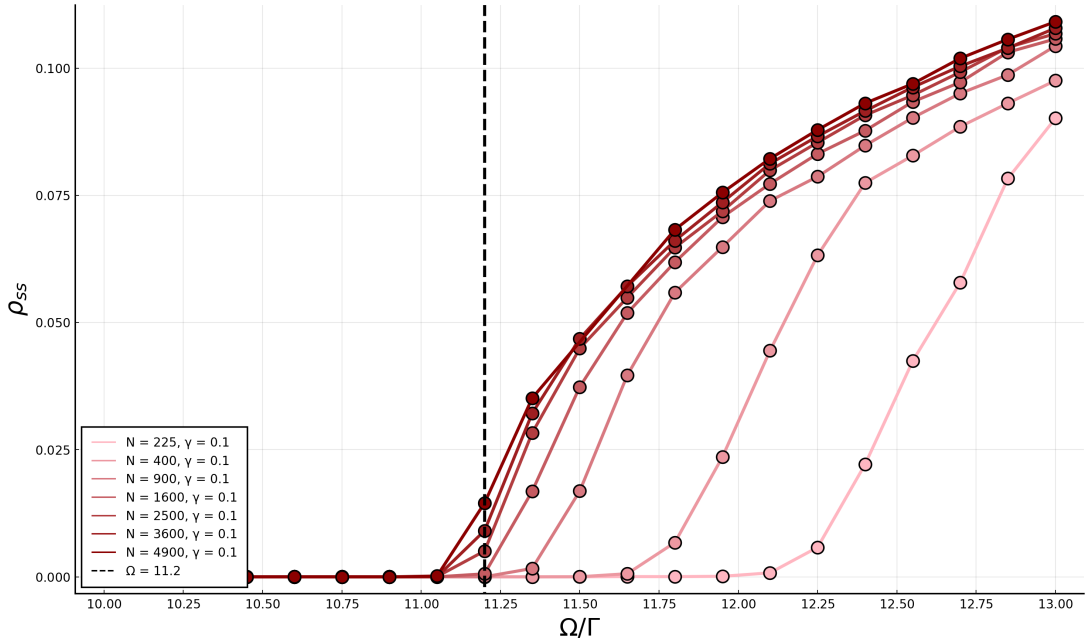


Figure 5.3: The phase transition from the absorbing state to the active state varies with different system sizes. Notably, the critical point that the transition shifts to the left with increasing number of spins. The higher the value of spins the sooner the facilitation happens. As the driving happens universally and simultaneously across all spins, the spreading occurs at a faster rate with smaller driving strength.

More importantly, selecting the correct solver is crucial for running the simulations. For the solver, SRIW1 is a great candidate, and some caveats of this method should be addressed. It is important to set both the absolute error and relative error to around 1×10^{-3} , which helps reduce the noise originating from the stochastic term. Moreover, the maximum number of iterations for the solver must be set above 5×10^9 to prevent the solver from returning incorrect results before proceeding to the next time step. Since SRIW1 is an adaptive solver, it can determine its own time step depending on how well the numerical results converge at each step. However, to avoid excessive computational time, the maximum time step should be restricted to about 1×10^{-4} ,

which allows the solver to converge efficiently without stalling.

Several numerical errors were observed when testing different solver parameter settings, most notably in the steady-state solution. For instance, when using a weaker solver such as EM, the computational time was approximately five times faster than with SRIW1. However, when the driving strength was close to zero, EM proved inefficient, as it produced steady-state values outside the physical range. Specifically, the physical range of the spins in the ensemble should lie between -1 and 1 , but with EM the solution dipped below -1 , an issue that SRIW1 resolves.

While both EM and SRIW1 are numerical solvers for systems with deterministic drift and random noise, such as those encountered in non-equilibrium phase transitions, they differ in design, accuracy, and applicability. EM discretizes the SDE using a first-order approximation, achieving a strong convergence order of 0.5 and a weak convergence order of 1.0 . It is computationally inexpensive but suffers from low accuracy. As a consequence, phase transition results obtained with EM often become unphysical, even when using a large number of trajectories. This issue is addressed by employing the SRIW1 solver, a higher-order adaptive Runge-Kutta-type method specifically designed for Itô SDEs. Derived from stochastic Taylor expansions, SRIW1 incorporates multiple stages to better approximate both drift and diffusion integrals, achieving a strong convergence order of 1.5 and a weak order of 2.0 . Unlike the fixed-step EM, SRIW1 is adaptive, automatically adjusting the time step Δt based on local error estimates to maintain a specified tolerance, thereby improving both efficiency and accuracy.

Nevertheless, even with SRIW1 another issue arises if the maximum number of iterations is set below the recommended threshold. For short simulation times, the solver works well. However, when the simulation time extends beyond $1 \times 10^4 \Gamma$, the steady-state solution for all driving strengths abruptly jumps to 1 , breaking the physical consistency. Setting the correct maximum number of iterations resolves this issue effectively.

The numerical procedure in this work is similar to that used in the equilibrium phase transition case, with a few caveats. First, the spins are initialized in the up state, where the solution is positive definite. Second, the solver differs: while EPT uses EM, the present model requires SRIW1. The Ising chain itself has no noise term and is not coupled to an external system. In contrast, this model includes dephasing γ and decay Γ . Here, $\Gamma = 1$ is used as a default value and as the standard unit for other system parameters, while the dephasing rate γ is varied to study the phase transition. There is also a detuning term, $\Delta = 2000\Gamma$, introduced to avoid accidental excitation of neighboring spins. This detuning is applied within the Monte Carlo method, where facilitation is randomly selected, similar to a “wheel of fortune” process. However, in TWA, such a high detuning may not be necessary. Once Δ is sufficiently large, it dominates the driving process of the spin Itô SDEs. Thus, Δ should instead be set high enough to satisfy $|\Delta| = V \gg \gamma$. If one chooses to use the EM solver for reduced computational cost, a smaller detuning value, $\Delta \approx 40$, is required to prevent the steady-state solution from becoming unphysical, as discussed above.

To reiterate, the solver specifications used in this work are: saveat = tSave, trajectories =

nTraj (note that the number of trajectories can be reduced for higher system sizes, yet still obtain high degree of accuracy because of taking the mean of the steady state for each spin in the ensemble), maxiters = 5×10^9 , abstol = 1×10^{-3} , reltol = 1×10^{-3} , and dtmax = 0.0001. These settings ensure robust simulation of the phase transition: saveat specifies the time points for data collection, trajectories sets the number of stochastic paths (typically 200 for reasonable accuracy), maxiters provides a sufficiently high iteration cap to handle long simulations, abstol and reltol control the absolute and relative error tolerances, and dtmax limits the maximum time step to ensure stability and precision.

In the case of finite-size systems, the number of spins is relatively large but remains far from the thermodynamic limit as $N \rightarrow \infty$. Simulations show that rather large system sizes are needed to achieve convergence. This means that results about critical behavior obtained from small lattices, e.g., by exact diagonalization, may not be reliable. However, TWA faces little to no challenges except for computational time as the number of spins increases. Additionally, one can observe that the convergence is not yet definite with 10^3 spins; it is recommended to increase the number of spins to approximately 5×10^4 to improve the results. Throughout the analysis, the model is consistently treated as a square lattice, and alternative system structures, such as extending the lattice at one end or introducing other geometric modifications, have not yet been investigated. These unexplored configurations could potentially influence the critical exponents and scaling behavior. Therefore, further detailed and comprehensive investigations into these aspects are recommended for future work to deepen our understanding of the system's dynamics under diverse lattice geometries and larger system sizes.

The phase transition is extracted by evaluating the final value of the active density in the steady state. With a simulation time of $t = 10^3\Gamma$, the system is allowed to equilibrate sufficiently to capture the long-term behavior. This time scale is chosen based on the relaxation dynamics of the 2D lattice, where interactions between neighboring atoms stabilize the active phase. The active density is then analyzed as a function of the driving strength Ω , enabling identification of the critical point at which the transition from the absorbing to the active phase occurs. This allows us to observe the system's critical behavior and validate the finite-size scaling results. Figure 5.6 shows the transition: the change is sharp, with the active state steadily increasing. TWA demonstrates its capability to simulate complex transitions even in the presence of small perturbations.

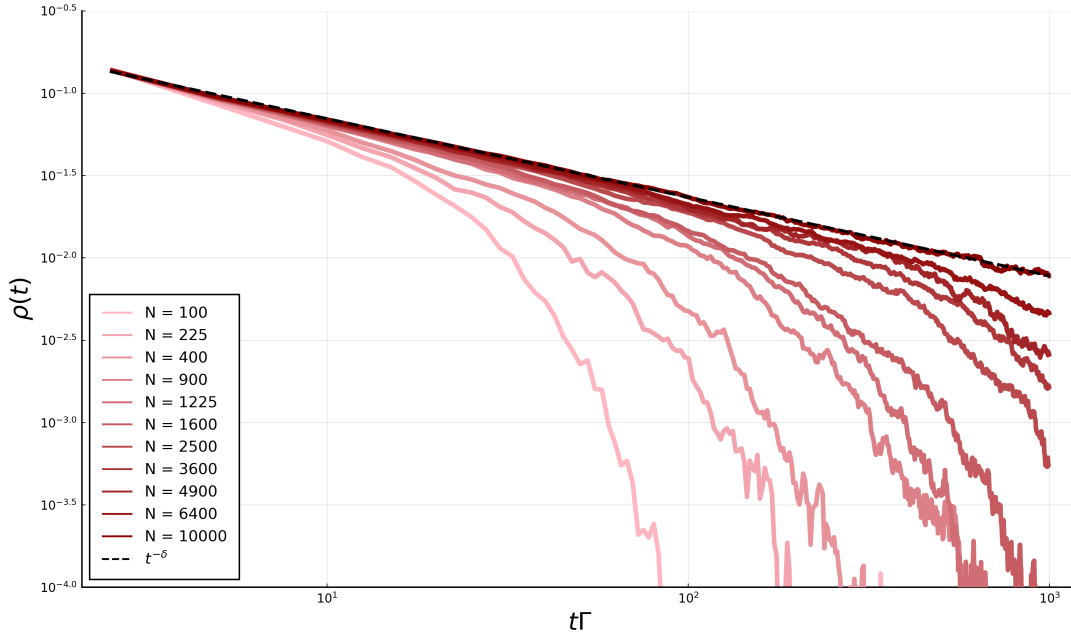


Figure 5.4: Finite size expansion of the critical exponent δ for a 2D square lattice with open boundary conditions. The number of atoms ranges from 100 to 10,000, with a driving strength $\Omega_c \approx 11.2\Gamma$. Convergence is observed at $\delta \approx 0.477$ for 10,000 atoms.

With the numerical parameters set, critical exponents can be extracted from the system. Finite-size scaling is employed to determine critical exponents, such as δ , by analyzing how physical quantities (e.g., the order parameter or susceptibility in the Ising chain) vary with system size near the critical point. For a 1D chain, 2D lattice, or 3D cube with open boundary conditions, the finite-size scaling procedure involves using the same critical point across different system sizes and observing the convergence of δ . As the system size N increases, the critical points converge toward a limiting value, allowing accurate extrapolation to the thermodynamic limit, where δ reflects the true critical exponent governing the system's critical behavior.

Fig. 5.4 shows the values of Ω_c for different system sizes converging toward the decay exponent δ . One key observation is the reduction of noise in the results as the system size increases. Because the active-state density is determined by the ensemble average, the logarithmic curves at the critical point become smoother, making the determination of δ much easier. Notice that for an ensemble with a system size of 10^2 , the curves exhibit significant fluctuations, whereas the ensemble with a size of 10^3 produces much clearer results. This is reflected by the fact that the scaling law of the decay (black dashed line) lies almost entirely on top of the decay curve at the critical point for the 10×10^3 system.

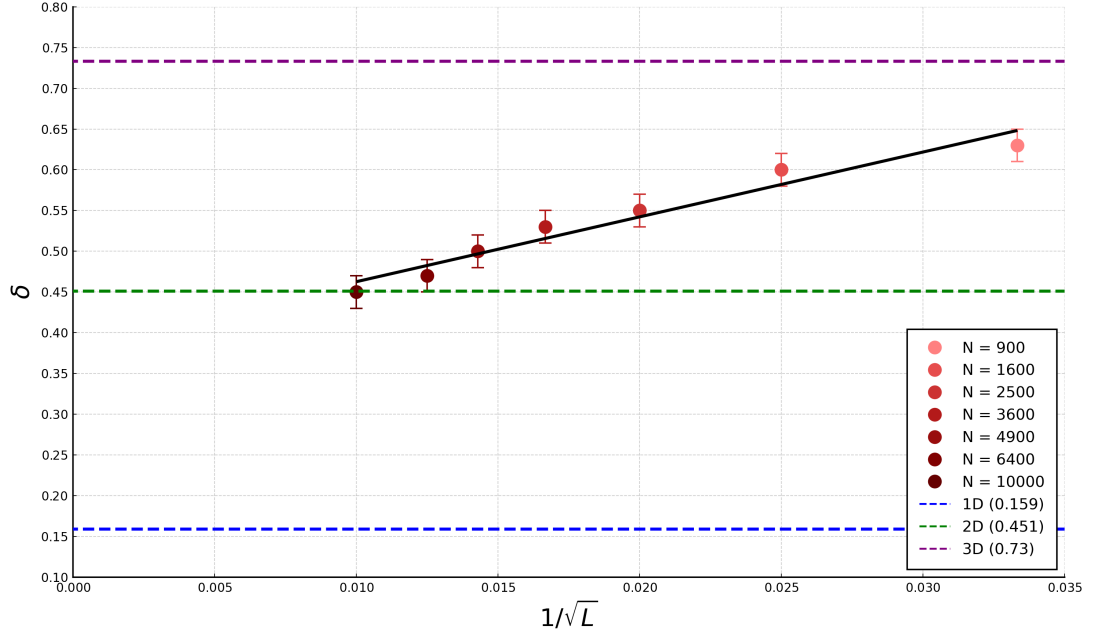


Figure 5.5: The critical exponent δ for each system size with the same driving parameter. The results show δ converging toward a value of 0.45 ± 0.002 . Note that the decay exponent limit for the 2D case is 0.451, since even with a large number of spins, such as 10^3 , the system remains far from the thermodynamic limit. This is indicated by the convergence of the critical point of the phase transition still shifting to the left Fig. 5.3. For this reason, in the presence of an even larger number of spins, the critical point would shift further. Here, the critical point is found to be $\Omega_c \approx 11.2$. Because of this, δ is shown to be close to the DP decay exponent. With an increased number of spins, the δ values for smaller system sizes also increase, in other words, because of the shift in critical point, δ s move upward along the fit line, and the fit line itself shifts to the left until it reaches $1/\sqrt{L} = 0$, causing the decay exponent in the thermodynamic limit to approach approximately 0.451 but cannot cross it.

To further verify the reliability of the results, the values of δ obtained for different system sizes were plotted against their inverse. Figure 5.5 shows that as the number of spins increases, the values of δ converge toward $\delta \approx 0.45$, with larger systems providing higher accuracy in estimating this critical exponent. Simulations involving large numbers of spins, however, present a significant challenge, as they demand enormous computational resources and long simulation times. Despite these difficulties, systems with up to 10^4 spins arranged on a square lattice are sufficient to demonstrate that the convergence of δ is both robust and reliable, giving confidence in the scaling analysis.

Nevertheless, when system sizes approach the thermodynamic limit, $N \rightarrow \infty$, the situation becomes more subtle. In this limit, the critical value for the phase transition can shift further below $\Omega_c \approx 11.2\Gamma$, leading δ to move below the apparent value of 0.4577. This finite-size drift of the critical point complicates the direct extraction of δ . However, the issue can still be addressed in a controlled manner by systematically shifting back the critical point to accommodate the increasing number of spins in the ensemble.

Numerical simulations confirm that, even under these adjustments, finite-size scaling continues to provide an agreeable and consistent estimate of $\delta \approx 0.45$.

Importantly, the thermodynamic limit of the phase transition can, in principle, still be reached through careful extrapolation. Yet, achieving a fully convincing and precise value requires simulations with system sizes well beyond 5×10^5 spins to make inverse square root system size well within the $1/\sqrt{L} \approx 0.001$ range, only at such scales does the critical point for each system size display clear and systematic convergence. Together, these findings emphasize both the power and the limitations of current computational methods: while large-scale simulations already capture the essential features of the transition, pushing further toward the true thermodynamic regime remains an open challenge that demands even greater computational resources.

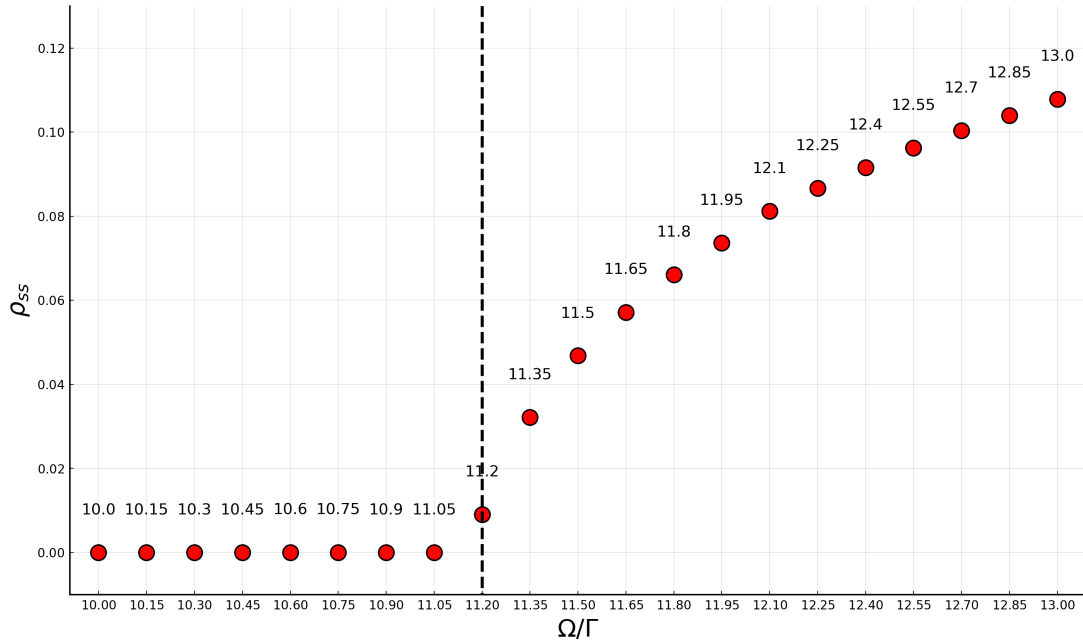


Figure 5.6: Phase transition diagram for a 2D lattice with a system size of 3600 atoms, illustrating the critical point around $\Omega_c \approx 11.2\Gamma$ with a margin of error of ± 0.02 . The transition from the absorbing state to the active phase is clearly evident, highlighting the facilitation mechanism across neighboring atoms. This behavior underscores the cooperative dynamics and interaction strengths that drive the phase transition, providing insight into the system’s critical phenomena.

To determine the critical exponent β from the phase transition plot, the process begins by taking the natural logarithm of the steady-state active density ρ (i.e., $\ln(\rho)$) and the difference between the driving strength Ω and the critical point Ω_c (i.e., $\ln(\Omega - \Omega_c)$). Using the finite size extrapolation data from the previous analysis, the critical driving strength Ω_c is extracted, typically identified as the point where the phase transition occurs (e.g., around $\Omega_c \approx 11.2\Gamma$ for a 3600-atom system). The relationship between $\ln(\rho)$ and $\ln(\Omega - \Omega_c)$ is then fitted to a linear model, where the slope of the line corresponds to $-\beta$. This fitting process leverages the power-law behavior $\rho \sim (\Omega - \Omega_c)^\beta$ near the

critical point, allowing β to be determined, giving the value of $\beta \approx 0.586$ for the given system size.

The accuracy of the method for extracting the critical exponent β may vary, as marginal differences in small values of Ω can lead to different values for the exponent. These variations are likely due to the difference in system sizes. The estimated error for this method is approximately in the range of ± 0.002 , determined through repeated simulations across the 2D models. This error margin suggests that while the method provides a reasonable estimate, further refinement or larger sample sizes may be necessary to enhance precision, especially for critical phenomena near phase transitions.

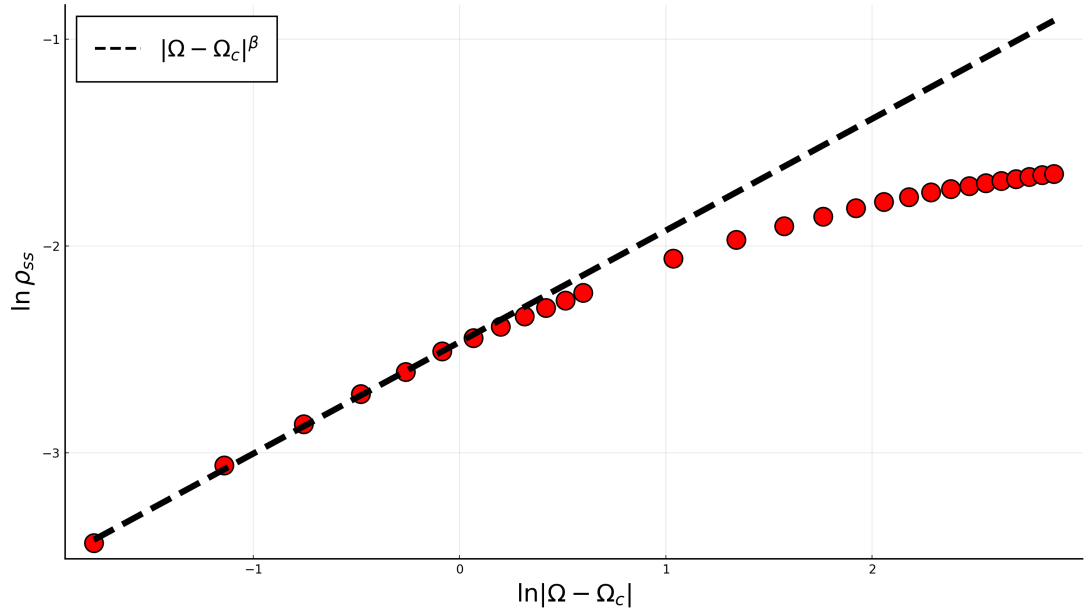


Figure 5.7: Power-law behavior illustrating the critical exponent $\beta = 0.586$ for a 2D lattice system with a size of 3600 atoms. The critical point is extracted through finite size scaling analysis, and this method is used to derive the value of β .

As a further check to understand the signatures of the transition, plotting the dynamics on a logarithmic scale reveals qualitative changes in the curves. Figure 5.8 illustrates that curves below the critical point decay to zero, while those above the critical point remain in the active state. The observed algebraic decay aligns with the scaling law of δ determined previously.

Traditional methods, such as those mentioned above, appear to serve well in extracting the statistical properties of the exponents. However, these methods do exhibit disagreements among themselves. For instance, the values of the critical Ω obtained from finite-size extractions sometimes do not agree with the critical point derived from the scaling power law. The component β is particularly questionable because, first, the steady-state points for different driving strengths must reach equilibrium after a long evolution time, and second, there is no such thing as a stable steady state, as all states decay during simulations with long running times. As a result, margins of error always

exist for critical exponents. Further checks could also be imposed, such as avalanches, as extensively demonstrated by Ohler, *et al.* [67]. Unlike the case for equilibrium phase transitions, the analytical approach reveals a trivial mapping between the quantum and classical cases. However, no analytical methods have been carried out to prove the validity of the TWA method itself. Still, reasonable results from β and δ show a one-to-one mapping to the classical case, unlike the equilibrium case of d and $d + 1$. These extractions could provide insight into the mapping analytically. The question of why such a mapping exists remains beyond the scope of this work.

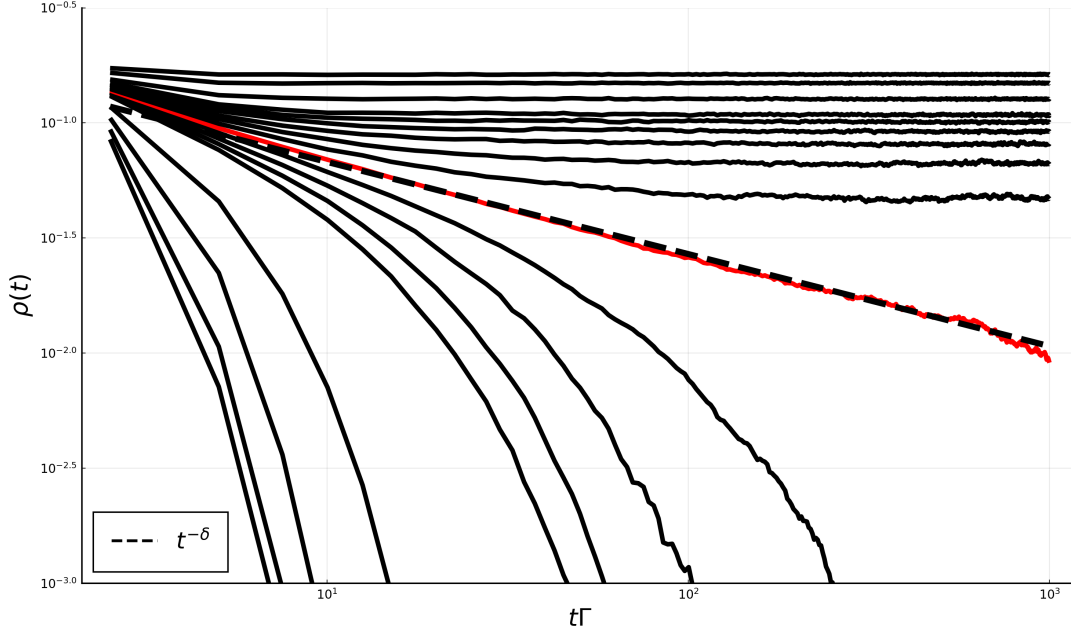


Figure 5.8: Logarithmic plot of the dynamics near the critical point, showing the transition from decay to active-state behavior. The algebraic decay corresponds to the scaling law of δ . Below the red curve, Ω increases up to the critical point of approximately 11.2Γ , indicating that everything below remains in the absorbing state. On the other hand, all curves above the critical point saturate and reach the active state.

After extracting β and δ , the collapse of the curves for different driving strengths reveals a clear decay to zero for curves below the critical point, and an exponential relaxation to a finite value with a characteristic bending downward and then upward. This behavior is plotted using $t^\delta \rho_{ss}$ as a function of $t|\Omega - \Omega_c|^{\beta/\delta}$. Figure 5.9 shows the shifting of the collapse as the driving strength is varied, approaching the critical point from below and moving out of criticality, with the critical point at $\Omega_c \approx 11.2\Gamma$.

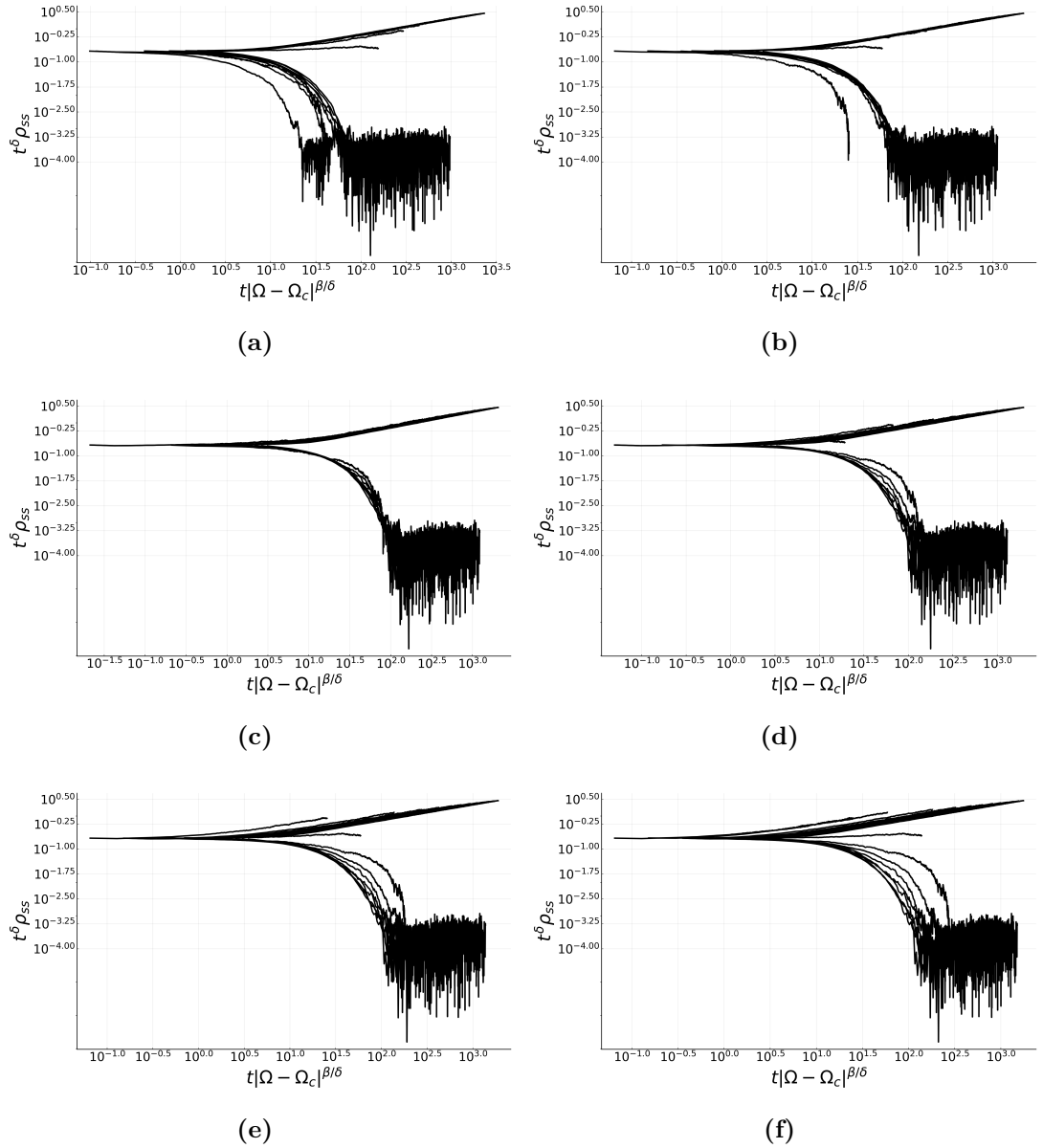


Figure 5.9: Selected collapse images from the 2D atomic simulation with 3600 spins and open boundary conditions. The subfigures show the data collapse of $t^\delta \rho_{ss}$ as a function of $t|\Omega - \Omega_c|^{\beta/\delta}$ for different driving strengths Ω : (a) $\Omega = 10.98\Gamma$, (b) $\Omega = 11.1\Gamma$, (c) $\Omega = 11.18\Gamma$, (d) $\Omega = 11.24\Gamma$, (e) $\Omega = 11.3\Gamma$, and (f) $\Omega = 11.4\Gamma$. The collapse illustrates the transition from decay to a finite steady-state density, highlighting the critical behavior in the 2D geometry.

Having determined the value of Ω_c , β and δ , a further estimation of another critical exponent is z , which is the ratio of the critical exponents for the spatial and temporal correlation lengths $\nu_\perp$ and $\nu_\parallel$, where $z = \nu_\parallel/\nu_\perp$ is called dynamic exponent. With this set of β , δ and z , a complete set of bulk exponents that labels the universality class. According to the equation

$$\rho(t) \simeq N^{-\delta} f(\Delta t^{1/\nu_\perp}, t^{d/z}/N), \quad (5.4)$$

$\rho(t)t^\delta$ is plotted against $t/N^{z/d}$ for various system sizes. By tuning z the data points collapse onto a single curve. Fig. 5.10 shows the transition of the curves from outside the critical region, approaching a value of $z = 1.86$, after which the curves align and lie on top of each other.

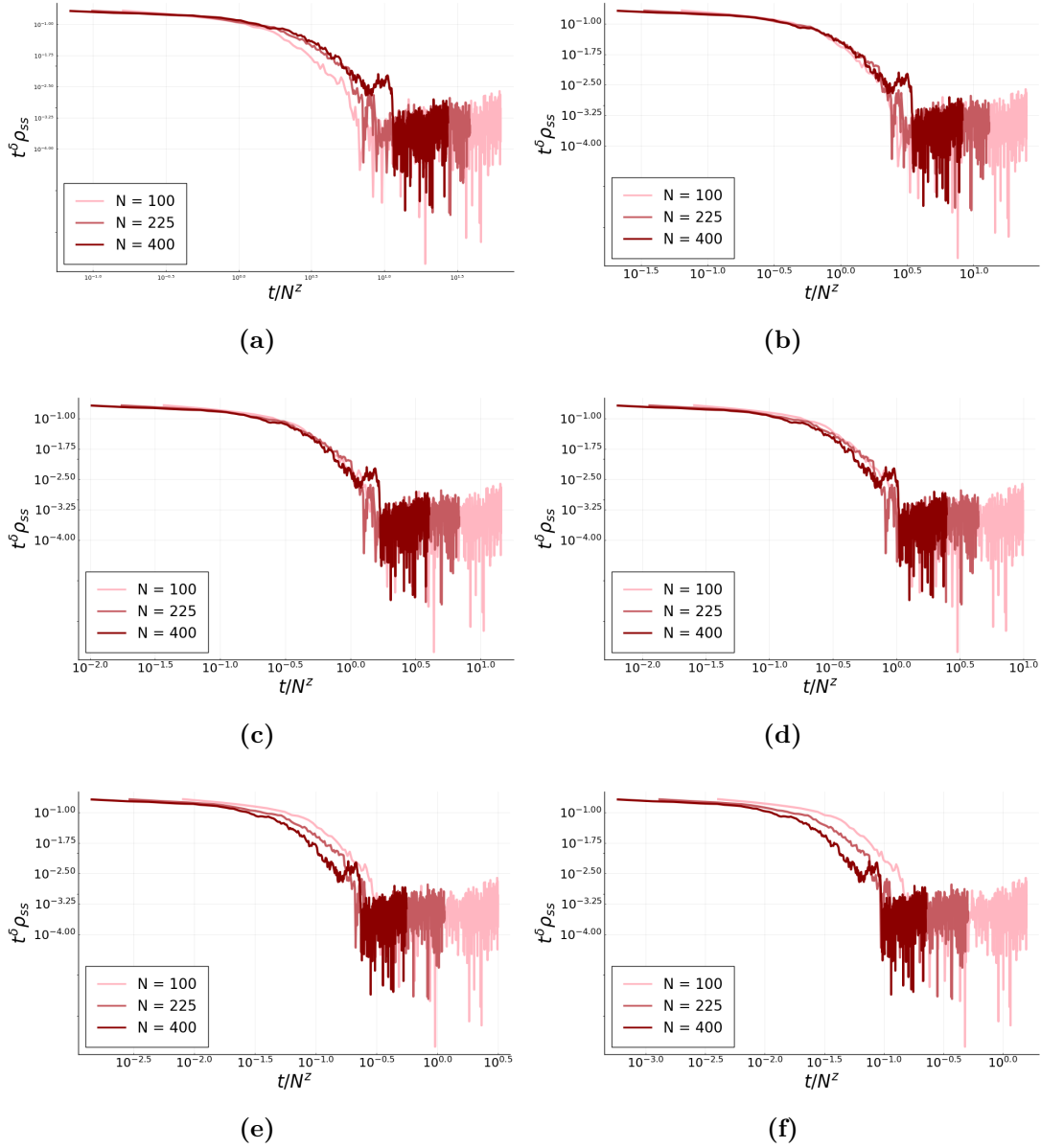


Figure 5.10: Transition of curves as a function of system size for different z values. Subfigures show $z = 1.20$ (a), $z = 1.60$ (b), $z = 1.84$ (c), $z = 2$ (d), $z = 2.5$ (e), and $z = 2.8$ (f). The curves converge to $z \approx 1.86$ near the critical point.

An interesting observation from extracting z is its behavior at high numbers of spins, where it takes a different value of $z \approx 2.2$ as shown in Fig. 5.11. For smaller systems,

previous literature reports $z = 1.86$. However, it appears that no current work has systematically explored the regime of large spin ensembles. Further investigation is therefore needed to understand why this shift occurs, especially since the values of β and δ remain consistent with those of the classical phase transition case. One key contributing factor is that, at the transition, the critical point $\Omega_c \approx 11.2\Gamma$ corresponds to the absorbing state for small system sizes, whereas for larger ensembles the system has already begun to transition into the active phase. As a result, the curves for small system sizes collapse onto each other because the absorbing state remains dominant, while in the case of larger systems, facilitation effects become apparent. This behavior ultimately alters how the system evolves: at high spin numbers the curves collapse when, in the small system case, they have already diverged. This distinction is important for future research, since small systems may yield values of the dynamic exponent in agreement with classical phase transition scenarios, while larger systems reveal qualitatively different behavior once the ensemble size increases significantly.

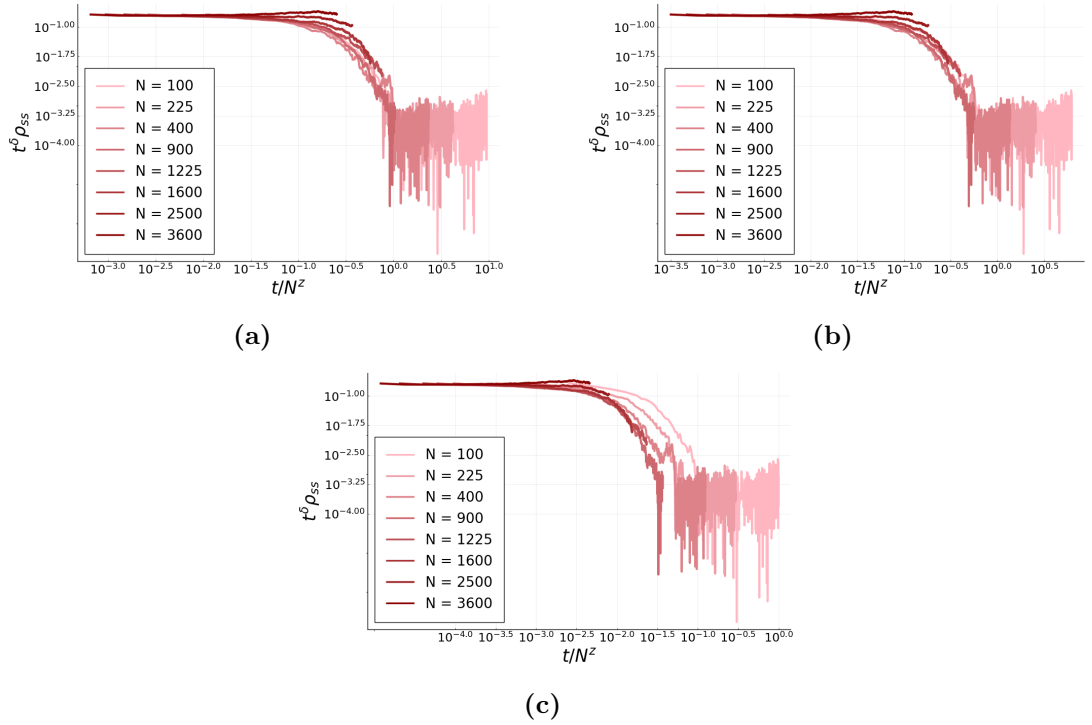


Figure 5.11: Transition of curves as a function of system size for larger z values. Subfigures show $z = 2.02$ (a), $z = 2.20$ (b), and $z = 3.00$ (c). The curves continue to converge toward $z \approx 1.86$ near the critical point.

Concluding for the 2D case, the critical exponents exhibit values similar to those of the classical case under conditions of low dephasing. This similarity suggests a potential mapping between the quantum and classical descriptions, particularly when dissipative effects vanish, as observed in the steady-state solutions and curve collapses (e.g., Figure 5.9). The extracted exponents β , δ , and z align reasonably well with classical percolation models, with the exception of the z value at large spin numbers. Deviations are attributed to finite-size effects and the first-order truncation in the TWA

method. Theoretical predictions highlight the influence of open system dynamics, including spontaneous decay and Doppler broadening, which may suppress quantum correlations. Despite these challenges, the method’s ability to capture the qualitative transition from absorbing to active phases underscores its utility.

However, the lack of analytical validation for TWA and the presence of error margins (± 0.002) suggest that further refinements are needed, such as increasing system size, adjusting dissipation rates, or incorporating higher-order interaction terms. Future studies could also explore the robustness of this mapping across different lattice dimensions or under varying initial conditions, to better understand the universality class of the non-equilibrium absorbing state phase transition.

Moreover, several open questions remain. An analytical mapping between the quantum and classical phase transitions, similar to that of the EPT, is still unknown. Another concern is whether the critical values of the extracted exponents genuinely lie in the quantum regime. Theoretically, at such low dephasing rates, the system is indeed expected to operate beyond the classical limit. Yet, whether this semi-classical method truly preserves the “quantumness” during the truncation of Ising interactions remains an open question worthy of investigation, especially since the results from EPT cast doubt on the validity of such an approach for this system.

Nevertheless, an encouraging conclusion can be drawn: a phase transition for the spin ensemble is clearly present in the results, and the extracted critical exponents yield reasonable values that are not inconsistent with classical benchmarks. From the results, the universality class remain the same when going from the classical case to the quantum case.

5.3 1D Chain

5.3.1 System Description

The 1D chain consists of 3000 spins with nearest-neighbor interactions and open boundary conditions. The two spins at the edges have only one neighbor, while all other spins have two. The interaction term $V \sum_{j \neq i} (1 + \sqrt{3} \cos \theta_j)$ involves fewer neighbors, and the corresponding dt term in the SDE requires a larger Ω to facilitate excitations between spins and their neighbors.

5.3.2 Simulation Setup

In terms of the numerical procedure, the same setup applies to the other parameters, in particular, dephasing $\gamma = 0.1\Gamma$, decay $\Gamma = 1$, detuning $\Delta = 2000\Gamma$, with a total simulation time of $t = 10^3$. All spins are initially set to the up state. The simulation uses the SRIW1 solver with the same solver parameters as for the 2D square lattice. This simulation time allows the solution to reach a reasonable steady state.

5.3.3 Phase Transition

The phase transition of the 1D chain occurs at a higher driving strength than in the 2-dimensional case. This is due to the reduced connectivity of the linear chain: each interior spin interacts with only two neighbors, while the edge spins have only one neighbor. Consequently, the critical point appears further from the value where the active state density ρ_{ss} becomes nonzero, compared to the 2-dimensional lattice. The critical point is determined at $\Omega_c \approx 29.85\Gamma$, with an error of 0.002.

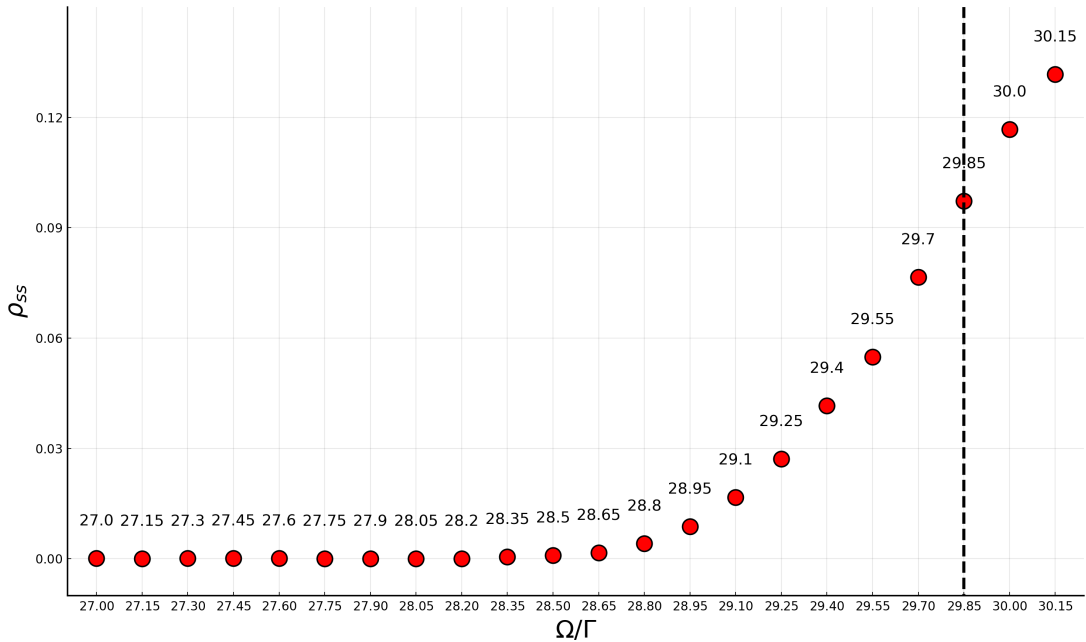


Figure 5.12: Phase transition curves for the 1D chain of 3000 spins, showing the shift from absorbing to active phases near $\Omega_c \approx 29.85\Gamma$.

5.3.4 Critical Exponents

With the critical point determined at $\Omega_c \approx 29.85\Gamma$, the scaling law fits well on the logarithmic phase transition plot. The scaling exponent $\beta = 0.277 \pm 0.002$ agrees with the classical 1D critical exponent in the high-dephasing regime. A slight shift in the critical point appears when fitting to match δ and β , as the relation $t^{-\beta}$ introduces a small bias toward the fitted values. Finite-size analysis further supports the reliability of this approach, with the steady-state density yielding $\delta \approx 0.159$.

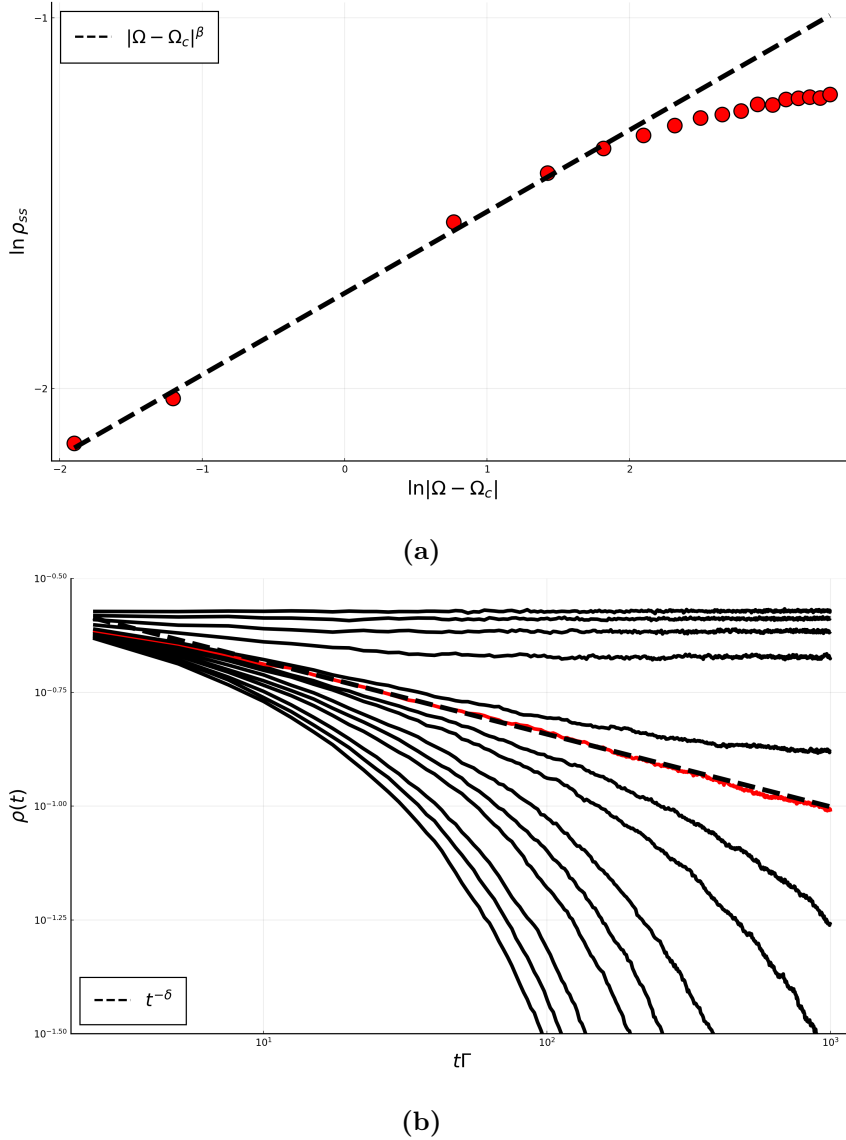


Figure 5.13: Power-law scaling of the 1D system: (a) order parameter yielding $\beta \approx 0.276$, (b) steady-state density yielding $\delta \approx 0.159$, with red curve being the critical point of $\Omega_c \approx 29.85\Gamma$, curves below it are in the absorbing state with $\Omega < \Omega_c$, while those above are in the active state, with $\Omega > \Omega_c$.

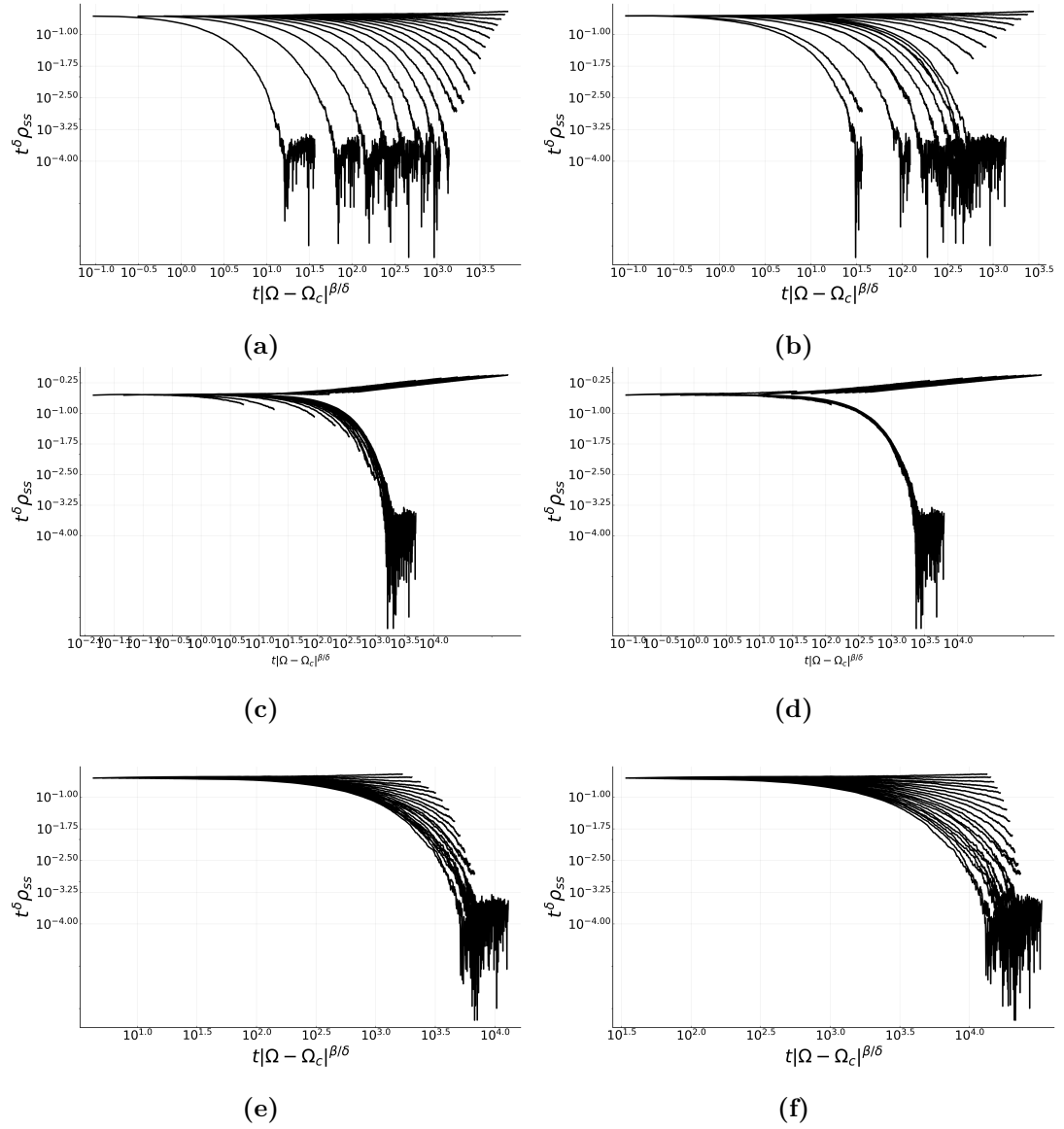


Figure 5.14: Selected collapse images from the 1D atomic simulation with 3000 spins and open boundary conditions. The subfigures show the data collapse of $t^\delta \rho_{ss}$ as a function of $t|\Omega - \Omega_c|^{\beta/\delta}$ for different driving strengths Ω : (a) $\Omega = 27.0\Gamma$, (b) $\Omega = 28.20\Gamma$, (c) $\Omega = 29.5\Gamma$, (d) $\Omega = 29.85\Gamma$, (e) $\Omega = 31.35\Gamma$, and (f) $\Omega = 34.45\Gamma$. The collapse illustrates the transition from decay to a finite steady-state density, with the critical behavior influenced by high dephasing and finite-size effects.

5.4 3D Cube

5.4.1 System Description

The 3D system consists of 3375 spins arranged in a $15 \times 15 \times 15$ cubic lattice with nearest-neighbor interactions and open boundary conditions. Corner spins have three neighbors, edge spins (excluding corners) have four, face spins (excluding edges and corners) have five, and interior spins have six. The interaction term $V \sum_{j \neq i} (1 + \sqrt{3} \cos \theta_j)$ captures this connectivity, with each interior spin interacting with six neighbors.

5.4.2 Simulation Setup

All spins are initially polarized in the up state. The simulation employs the SRIW1 method with a maximum time step of $dt_{\max} = 0.0001$ and a total duration of 10^3 time units, allowing the system to reach a steady state under a driving strength Ω . To ensure the dynamics remain in the quantum regime, a small dephasing rate of $\gamma = 0.0001$ is used, smaller than in previous cases.

5.4.3 Phase Transition

The cubic geometry, with a coordination number of six for interior spins, enhances connectivity and accelerates correlation spreading compared to the 1D lattice. Consequently, a lower driving strength Ω is sufficient to induce the transition between steady states. The critical point is estimated at $\Omega_c \approx 7.56\Gamma$.

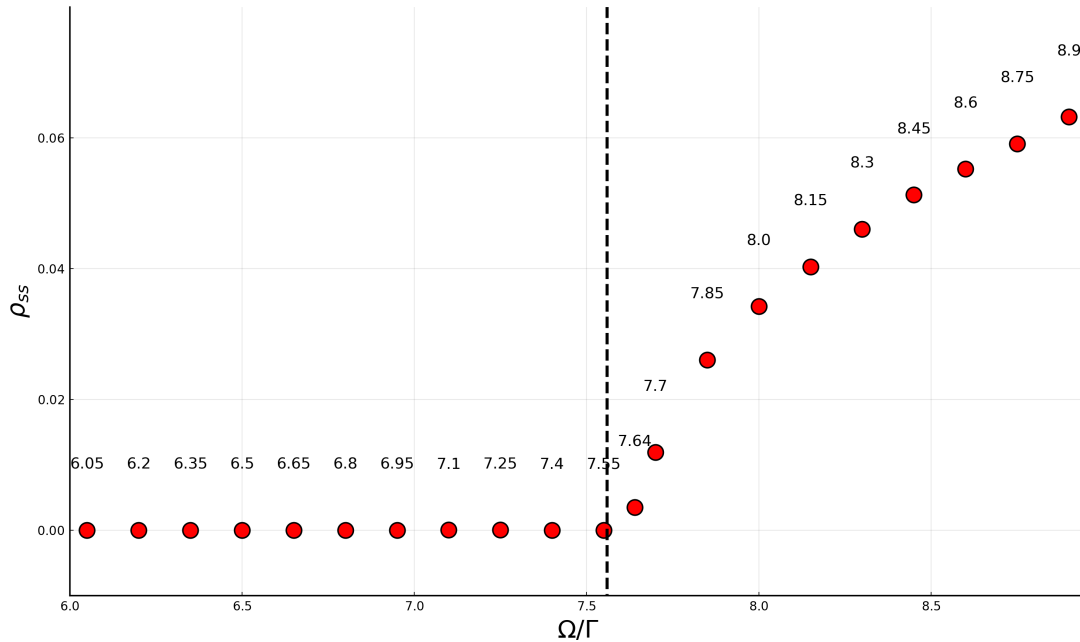


Figure 5.15: Phase transition curves for the 3D cube of 3375 spins, showing the shift from absorbing to active phases near $\Omega_c \approx 7.56\Gamma$.

5.4.4 Critical Exponents

The collapse of the curves, which depends on β and δ , yields results consistent with expectations. This collapse occurs around $\Omega \approx 7.56\Gamma$ and shifts outward from either side of the critical point. The order parameter yields $\beta \approx 0.812$, consistent with mean-field predictions for high dephasing. The steady-state density yields $\delta \approx 0.733$, reflecting the 3D critical behavior under high dephasing.

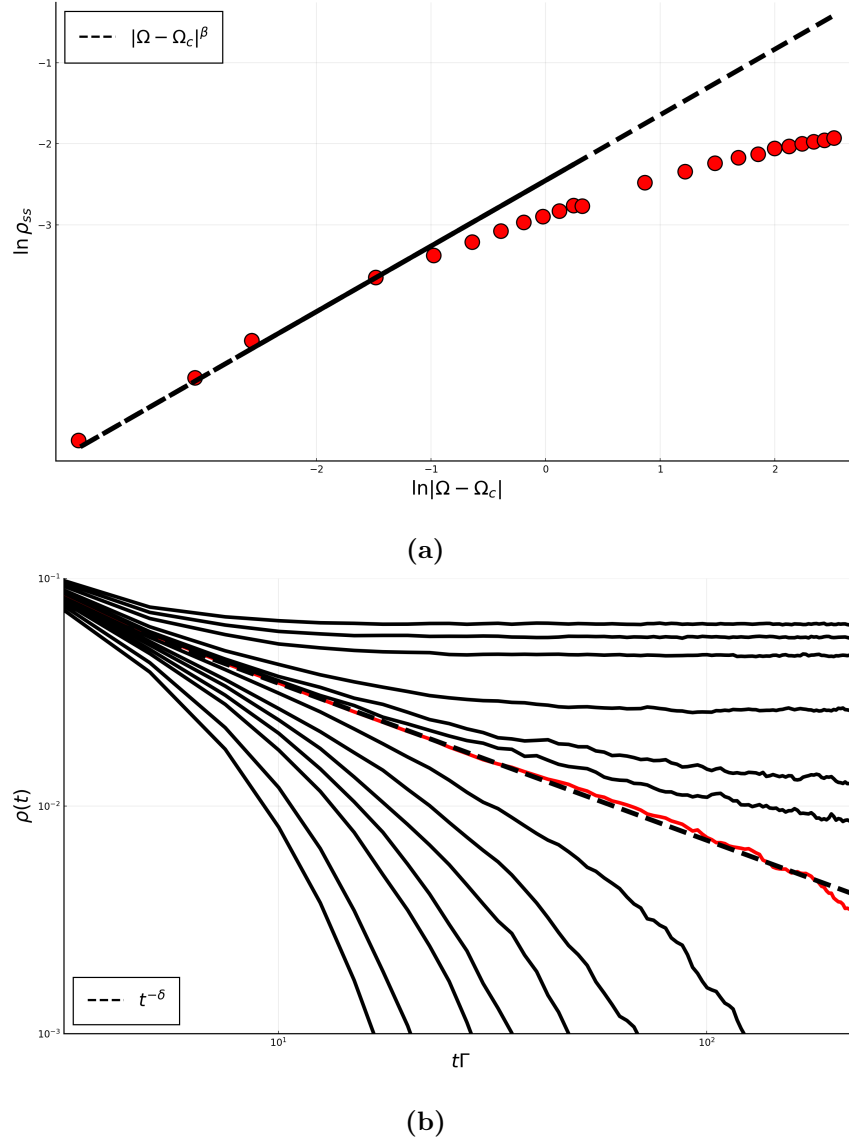


Figure 5.16: Power-law scaling of the 3D system: (a) order parameter yielding $\beta \approx 0.812$, consistent with mean-field predictions for high dephasing; (b) steady-state density yielding $\delta \approx 0.733$, with red curve being the critical point of $\Omega_c \approx 7.56\Gamma$, curves below it are in the absorbing state with $\Omega < \Omega_c$, while those above are in the active state, with $\Omega > \Omega_c$.

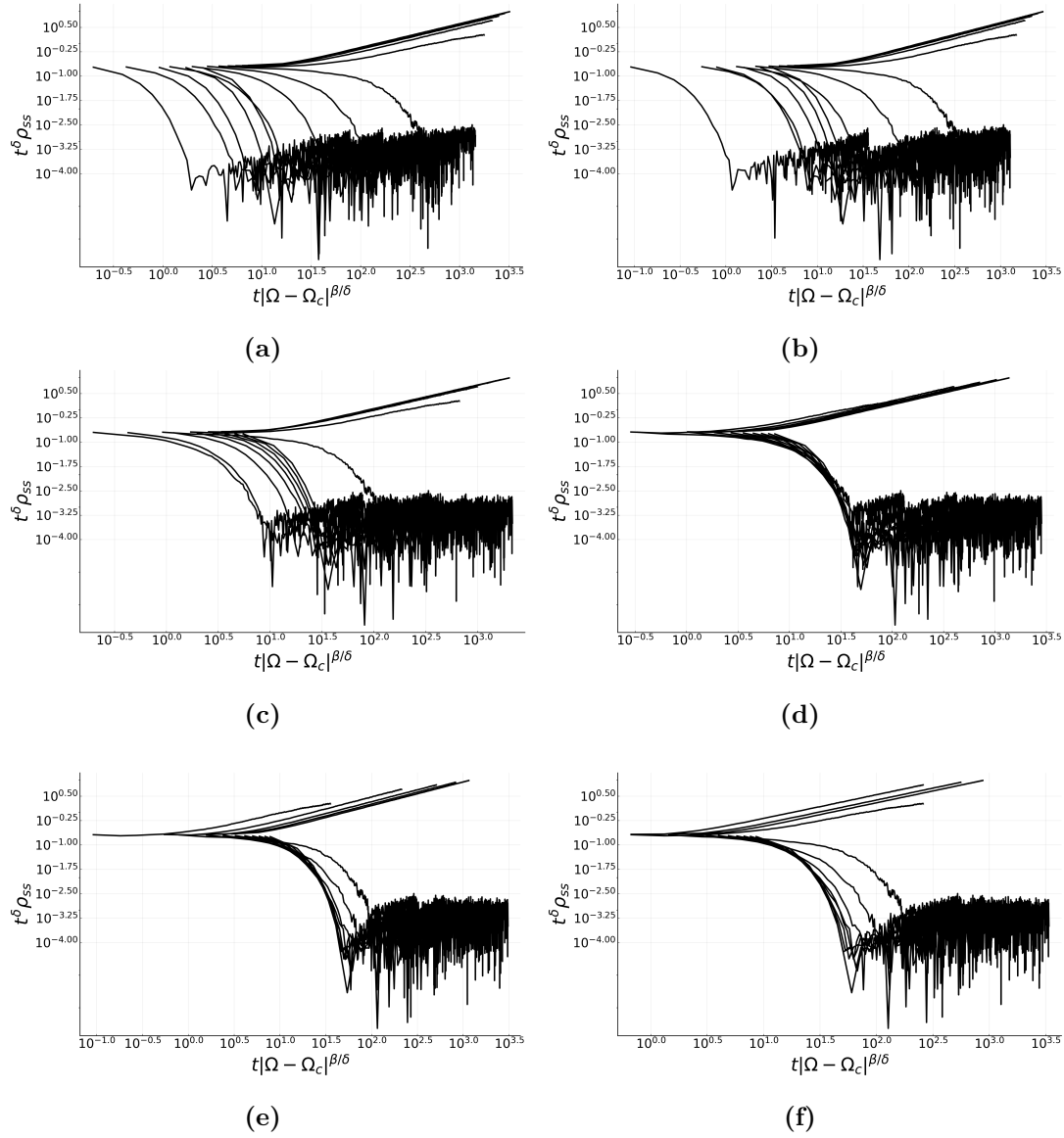


Figure 5.17: Selected collapse images from the 3D atomic simulation with 3375 spins and open boundary conditions. The subfigures show the data collapse of $t^\delta \rho_{ss}$ as a function of $t|\Omega - \Omega_c|^{\beta/\delta}$ for different driving strengths Ω : (a) $\Omega = 6.0\Gamma$, (b) $\Omega = 6.25\Gamma$, (c) $\Omega = 7.0\Gamma$, (d) $\Omega = 7.56\Gamma$, (e) $\Omega = 7.75\Gamma$, and (f) $\Omega = 8.0\Gamma$. The collapse illustrates the transition from decay to a finite steady-state density, with the critical behavior influenced by high dephasing and the enhanced connectivity of the 3D structure.

The critical exponents for the 1D chain, 2D lattice, and 3D cube systems are summarized in the following table:

For the three cases considered, the extracted critical exponents fit reasonably well with the simulation results. Nevertheless, several concluding remarks are in order. First, when comparing the fitted exponents to those of classical phase transitions, the 3D

System	β	δ	β_{DP}	δ_{DP}
1D Chain	0.277	0.159	0.276	0.159
2D Lattice	0.586	0.4577	0.584	0.451
3D Cube	0.812	0.733	0.81	0.73

Table 5.1: Critical exponents β and δ obtained from the TWA method for 1D, 2D, and 3D systems and from DP universality class critical exponents.

case exhibits the closest agreement in the phase transition plot. In contrast, the fits for the 2D and 1D cases deviate progressively further from the transition point of the active density states. Still, the critical points in those cases remain valid regardless of their positions in the phase transition. From the logarithmic decay plots in all three cases, the decay exponent δ aligns directly with its respective critical point, indicating a valid onset of the phase transition.

It is possible that the number of atoms used in the simulations was not sufficient to reach the thermodynamic limit, or that the simulation time was not long enough for the systems to reach a steady state. As a result, the true critical point may not exactly coincide with the position where the active density first becomes nonzero, but rather lies slightly further to the right. Nonetheless, scaling laws in the case of NEPT demonstrate that one can still recover reasonable values of β , δ , and z .

Additionally, during the process of extracting exponents, several approaches did not consistently yield compatible results. In particular, values of β and δ obtained using the same critical point Ω_c were sometimes in disagreement. A viable method to address this issue is finite-size extrapolation with a larger number of spins, on the order of 10^5 . Indeed, the results for δ show convergence toward a reasonable value; however, larger systems with different lattice structures should be investigated. For example, lattices that extend only in one direction or honeycomb lattices, where spin interactions can be more complex. Furthermore, dipole–dipole and van der Waals interactions were not considered; including these could provide valuable insight into long-range correlations between spins.

6 Transition from Classical to Quantum Nonequilibrium Phase Transitions

This chapter explores the differences between quantum and classical nonequilibrium phase transitions using the TWA method. Following the procedure outlined previously, critical exponents for the 2D case are extracted, focusing on the critical points of facilitation in the classical regime. Finally, the chapter investigates the one-to-one mapping between classical and quantum phase transitions from a numerical perspective.

In the high-dephasing limit, where $\gamma \gg \Gamma$, the lack of coherence in the density matrix reduces the effective dimensionality of each spin. Entanglement between spins is minimal, allowing each spin to be treated as nearly independent.

The simulation for the 2D classical case adopts the same structure as the quantum case, with the only difference being the dephasing term, with range of $\gamma = 0.1, 5, 10, 20\Gamma$. Notably, despite the high dephasing, the EM solver remains unsuitable, and the SRIW1 solver is preferred due to its robustness.

In the high-dephasing regime, the phase transition exhibits a less distinct critical point compared to its quantum counterpart. This is attributed to the Wiener noise term in the SDE, where γ significantly influences the noise behavior. As the driving strength increases, the low correlation between neighboring spins causes the phase transition to occur prematurely or with reduced precision.

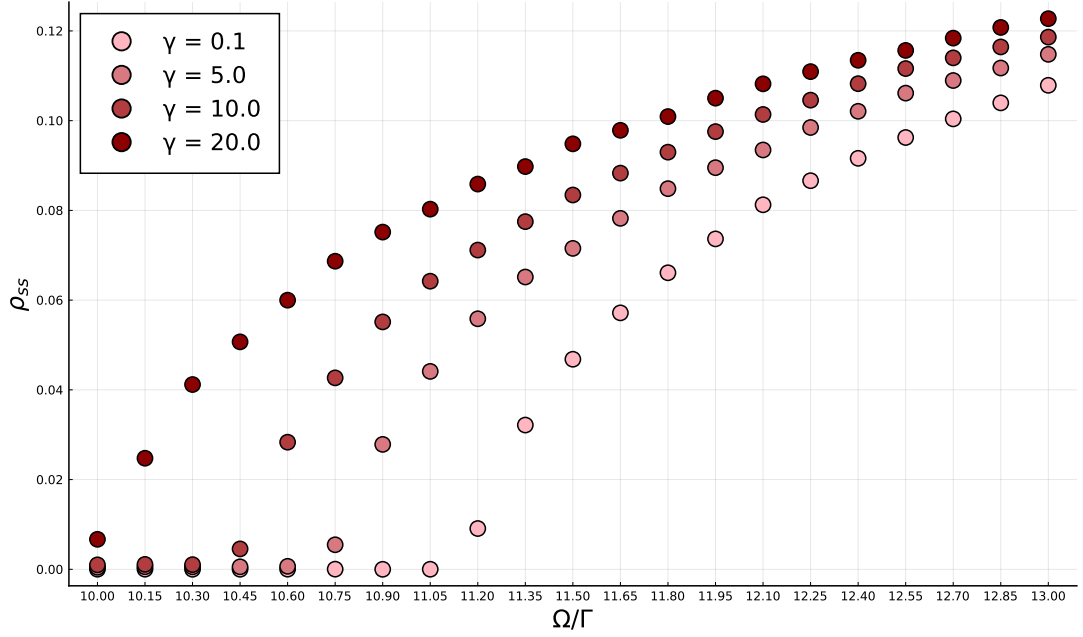


Figure 6.1: Comparison of phase transitions between the classical and quantum cases. With decreasing dephasing $\gamma \rightarrow 0$, the critical point shifts to the right and appears to be sharper.

Figure 6.1 shows the phase transitions of the same system using 3600 spins in a 2D lattice. With higher dephasing, the critical point shifts leftward until reaching a point of saturation at extremely large dephasing values. For larger γ , the phase transition is not as sharp as in the low-dephasing case. While finite-size extrapolation could be used here, a simple scaling-law fit is sufficient since the classical values of the DP universality class have already been observed. The results obtained are consistent with the literature: $\delta \approx 0.47$ and $\beta \approx 0.588$, both within a reasonable error of 0.002. These results are shown in Figure 6.2.

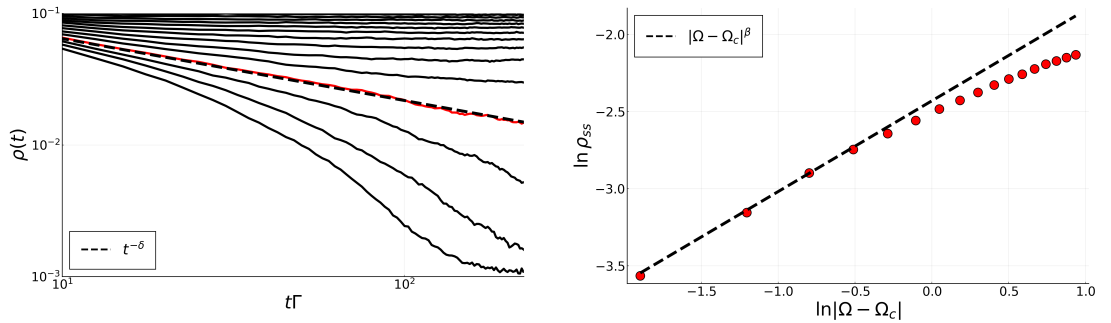


Figure 6.2: The particle density as a function of time (left) found $\delta = 0.47$ and the logarithmic plot (right) of the active phase with the distance from the critical point found $\beta = 0.588$.

The differences between the quantum and classical nonequilibrium phase transitions

cases requires further analysis, particularly in the context of Monte Carlo and TWA methods. For TWA, simulations show a good fit for both δ and β with much more accuracy than Marcuzzi, *et al.*. This is due to the fact that Marcuzzi only uses 4×10^2 spins for their system, while this is an order of magnitude higher.

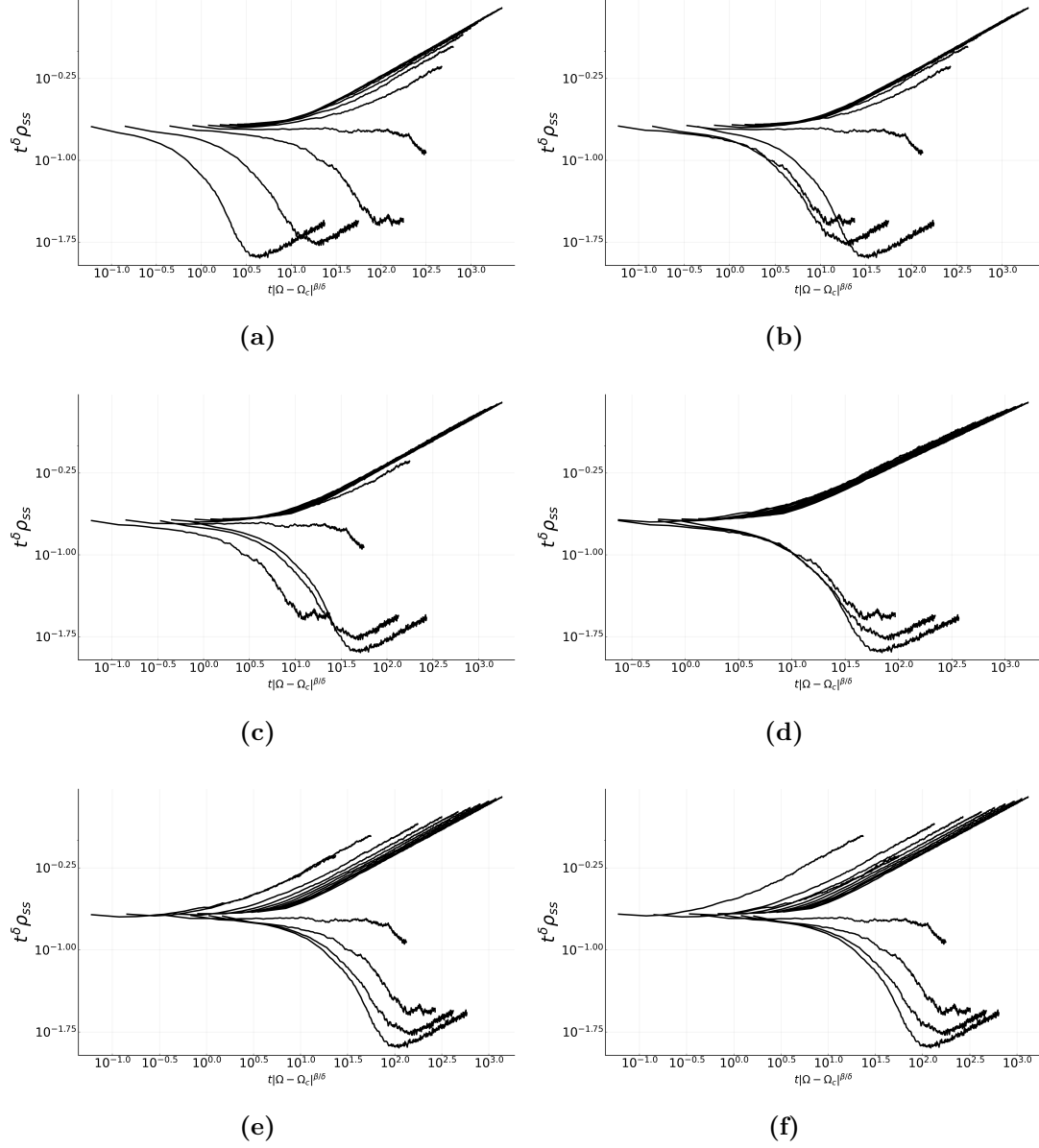


Figure 6.3: Selected collapse images from the 2D atomic simulation with 3600 spins and open boundary conditions. The subfigures show the data collapse of $t^\delta \rho_{ss}$ as a function of $t|\Omega - \Omega_c|^{\beta/\delta}$ for different driving strengths Ω : (a) $\Omega = 10.05$, (b) $\Omega = 10.25$, (c) $\Omega = 10.35$, (d) $\Omega = 10.45$, (e) $\Omega = 10.65$, and (f) $\Omega = 10.7$. The collapse illustrates the transition from decay to a finite steady-state density, highlighting the critical behavior in the 2D geometry.

The collapse figures show characteristic dips followed by upward trends for curves below

the critical point. For values above the critical point, the curves collapse onto each other and point upward. These results show similarities with those reported by Marcuzzi. One particular difference between the quantum regime and the classical is not from the critical exponents extracted, but rather the subtlety in the shape of the curves. In the classical case, the curves below the critical point are visibly clear with little noise, and each shows an upward trend at the end. For the quantum case, the end is noisy and does not point upward like its counterpart. Numerically, this can be explained simply because the phase transition for the quantum case is sharp and abrupt. Around the critical point, the dynamics of the system either approach an active state and remain there or the absorbing one where the density is zero. Because the dynamics of those below the critical are zero, the tail end of the dips has no upward trends. In the classical case, the dephasing is higher, thus making the phase transition a bit less sharp. Around the critical point, the active density for the ones below the critical point is not zero; as a consequence, the upward trends from the tail ends are present. Overall, the TWA method shows that it can indeed extract the correct critical exponents for the classical Rydberg DP universality class. The behavior of the collapse resembles that of previous works, making this method a promising candidate to benchmark against the Monte Carlo approach for future work if one wishes to study the classical non-equilibrium absorbing state phase transition.

In regard to the mapping, numerical simulations using TWA show that there exists a one-to-one correspondence between the two cases. From the study, the power law remains invariant under different values of dephasing γ . One notable observation is that with smaller dephasing, the critical point becomes sharper, making it easier to identify. Nevertheless, facilitation trends appear to behave in a similar manner in both scenarios.

An important issue to address is the very definition of the mapping itself: should such a mapping exist in the first place, or more precisely, are the specified conditions sufficient for it to be considered “quantum”? More specifically, in the equilibrium phase transition of the TFIM, classical phase transitions occur under the influence of thermal fluctuations. At finite temperature T , thermal energy disrupts long-range order. In the 1D TFIM, no phase transition occurs at $T > 0$. In 2D or higher dimensions, however, a transition occurs at a critical temperature T_c , e.g., from the ferromagnetic to the paramagnetic phase. In contrast, quantum phase transitions occur due to the vanishing of thermal fluctuations.

For the NEPT, the assumption $\gamma \ll \Gamma$ enhances the coherence in the density matrix of a single spin, thereby giving the phase transition a quantum character. Simulations show that as γ decreases, the critical point becomes sharper. However, this change has no influence on the power-law scaling of the system. This raises an important concern: does decreasing γ alone truly place the system in the quantum regime, or must additional conditions be imposed? More specifically, should temperature be explicitly considered, since thermal fluctuations universally affect spin dynamics and may play a decisive role in distinguishing between classical and quantum behavior? Though this hypothesis can be refuted by assuming that when the system is approach a high dephasing, the quantum coherences are effectively suppressed, and the dynamics become dominated by classical stochastic noise. In this limit, thermal fluctuations and dephasing have equivalent ef-

fects on the spin dynamics — both destroy phase coherence and wash out any quantum correlations. Therefore, increasing γ naturally drives the system toward the classical regime, even without explicitly introducing temperature.

Another consideration should be addressed here is the universal coherent driving for the entire system, alternatively, one should consider the case of incoherent or locally varying driving, where each spin is subject to an independent or spatially fluctuating phase. This breaks the global phase coherence and can significantly alter the collective behavior, potentially suppressing or broadening the phase transition. Alternatively, one should consider the case of spatially disordered driving, where the driving strength or phase varies across the lattice. Such disorder can lead to nontrivial effects such as Griffiths phases or smeared transitions.

Conclusion and Outlook

Rydberg physics is a fascinating topic that has currently garnered much attention in the field of many-body physics from both theoretical and experimental communities, with new discoveries happening every decade. The main objective of this research is to determine the critical exponents of a quantum absorbing-state phase transition in the low dephasing regime and to gain more insight into the universality class of Directed Percolation within the quantum framework. While classical DP models are well-characterized, with critical exponents extracted through numerical simulations like Monte Carlo methods, the quantum regime remains largely a mystery due to expensive computational resources arising from quantum fluctuations, spin entanglement, and a lack of a unified analytical framework. By focusing on the low dephasing regime, where coherence contributes significantly to the behavior of the system, the research aims to bridge the gap between the classical and quantum understanding, extracting critical exponents such as β , related to the order parameter, and δ , related to survival probability, to evaluate whether the quantum regime belongs to the same universality class and to establish the mapping between quantum and classical ASPT.

To achieve this goal, Rydberg physics, and in particular Rydberg facilitation, is a good candidate as a starting point. Rydberg facilitation is a process in which strongly interacting Rydberg atom systems, where excitation of one atom to a high-energy Rydberg state shifts the energy levels of neighboring atoms, effectively detuning them into resonance with an external driving laser. This facilitation occurs when the laser detuning matches the van der Waals interaction-induced level shift, the C_6 interaction, allowing facilitation to occur. In this thesis, Rydberg facilitation serves as the key mechanism for activity spreading in the lattice model, mimicking the percolation process in DP. This particular system naturally realizes ASPT, where the system can transition between the absorbing state (where all atoms are inactive with no activation) to the active one (sustained activation occurs across the lattice sites). In this case, facilitation acts as a spreading mechanism for the system, analogous to classical DP, where activity might either spread with probability p or die out. In this Rydberg setup, a coherent laser drives the system with external factors from the environment, i.e., dephasing and decay terms. By observing the behavior of the system near the critical point, one can study the properties of the system through power-law scaling in the density of active sites $\rho(t)$ to understand the quantum universality class beyond the classical limit.

To tackle a task that Monte Carlo fails to perform, one primary, promising candidate lies in the Truncated Wigner Approximation for spins. The TWA simulates the dynamics of the open quantum Rydberg system. TWA maps the quantum master equation in Hilbert space to a semi-classical phase space representation using phase-point operators. Specifically, for spin-1/2 systems, the density matrix ρ is represented in a discrete Wigner function over phase space parameterized by spherical coordinates (θ, ϕ) , where

quantum operators such as the Pauli matrices σ^x , σ^y , σ^z are mapped to classical variables S^x , S^y , S^z with added noise to capture quantum fluctuations. The mapping uses kernel operators, such as the PPO for spins, which transform the Lindblad master equation $d\rho/dt = -i[H, \rho] + \sum L[\rho]$ into a set of SDE in phase space. The dynamics become the contribution from the drift term and the noise term, solved with numerical solvers like SRIW1 or EM.

Truncation is the key approximation in TWA; this neglects higher-order derivatives in the phase space expansion. The benefits of truncation include computational efficiency, since TWA scales polynomially with system size, making it feasible for large lattices where exact methods like density matrix evolution fail due to exponential Hilbert space growth. It also captures quantum effects like fluctuations and coherence with a high degree of accuracy. However, the losses are notable; truncation can underestimate long-time correlations or entanglement, leading to inaccuracies in strongly quantum regimes (e.g., as seen in the benchmark for the TFIM, where TWA fails to capture the correct critical point).

The simulations using TWA reveal that the critical exponents in the low dephasing regime are similar to those in the high dephasing regime. These found values align closely with the classical DP universality class exponents. No significant differences in exponents were observed between low and high dephasing, except for the z exponent. This suggests that quantum DP falls into the same universality class and corresponds to a direct mapping with no dimensional addition, as in the case of EPT. However, a key distinction is that the critical point is sharper in the low dephasing case. In high dephasing, the transition curve is smoother and broader, with less abrupt shifts from the absorbing to the active phase. In low dephasing, coherence enhances correlations, leading to a sharper critical point.

Overall, TWA proves sufficient for capturing the quantum part of the Hamiltonian, as demonstrated in the case of a single driven atom and the Dicke model with dissipation. However, in the case of NEPT, in the low dephasing regime, quantum effects dominate, and TWA's semi-classical nature might overlook coherence and spin entanglement. To better understand TWA and NEPT, future research should include direct comparisons with Monte Carlo methods. Extending to higher-order truncations could improve accuracy. For more complex behavior, incorporating long-range interactions, disorder in lattice spacing, or time-dependent driving could reveal phenomena like Griffiths phases or non-DP universality [68–70]. Further experimental validation using Rydberg atoms would test these predictions. Ultimately, this work breaks new ground for improving quantum simulation for NEPT, with potential applications in quantum computing and many-body physics.

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Declaration of Authorship

I hereby declare that I have written this Master's thesis independently and have used only the sources and aids indicated. All material that is quoted directly or paraphrased from published or unpublished sources have been clearly cited. I further acknowledge that AI-based Large Language Models, were solely used to help improve grammar and phrasing. Any assistance received during the research process has been acknowledged. This thesis has not been submitted, either in part or in full, for the award of any other degree or diploma at this or any other institution. This work was carried out at RPTU Kaiserslautern–Landau, in partial fulfillment of the requirements of the Erasmus Mundus Joint Master Degree Programme QuanTEEM and the Advanced Quantum Physics Master's program at RPTU.

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